Molecular Biology & Genetics

Career Avenues

Contents

MOLECULAR STRUCTURE OF GENES AND CHROMOSOMES 4
DNA REPLICATION AND ITS CONTROL 12
TRANSCRIPTION AND ITS CONTROL 40
TRANSLATIONAL PROCESS 57
REGULATORY CONTROL IN PROKARYOTES AND EUKARYOTES 66
MENDELIAN INHERITANCE 73
GENE INTERACTION 85
LINKAGE, RECOMBINATION AND CHROMOSOME MAPPING 92
EXTRACHROMOSMAL INHERITANCE 99
CHROMOSOMAL VARIATION 110
POPULATION GENETICS 113
TRANSPOSABLE ELEMENTS 119
MOLECULAR BASIS OF GENETIC DISEASES AND ITS APPLICATIONS 126

Career Avenues

MOLECULAR STRUCTURE OF GENES AND CHROMOSOMES

Eukaryotic genome can be split into two components: the nuclear genome and the mitochondrial genome. This is the typical pattern for most eukaryotes, the bulk of the genome being contained in the chromosomes in the cell nucleus and a much smaller part located in the mitochondria and, in the case of photosynthetic organisms, in the chloroplasts.

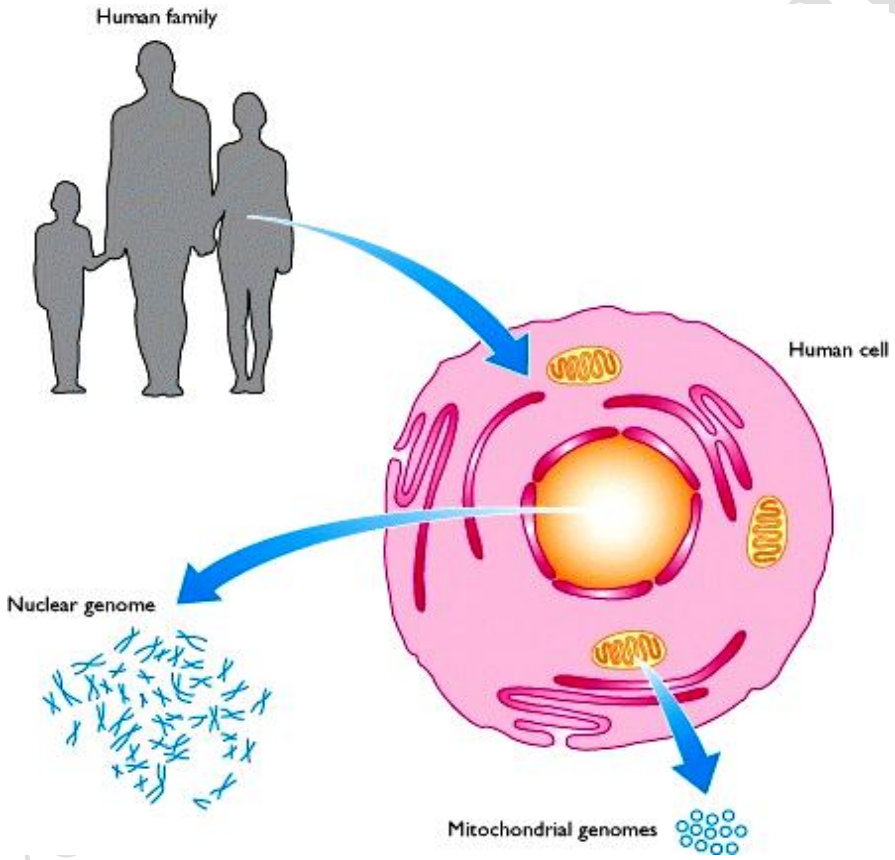


Figure: The nuclear and mitochondrial components of the human genome

EUKARYOTIC NUCLEAR GENOMES

The nuclear genome is split into a set of linear DNA molecules, each contained in a chromosome. No exceptions to this pattern are known: all eukaryotes that have been studied have at least two chromosomes and the DNA molecules are always linear. The only variability at this level of eukaryotic genome structure lies with chromosome number, which appears to be

unrelated to the biological features of the organism. For example, yeast has 16 chromosomes, four times as many as the fruit fly. Nor is chromosome number linked to genome size: some salamanders have genomes 30 times bigger than the human version but split into half the number of chromosomes. These comparisons are interesting but at present do not tell us anything useful about the genomes themselves; they are more a reflection of the non-uniformity of the evolutionary events that have shaped genome architecture in different organisms.

EUKARYOTIC NUCLEAR GENOMES

The nuclear genome is split into a set of linear DNA molecules, each contained in a chromosome. No exceptions to this pattern are known: all eukaryotes that have been studied have at least two chromosomes and the DNA molecules are always linear. The only variability at this level of eukaryotic genome structure lies with chromosome number, which appears to be unrelated to the biological features of the organism. For example, yeast has 16 chromosomes, four times as many as the fruit fly. Nor is chromosome number linked to genome size: some salamanders have genomes 30 times bigger than the human version but split into half the number of chromosomes. These comparisons are interesting but at present do not tell us anything useful about the genomes themselves; they are more a reflection of the non-uniformity of the evolutionary events that have shaped genome architecture in different organisms.

PACKAGING OF DNA INTO CHROMOSOMES

Chromosomes are much shorter than the DNA molecules that they contain: the average human chromosome has just under 5 cm of DNA. A highly organized packaging system is therefore needed to fit a DNA molecule into its chromosome. We must understand this packaging system before we start to think about how genomes function because the nature of the packaging has an influence on the processes involved in expression of individual genes

The important breakthroughs in understanding DNA packaging were made in the early 1970s by a combination of biochemical analysis and electron microscopy. It was already known that nuclear DNA is associated with DNA-binding proteins called histones but the exact nature of the association had not been delineated. In 1973-74 several groups carried out nuclease protection experiments on chromatin (DNA-histone complexes) that had been gently extracted from nuclei by methods designed to retain as much of the chromatin structure as possible. In a nuclease protection experiment the complex is treated with an enzyme that cuts the DNA at positions that are not ‘protected’ by attachment to a protein. The sizes of the resulting DNA fragments indicate the positioning of the protein complexes on the original DNA molecule (Figure 2.4). After limited nuclease treatment of purified chromatin, the bulk of the DNA fragments have lengths of approximately 200 bp and multiples thereof, suggesting a regular spacing of histone proteins along the DNA.

Career Avenues

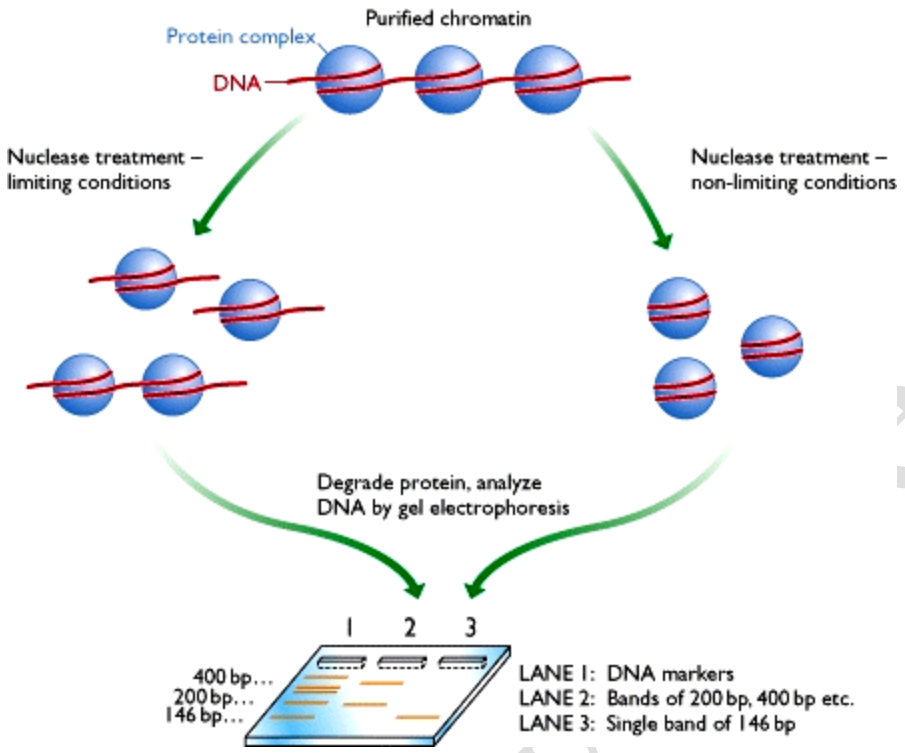


Figure: Nuclease protection analysis of chromatin from human nuclei

Chromatin is gently purified from nuclei and treated with a nuclease enzyme. On the left, the nuclease treatment is carried out under limiting conditions so that the DNA is cut, on average, just once in each of the linker regions between the bound proteins. After removal of the protein, the DNA fragments are analyzed by agarose gel electrophoresis and found to be 200 bp in length, or multiples thereof. On the right, the nuclease treatment proceeds to completion, so all the DNA in the linker regions is digested. The remaining DNA fragments are all 146 bp in length. The results show that in this form of chromatin, protein complexes are spaced along the DNA at regular intervals, one for each 200 bp, with 146 bp of DNA closely attached to each protein complex.

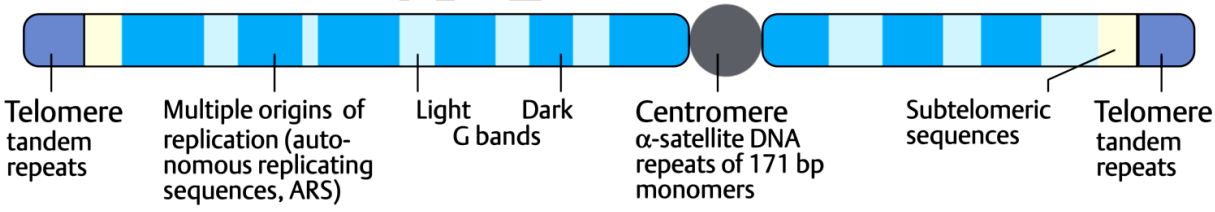
In 1974 these biochemical results were supplemented by electron micrographs of purified chromatin, which enabled the regular spacing inferred by the protection experiments to be visualized as beads of protein on the string of DNA (Figure 2.5A). Further biochemical analysis indicated that each bead, or nucleosome, contains eight histone protein molecules, these being two each of histones H2A, H2B, H3 and H4. Structural studies have shown that these eight proteins form a barrel-shaped core octamer with the DNA wound twice around the outside (Figure 2.5B). Between 140 and 150 bp of DNA (depending on the species) are associated with the nucleosome particle, and each nucleosome is separated by 50–70 bp of linker DNA, giving the repeat length of 190–220 bp previously shown by the nuclease protection experiments.

Career Avenues

As well as the proteins of the core octamer, there is a group of additional histones, all closely related to one another and collectively called linker histones. In vertebrates these include histones H1a-e, H1°, H1t and H5. A single linker histone is attached to each nucleosome, to form the chromatosome, but the precise positioning of this linker histone is not known. Structural studies support the traditional model in which the linker histone acts as a clamp, preventing the coiled DNA from detaching from the nucleosome (Figure 2.5C; Zhou et al., 1998; Travers, 1999). However, other results suggest that, at least in some organisms, the linker histone is not located on the extreme surface of the nucleosome-DNA assembly, as would be expected if it really were a clamp, but instead is inserted between the core octamer and the DNA (Pruss et al., 1995; Pennisi, 1996).

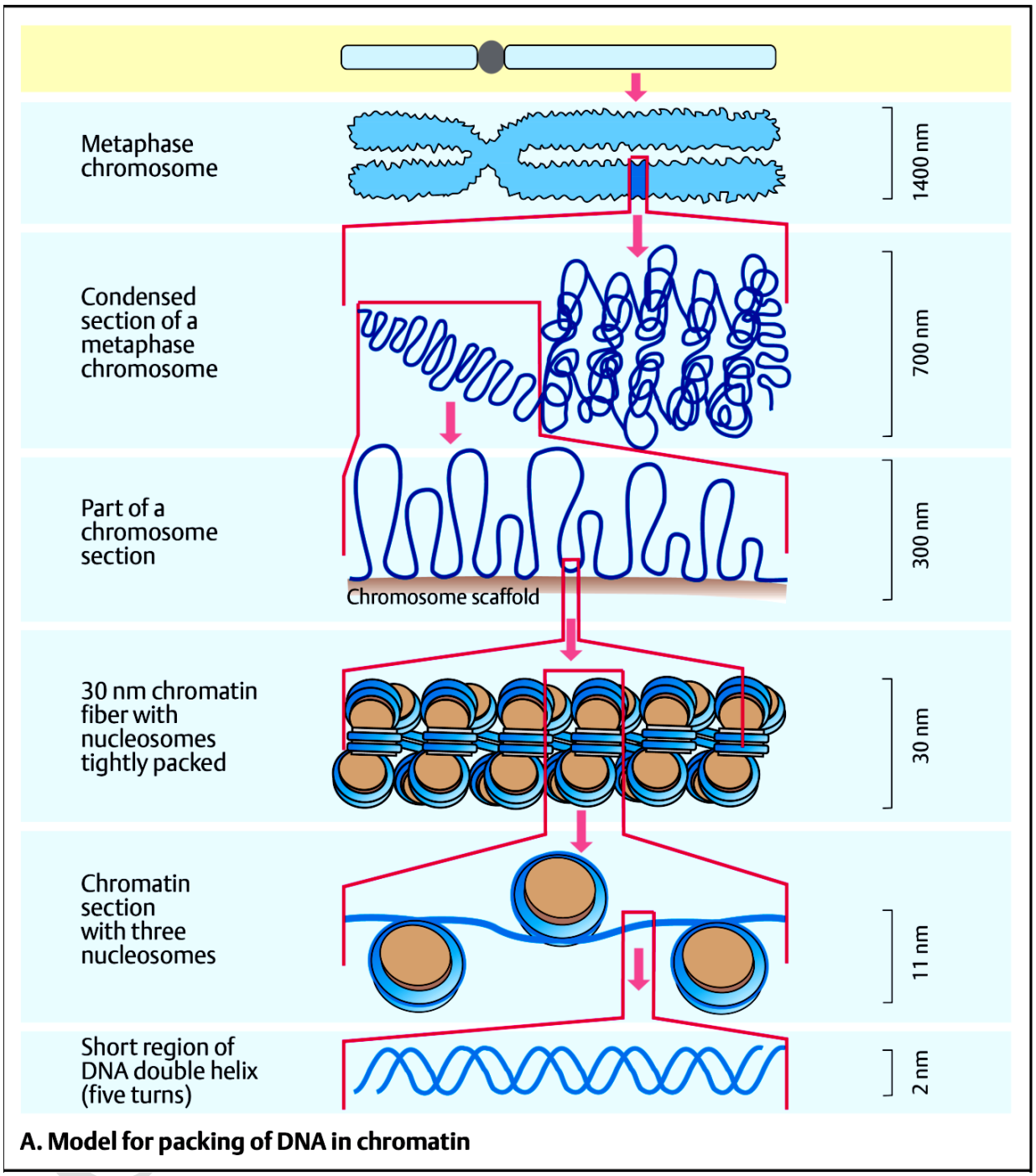
The 'beads-on-a-string' structure shown in Figure 2.5A is thought to represent an unpacked form of chromatin that occurs only infrequently in living nuclei. Very gentle cell breakage techniques developed in the mid-1970s resulted in a more condensed version of the complex, called the 30 nm fiber (it is approximately 30 nm in width). The exact way in which nucleosomes associate to form the 30 nm fiber is not known, but several models have been proposed, the most popular of which is the solenoid structure shown in Figure.

The individual nucleosomes within the 30 nm fiber may be held together by interactions between the linker histones, or the attachments may involve the core histones, whose protein 'tails' extend outside the nucleosome. The latter hypothesis is attractive because chemical modification of these tails results in the 30 nm fiber opening up, enabling genes contained within it to be activated.

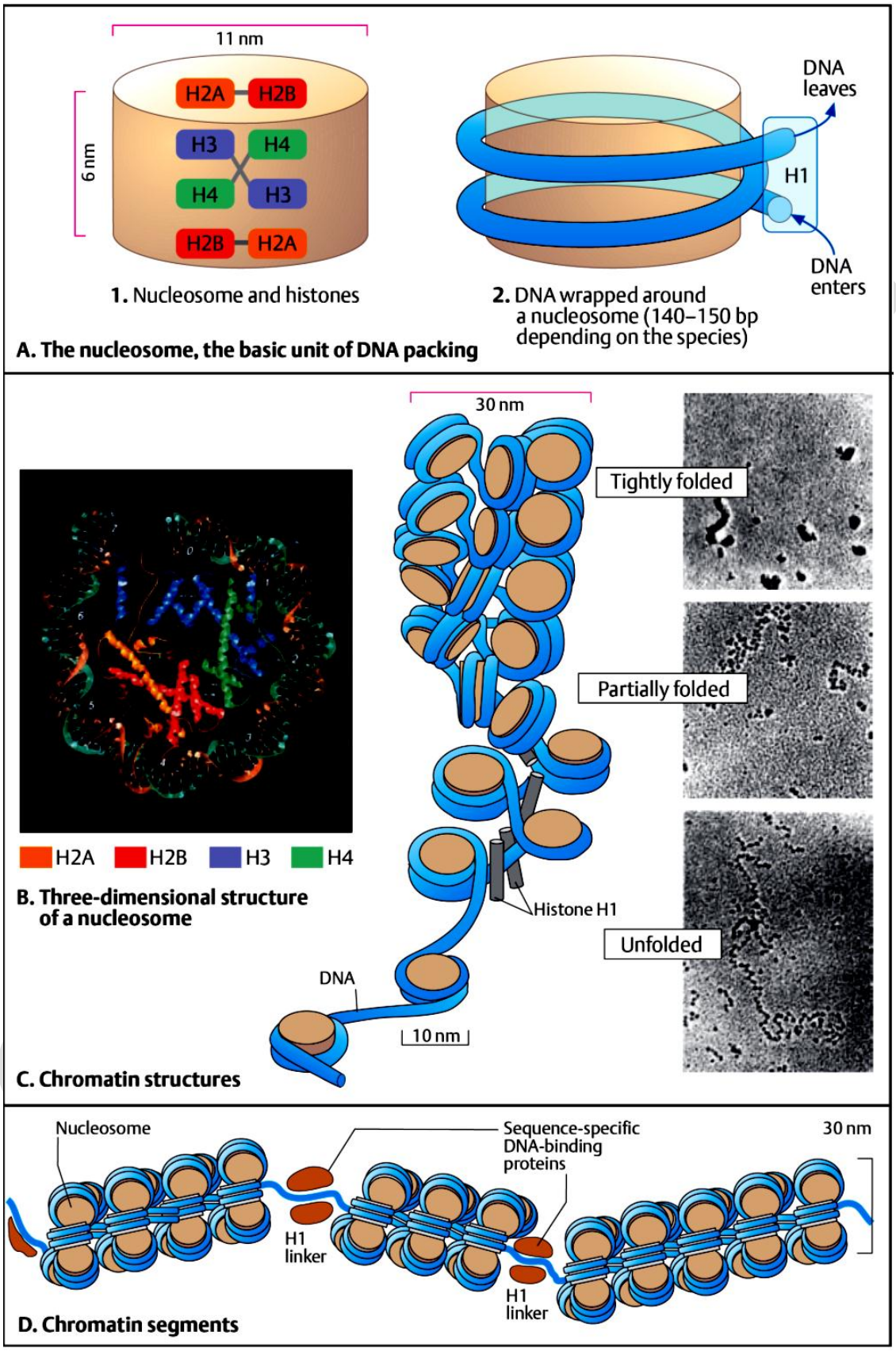


A. Basic features of a eukaryotic chromosome

Career Avenues



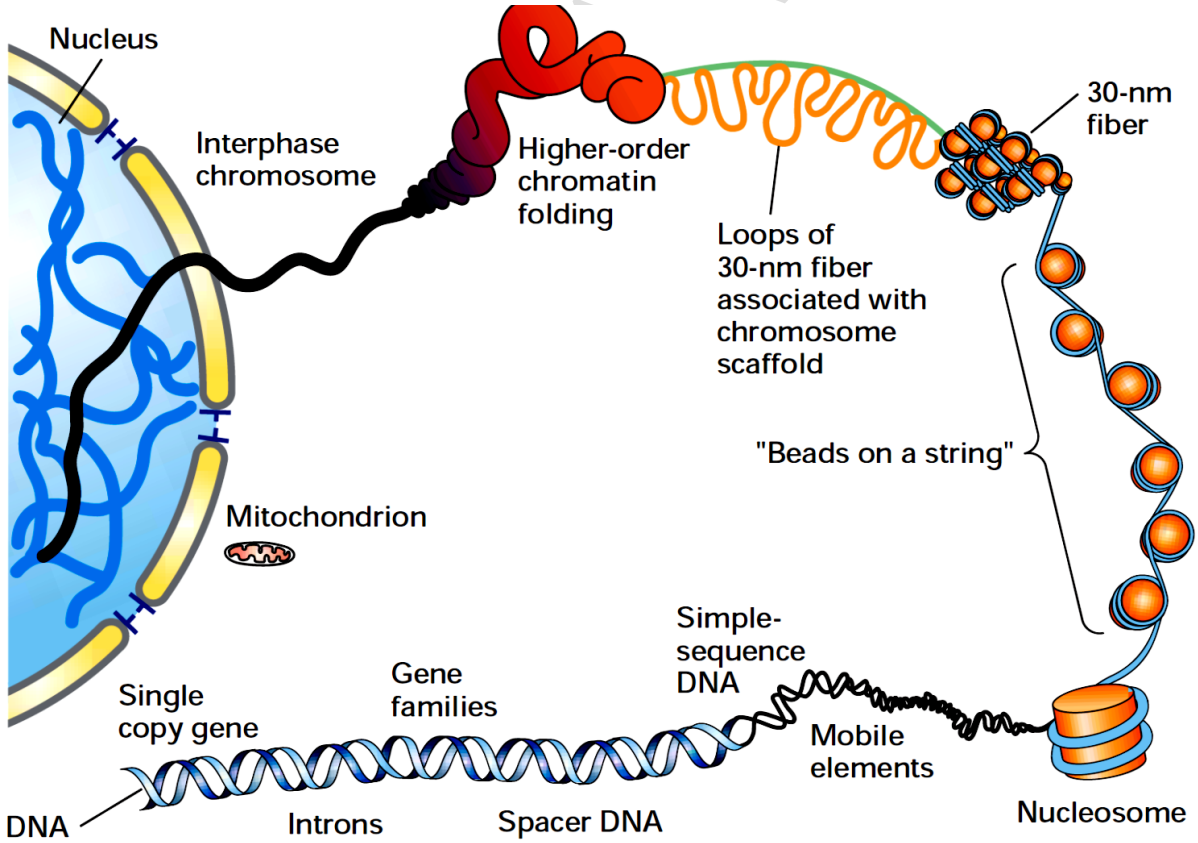
A. Model for packing of DNA in chromatin


11 nm
6 nm
H2A
H2B
H3
H4
H4
H3
H2B
H2A
1. Nucleosome and histones
DNA leaves
H1
DNA enters
2. DNA wrapped around a nucleosome (140–150 bp depending on the species)
A. The nucleosome, the basic unit of DNA packing
H2A
H2B
H3
H4
B. Three-dimensional structure of a nucleosome
30 nm
Tightly folded
Partially folded
Unfolded
Histone H1
DNA
10 nm
C. Chromatin structures
Nucleosome
Sequence-specific DNA-binding proteins
30 nm
H1 linker
H1 linker
D. Chromatin segments

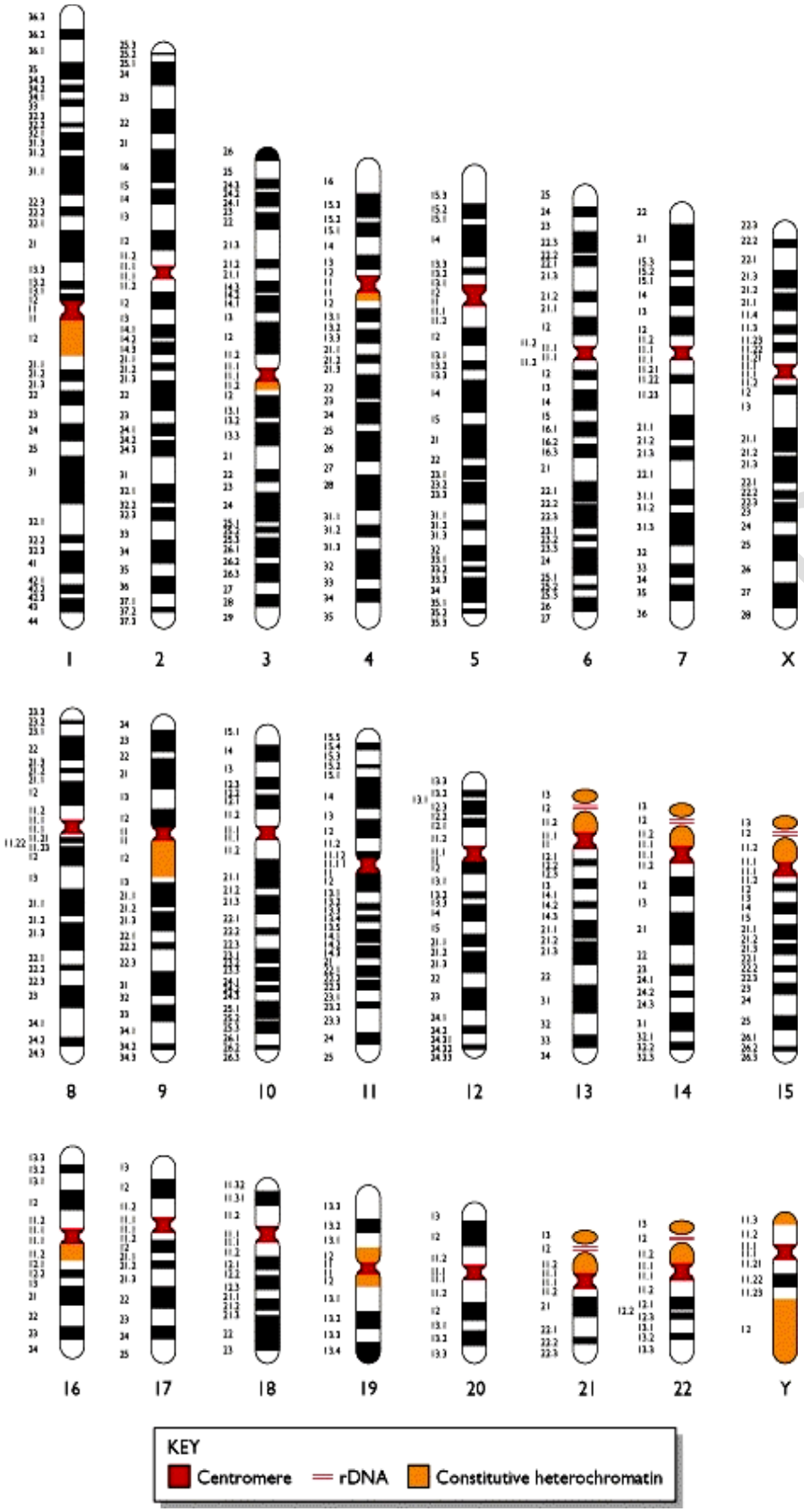
THE SPECIAL FEATURES OF METAPHASE CHROMOSOMES

The 30 nm fiber is probably the major type of chromatin in the nucleus during interphase, the period between nuclear divisions. When the nucleus divides, the DNA adopts a more compact form of packaging, resulting in the highly condensed metaphase chromosomes that can be seen with the light microscope and which have the appearance generally associated with the word 'chromosome'. The metaphase chromosomes form at a stage in the cell cycle after DNA replication has taken place and so each one contains two copies of its chromosomal DNA molecule. The two copies are held together at the centromere, which has a specific position within each chromosome. Individual chromosomes can therefore be recognized because of their size and the location of the centromere relative to the two ends. Further distinguishing features are revealed when chromosomes are stained. There are a number of different staining techniques (Table 2.4), each resulting in a banding pattern that is characteristic for a particular chromosome. This means that the set of chromosomes possessed by an organism can be represented as a karyogram, in which the banded appearance of each one is depicted.

The human karyogram is shown in figure on the next page.



© Career Avenues

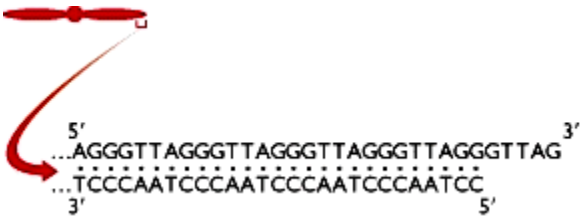


Career Avenues

Both the DNA in the centromere regions, and the proteins attached to it, have special features. The nucleotide sequence of centromeric DNA is best understood in the plant Arabidopsis thaliana, whose amenity to genetic analysis has enabled the positions of the centromeres on the DNA sequence to be located with some precision. Also, a special effort was made to sequence these centromeric regions, which are frequently excluded from draft genome sequences because of problems in obtaining an accurate reading through the highly repetitive structure that characterize these regions. Arabidopsis centromeres span 0.9–1.2 Mb of DNA and each one is made up largely of 180-bp repeat sequences. In humans the equivalent sequences are 171 bp and are called alphoid DNA. Before the Arabidopsis sequences were obtained it was thought that these repeat sequences were by far the principal component of centromeric DNA. However, Arabidopsis centromeres also contain multiple copies of genome-wide repeats, along with a few genes, the latter at a density of 7–9 per 100 kb compared with 25 genes per 100 kb for the non-centromeric regions of Arabidopsis chromosomes (Copenhaver et al., 1999). The discovery that centromere DNA contains genes was a big surprise because it was thought that these regions were genetically inactive.

A second important part of the chromosome is the terminal region or telomere. Telomeres are important because they mark the ends of chromosomes and therefore enable the cell to distinguish a real end from an unnatural end caused by chromosome breakage - an essential requirement because the cell must repair the latter but not the former. Telomeric DNA is made up of hundreds of copies of a repeated motif, 5′-TTAGGG-3′ in humans, with a short extension of the 3′ terminus of the double-stranded DNA molecule (Figure). Two special proteins bind to the repeat sequences in human telomeres. These are called TRF1, which helps to regulate the length of the telomere, and TRF2, which maintains the single-strand extension. If TRF2 is inactivated then this extension is lost and the two polynucleotides fuse together in a covalent linkage (van Steensel et al., 1998). Other telomeric proteins are thought to form a linkage between the telomere and the periphery of the nucleus, the area in which the chromosome ends are localized (Tham and Zakian, 2000). Still others mediate the enzymatic activity that maintains the length of each telomere during DNA replication. We will return to this last activity in Section 13.2.4: it critical to the survival of the chromosome and may be a key to understanding cell senescence and death.

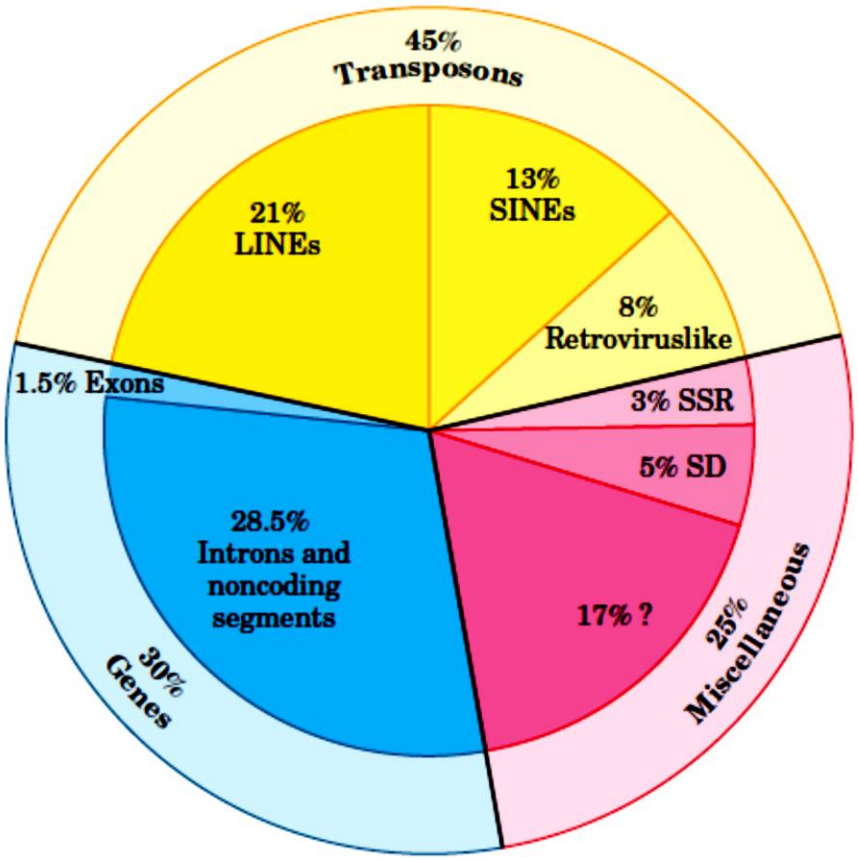
Fig: Telomeres



Career Avenues

Types of sequences in the human genome:

This pie chart divides the genome into transposons (transposable elements), genes, and miscellaneous sequences. There are four main classes of transposons. Long interspersed elements (LINEs), 6 to 8 kbp long (1 kbp = 1,000 bp), typically include a few genes encoding proteins that catalyze transposition. The genome has about 850,000 LINEs. Short interspersed elements (SINEs) are about 100 to 300 bp long. Of the 1.5 million in the human genome more than 1 million are Alu elements, so called because they generally include one copy of the recognition sequence for *Alu*I, a restriction endonuclease. The genome also contains 450,000 copies of retroviruslike transposons, 1.5 to 11 kbp long. Although these are "trapped" in the genome and cannot move from one cell to another, they are evolutionarily related to the retroviruses (Chapter 26), which include HIV. A final class of transposons (making up _1% and not shown here) consists of a variety of transposon remnants that differ greatly in length. About 30% of the genome consists of sequences included in genes for proteins, but only a small fraction of this DNA is in exons (coding sequences). Miscellaneous sequences include simple-sequence repeats (SSR) and large segmental duplications (SD), the latter being segments that appear more than once in different locations. Among the unlisted sequence elements (denoted by a question mark) are genes encoding RNAs (which can be harder to identify than genes for proteins) and remnants of transposons that have been evolutionarily altered so that they are now hard to identify.



Molecular structure of genes

In molecular terms, a gene commonly is defined as the entire nucleic acid sequence that is necessary for the synthesis of a functional gene product (polypeptide or RNA). According to this definition, a gene includes more than the nucleotides encoding the amino acid sequence of a protein, referred to as the coding region. A gene also includes all the DNA sequences required for synthesis of a particular RNA transcript. In eukaryotic genes, transcription-control regions known as enhancers can lie 50 kb or more from the coding region. Other critical noncoding regions in eukaryotic genes are the sequences that specify 3' cleavage and polyadenylation, known as poly(A) sites, and splicing of primary RNA transcripts, known as splice sites.

Although most genes are transcribed into mRNAs, which encode proteins, clearly some DNA sequences are transcribed into RNAs that do not encode proteins (e.g., tRNAs and rRNAs). However, because the DNA that encodes tRNAs and rRNAs can cause specific phenotypes when it is mutated, these DNA regions generally are referred to as tRNA and rRNA genes, even though the final products of these genes are RNA molecules and not proteins. Many other RNA molecules described in later chapters also are transcribed from non-protein-coding genes.

Most Eukaryotic Genes Produce Monocistronic mRNAs and Contain Lengthy Introns, on the other hand many bacterial mRNAs are polycistronic; that is, a single mRNA molecule (e.g., the mRNA encoded by the trp operon) includes the coding region for several proteins that function together in a biological process. In contrast, most eukaryotic mRNAs are monocistronic; that is, each mRNA molecule encodes a single protein. This difference between polycistronic and monocistronic mRNAs correlates with a fundamental difference in their translation.

Unlike bacterial and yeast genes, which generally lack introns, most genes in multicellular animals and plants contain introns, which are removed during RNA processing. In many cases, the introns in a gene are considerably longer than the exons. For instance, of the ≈50,000 base pairs composing many human genes encoding average-size proteins, more than 95 percent are present in introns and noncoding 5' and 3' regions. Many large proteins in higher organisms have repeated domains and are encoded by genes consisting of repeats of similar exons separated by introns of variable length. An example of this is fibronectin, a component of the extracellular matrix that is encoded by a gene containing multiple copies of three types of exons

Career Avenues

DNA REPLICATION AND ITS CONTROL

General Features of DNA Replication

The double-helical model for DNA includes the concept that the two strands are complementary. Thus, each strand can in principle serve as the template for making its own partner.

This is called **semiconservative** replication because each daughter duplex has one parental strand and one new strand.

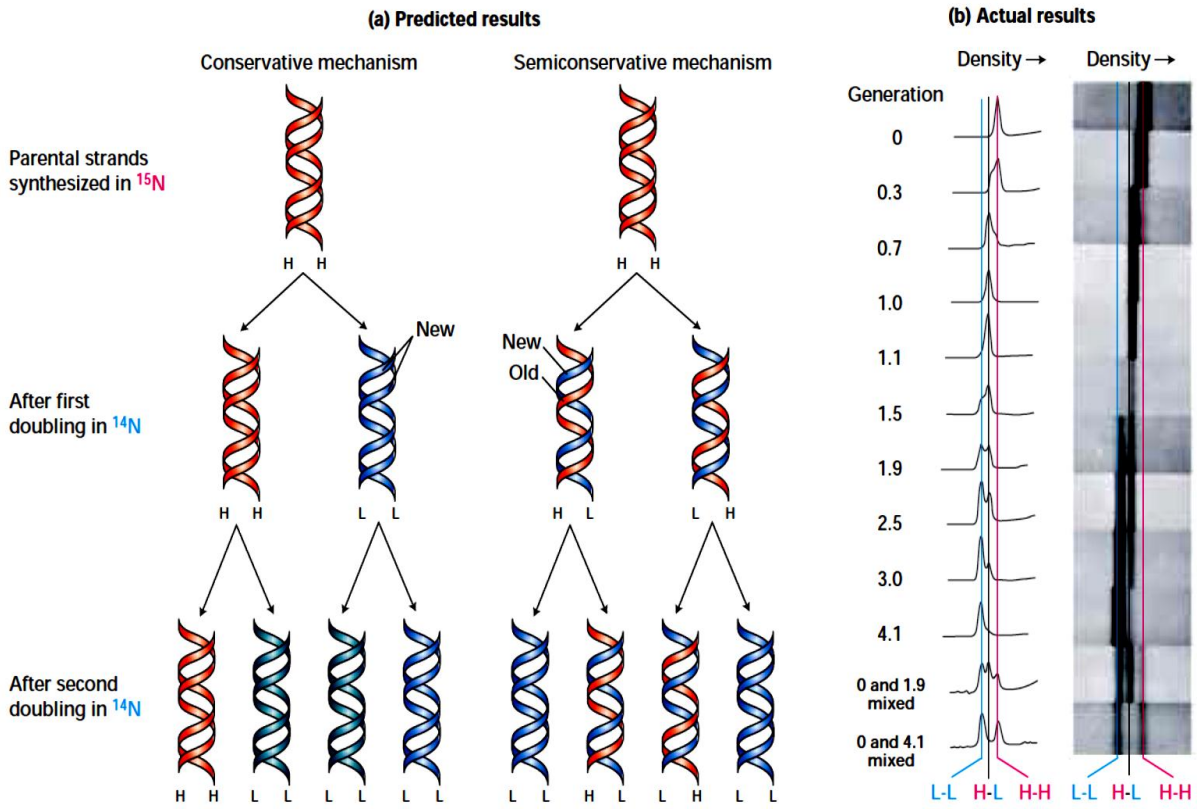
Another potential mechanism is **conservative** replication, in which the two parental strands stay together and somehow produce another daughter helix with two completely new strands. Yet another possibility is random **dispersive** replication, in which the DNA becomes fragmented so that new and old DNAs coexist in the same strand after replication.

In 1958, Matthew Meselson and Franklin Stahl performed an experiment to distinguish among these three possibilities.

The Meselson-Stahl experiment:

(a)- Cells were grown for many generations in a medium containing only heavy nitrogen, 15N, so that all the nitrogen in their DNA was 15N, as shown by a single band (blue) when centrifuged in a CsCl density gradient.
(b)- Once the cells had been transferred to a medium containing only light nitrogen, 14N, cellular DNA isolated after one generation equilibrated at a higher position in the density gradient (purple band).
(c)- Continuation of replication for a second generation yielded two hybrid DNAs and two light DNAs (red), confirming semiconservative replication.

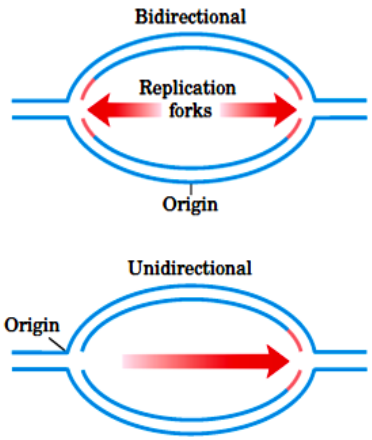
Career Avenues



Replication begins at an origin and usually proceeds bidirectionally. Replication is a highly coordinated process in which the parent strands are simultaneously unwound and replicated. The evidence was provided by John Cairns, using autoradiography.

In this experiment, DNA was radiolabelled by using radiolabeled thiamine with tritium ${}_1H^{3.}$

Addition of ${}_1H^3$ for a short period just before the reaction is stopped allows a distinction to be made between unidirectional and bidirectional replication, by determining whether label (red) is found at one or both replication forks in autoradiogram. This technique has revealed bidirectional replication in *E. coli.*

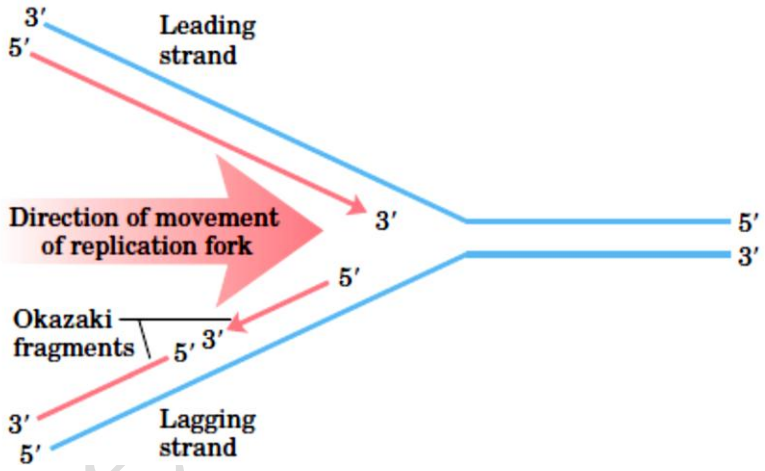


Career A

DNA synthesis proceeds in a 5' to 3' direction and is Semidiscontinuous. A new strand of DNA is always synthesized in the 5' to 3' direction, with the free 3'-OH as the point at which the DNA is elongated. The replication requires a template and RNA primer.

If synthesis always proceeds in the 5' to 3' direction, how can both strands be synthesized simultaneously?

Okazaki found that one of the new DNA strands is synthesized in short pieces, now called Okazaki fragments. This work ultimate layed to the conclusion that one strand is synthesized continuously and the other discontinuously. The continuous strand, or leading strand, is the one in which 5' to 3' synthesis proceeds in the same direction as replication fork movement. The discontinuous strand, or lagging strand, is the one in which 5' to 3' synthesis proceeds in the direction opposite to the direction of fork movement. Okazaki fragments range in length from a few hundred to a few thousand nucleotides, depending on the cell type. In bacteria, Okazaki fragments are ~1,000 to 2,000 nucleotides long. In eukaryotic cells, they are 150 to200 nucleotides long.



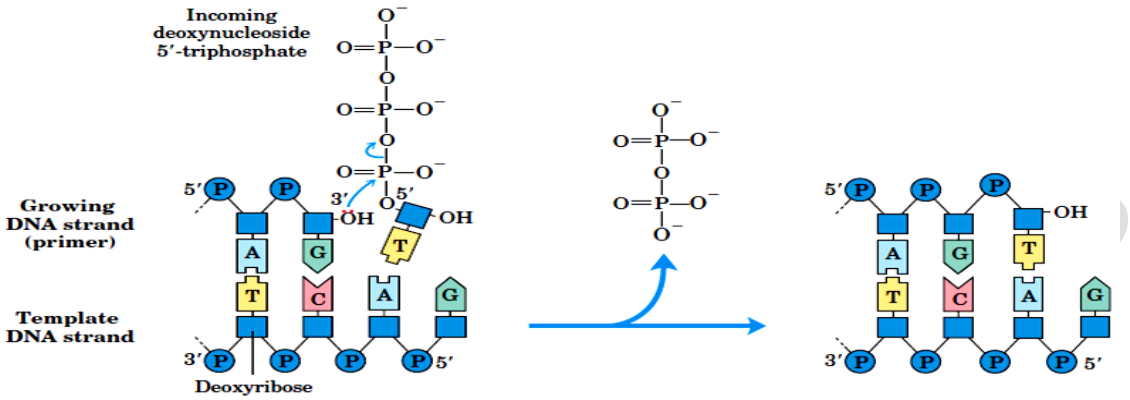
DNA synthesis and DNA polymerases:

DNA is synthesized by DNA polymerases. In 1955, Arthur Kornberg and colleagues did purification and characterization of DNA polymerase from *E. coli* cells, a single-polypeptide enzyme now called DNA polymerase I.

The fundamental reaction of DNA polymerisation is a phosphoryl group transfer. The nucleophile is the 3'-hydroxyl group of the nucleotide at the 3' end of the growing strand. Nucleophilic attack occurs at the phosphorus of the incoming deoxynucleoside 5'-triphosphate. Inorganic pyrophosphate is released in the reaction. The general reaction is:

$$(dNMP)_n + dNTP \longrightarrow (dNMP)_{n+1} + PP_i$$

DNA → Lengthened DNA



Replication Fidelity:

Replication Fidelity means 'How accurate the DNA replication?' In *E. coli*, a mistake is made only once for every 10^9 to 10^{10} nucleotides added. For the *E. coli* chromosome of ~4.6 10^6 bp, this means that an error occurs only once per 1,000 to 10,000 replications. The discrimination between correct and incorrect base pairing is based on:

- Hydrogen bonding behavior
- Geometry of standard base paring

One mechanism intrinsic to virtually all DNA polymerases is a separate 3'-5' exonuclease activity that double-checks each nucleotide after it is added. This nuclease activity permits the enzyme to remove a newly added nucleotide and is highly specific for mismatched base pairs. If the polymerase has added the wrong nucleotide, translocation of the enzyme to the position where the next nucleotide is to be added is inhibited. This kinetic pause provides the opportunity for a correction. The 3'-5' exonuclease activity removes the mispaired nucleotide, and the polymerase begins again. This activity is known as **proofreading.** Proofreading is not simply the reverse of the polymerization reaction, because pyrophosphate is not involved. The polymerizing and proofreading activities of a DNA polymerase can be measured separately. Proofreading improves the inherent accuracy of the polymerization reaction 10^2 to 10^3 fold. In the monomeric DNA polymerase I, the polymerizing and proofreading activities have separate active sites within the same polypeptide.

Career Avenues

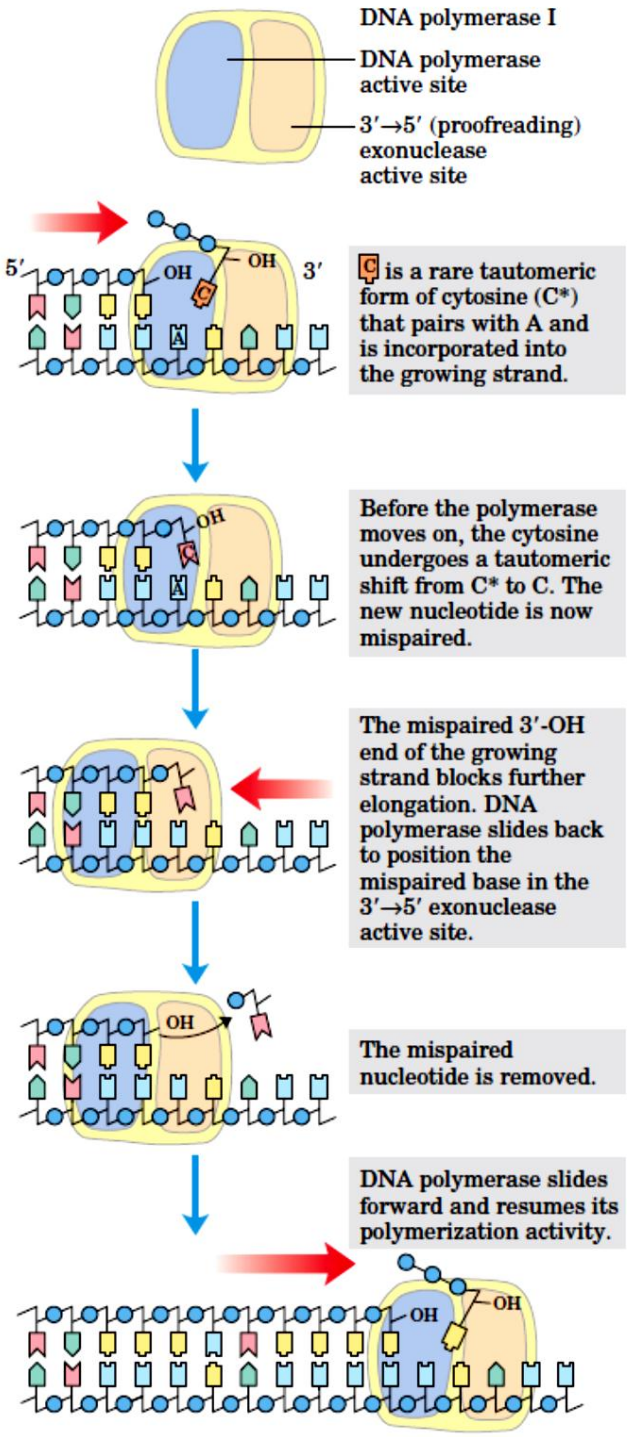


FIGURE: An example of error correction by the 3'-5' exonuclease activity of DNA polymerase I.

Types of DNA polymerases:

Polymerase Name	Gene	Function
DNA Pol I	PolA	Major repair enzyme
DNA Pol II	Pol B	Minor repair enzyme
DNA Pol III	Pol C	True replicase
DNA Pol IV	Din B	SOS repair
DNA Pol V	D_2C	SOS repair

Comparison of DNA Polymerases of *E. coli:*

	DNA polymerase		
	I	II	III
Structural gene*	*polA*	*polB*	*polC* (*dnaE*)
Subunits (number of different types)	1	7	≥10
M_r	103,000	88,000†	791,500
3′→5′ Exonuclease (proofreading)	Yes	Yes	Yes
5′→3′ Exonuclease	Yes	No	No
Polymerization rate (nucleotides/s)	16–20	40	250–1,000
Processivity (nucleotides added before polymerase dissociates)	3–200	1,500	≥500,000

\

DNA polymerase I:

Unique property of DNA pol I is-

- 5’ to 3’ exonuclease activity
- Nick translation

Klenow fragment is C-terminal 2/3 rd part of DNA pol I polypeptide has polymerization and proof reading activity.

Career Avenues

Nick translation:

In this process, an RNA or DNA strand paired to a DNA template is simultaneously degraded by the 5'-3' exonuclease activity of DNA polymerase I and replaced by the polymerase activity of the same enzyme. These activities have a role in both DNA repair and the removal of RNA primers during replication.

The strand of nucleic acid to be removed (either DNA or RNA) is shown in green, the replacement strand in red. DNA synthesis begins at a nick (a broken phosphodiester bond, leaving a free 3' hydroxyl and a free 5' phosphate). Polymerase I extends the nontemplate DNA strand and moves the nick along the DNA—a process called nick translation. A nick remains where DNA polymerase I dissociates, and is later sealed by another enzyme.

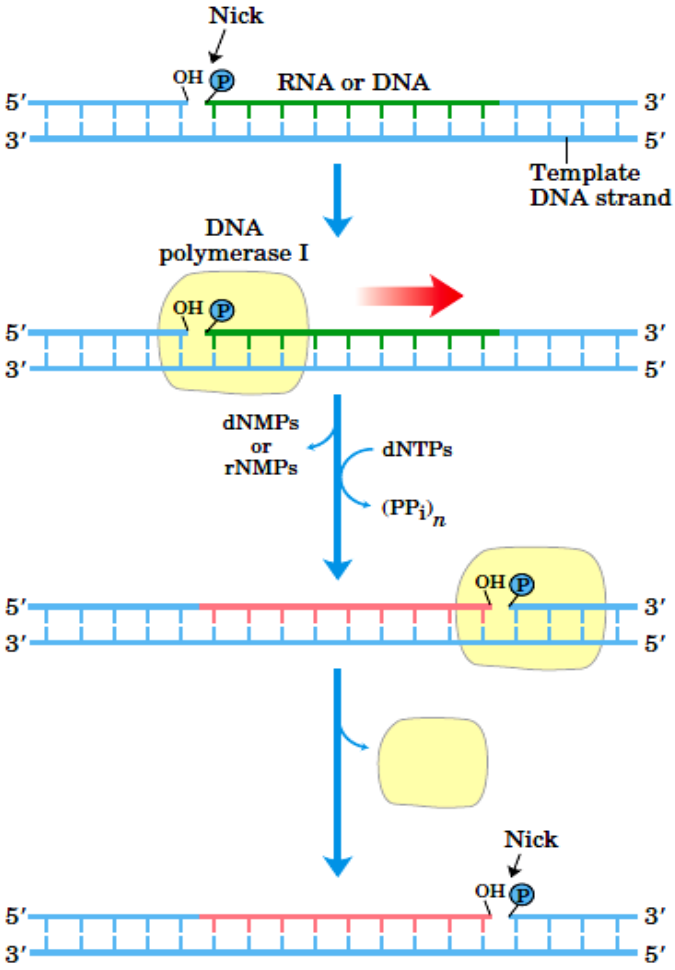


Fig: Nick translation

DNA polymerase III:

DNA polymerase III is a complex enzyme having ten types of subunits. Its polymerization and proofreading activities reside in its α and ε (epsilon) subunits, respectively. The θ subunit associates with α and ε to form a core polymerase, which can polymerize DNA but with limited processivity. Two core polymerases can be linked by another set of subunits, a clamp-loading complex, or γ complex, consisting of five subunits of four different types, $\tau_2\gamma\delta\delta'$. The core polymerases are linked through the τ (tau) subunits. Two additional subunits, χ (chi) and φ (psi), are bound to the clamp-loading complex. The entire assembly of 13 protein subunits (nine different types) is called DNA polymerase III.

DNA polymerase III can polymerize DNA, but with a much lower processivity that will not be sufficient for replication of an entire chromosome. The necessary increase in processivity is provided by the addition of the β subunits, four of which complete the DNA polymerase III holoenzyme. The β subunits associate in pairs to form donut-shaped structures that encircle the DNA and act like clamps. Each dimer associates with a core subassembly of polymerase III (one dimeric clamp per core subassembly) and slides along the DNA as replication proceeds. The β sliding clamp prevents the dissociation of DNA polymerase III from DNA, dramatically increasing processivity—to greater than 500,000.

- DNA polymerase III is true replicase.
- In contains 10 types of subunits.
  - Polymerisation activty by α subunit.
  - Proof reading activity by ε (epsilon) subunit.
  - Core polymerase will be "α,θ,ε".

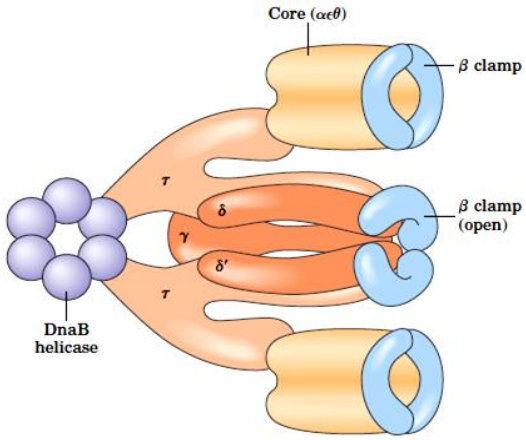


FIGURE: DNA polymerase III

Career Avenues

Replication of the *E. coli* chromosome:

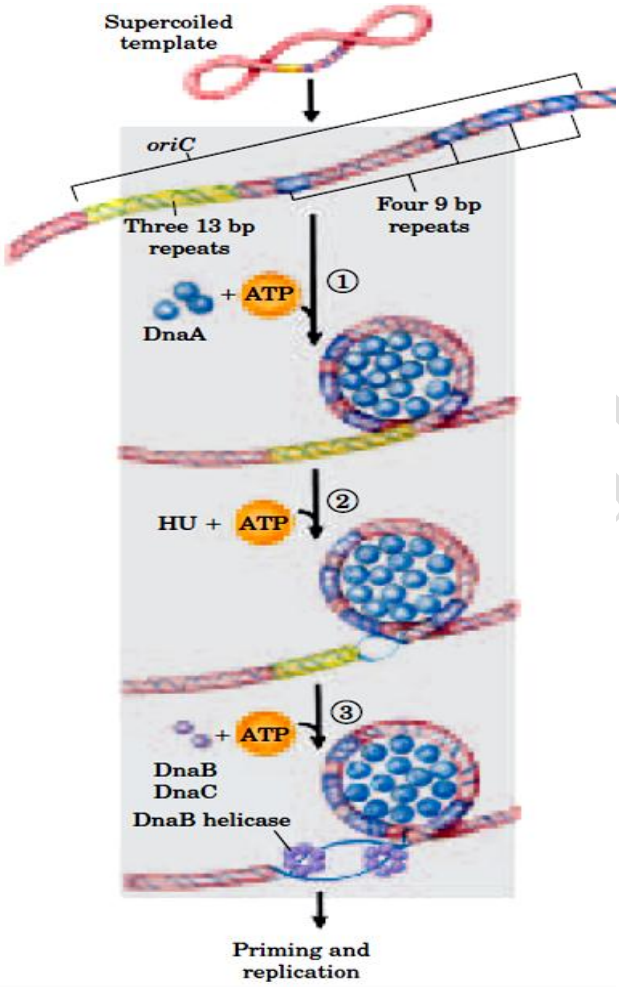
The synthesis of a DNA molecule can be divided into three stages: initiation, elongation, and termination.

Initiation:

- The *E. coli* replication origin, oriC, consists of 245 bp; it bears DNA sequence elements that are highly conserved among bacterial replication origins.
- The conserved sequences are two series of short repeats: three repeats of a 13 bp sequence and four repeats of a 9 bp sequence.
- At least nine different enzymes or proteins participate in the initiation phase of replication.
- The crucial component in the initiation process is the DnaA protein. A single complex of four to five DnaA protein molecules binds to the four 9 bp repeats in the origin (step-1), then recognizes and successively denatures the DNA in the region of the three 13 bp repeats, which are rich in A=T pairs (step 2).
- This process requires ATP and the bacterial histone like protein HU. The DnaC protein then loads the DnaB protein onto the unwound region.
- Two ring shaped hexamers of DnaB, one loaded onto each DNA strand, act as helicases, unwinding the DNA bi-directionally and creating two potential replication forks.
- Many molecules of SSB bind cooperatively to single stranded DNA, stabilizing the separated strands and preventing renaturation while gyrase relieves the topological stress produced by the DnaB helicase.
- When additional replication proteins are included in the in vitro system, the DNA unwinding mediated by DnaB is coupled to replication.
- Initiation is the only phase of DNA replication that is known to be regulated, and it is regulated such that replication occurs only once in each cell cycle.
- The timing of replication initiation is affected by DNA methylation and interactions with the bacterial plasma membrane.

Career Avenues

Model for initiation of replication at the *E. coli origin, oriC.*



Proteins Required to Initiate Replication at the *E. coli* Origin:

Protein	M_r	*Number of subunits*	*Function*
DnaA protein	52,000	1	Recognizes ori sequence; opens duplex at specific sites in origin
DnaB protein (helicase)	300,000	6*	Unwinds DNA
DnaC protein	29,000	1	Required for DnaB binding at origin
HU	19,000	2	Histonelike protein; DNA-binding protein; stimulates initiation
Primase (DnaG protein)	60,000	1	Synthesizes RNA primers
Single-stranded DNA–binding protein (SSB)	75,600	4*	Binds single-stranded DNA
RNA polymerase	454,000	5	Facilitates DnaA activity
DNA gyrase (DNA topoisomerase II)	400,000	4	Relieves torsional strain generated by DNA unwinding
Dam methylase	32,000	1	Methylates (5′)GATC sequences at *oriC*

Elongation stage of replication:

- The elongation phase of replication includes two distinct but related operations: leading strand synthesis and lagging strand synthesis.
- Leading strand synthesis begins with the synthesis by primase (DnaG protein) of a short (10 to 60 nucleotide) RNA primer at the replication origin.
- Deoxyribonucleotides are added to this primer by DNA polymerase III.
- Leading strand synthesis then proceeds continuously.

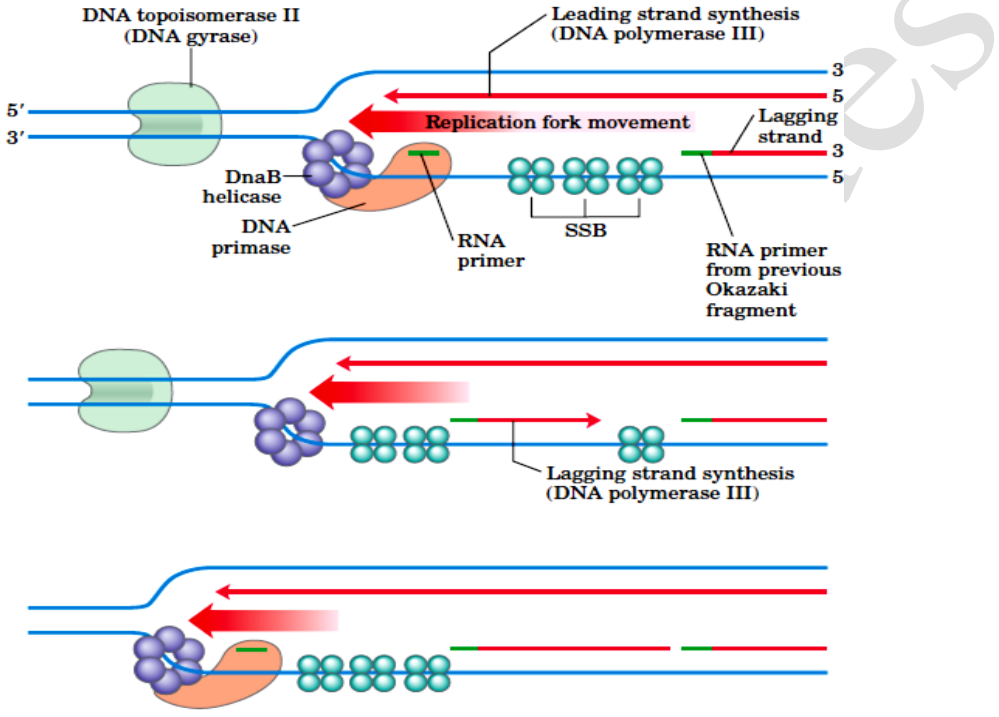


FIGURE: Synthesis of Okazaki fragments.

- Lagging strand synthesis is accomplished in short Okazaki fragments.
- First, an RNA primer is synthesized by primase and, as in leading strand synthesis, DNA polymerase III binds to the RNA primer and adds deoxyribonucleotides.
- On this level, the synthesis of each Okazaki fragment seems straight forward, but the reality is quite complex.
- The complexity lies in the coordination of leading and lagging strand synthesis: both strands are produced by a single asymmetric *DNA polymerase III* dimer, which is accomplished by looping the DNA of the lagging strand bringing together the two points of polymerization.
- The DnaB helicase and DnaG primase constitute a functional unit within the replication complex, the **primosome.**
- The DnaB helicase unwinds the DNA at the replication fork as it travels along the lagging strand template in the 5'-3' direction.

- DNA primase occasionally associates with DnaB helicase and synthesizes a short RNA primer.
- A new sliding clamp is then positioned at the primer by the clamp-loading complex of DNA polymerase III.
- When synthesis of an Okazaki fragment has been completed, replication halts, and the core subunits of DNA polymerase III dissociate from their sliding clamp (and from the completed Okazaki fragment) and associate with the new clamp. This initiates synthesis of a new Okazaki fragment.
- The entire complex responsible for coordinated DNA synthesis at a replication fork is a **replisome.**
- The replisome promotes rapid DNA synthesis, adding ~1,000 nucleotides/s to each strand (leading and lagging).
- Once an Okazaki fragment has been completed, its RNA primer is removed and replaced with DNA by DNA polymerase I, and the remaining nick is sealed by DNA ligase.
- DNA ligase catalyzes the formation of a phosphodiester bond between a 3' hydroxyl at the end of one DNA strand and a 5' phosphate at the end of another strand.
- The phosphate must be activated by adenylylation. DNA ligases isolated from viruses and eukaryotes use ATP for this purpose. DNA ligases from bacteria are unusual in that use NAD^+.

Career Avenues

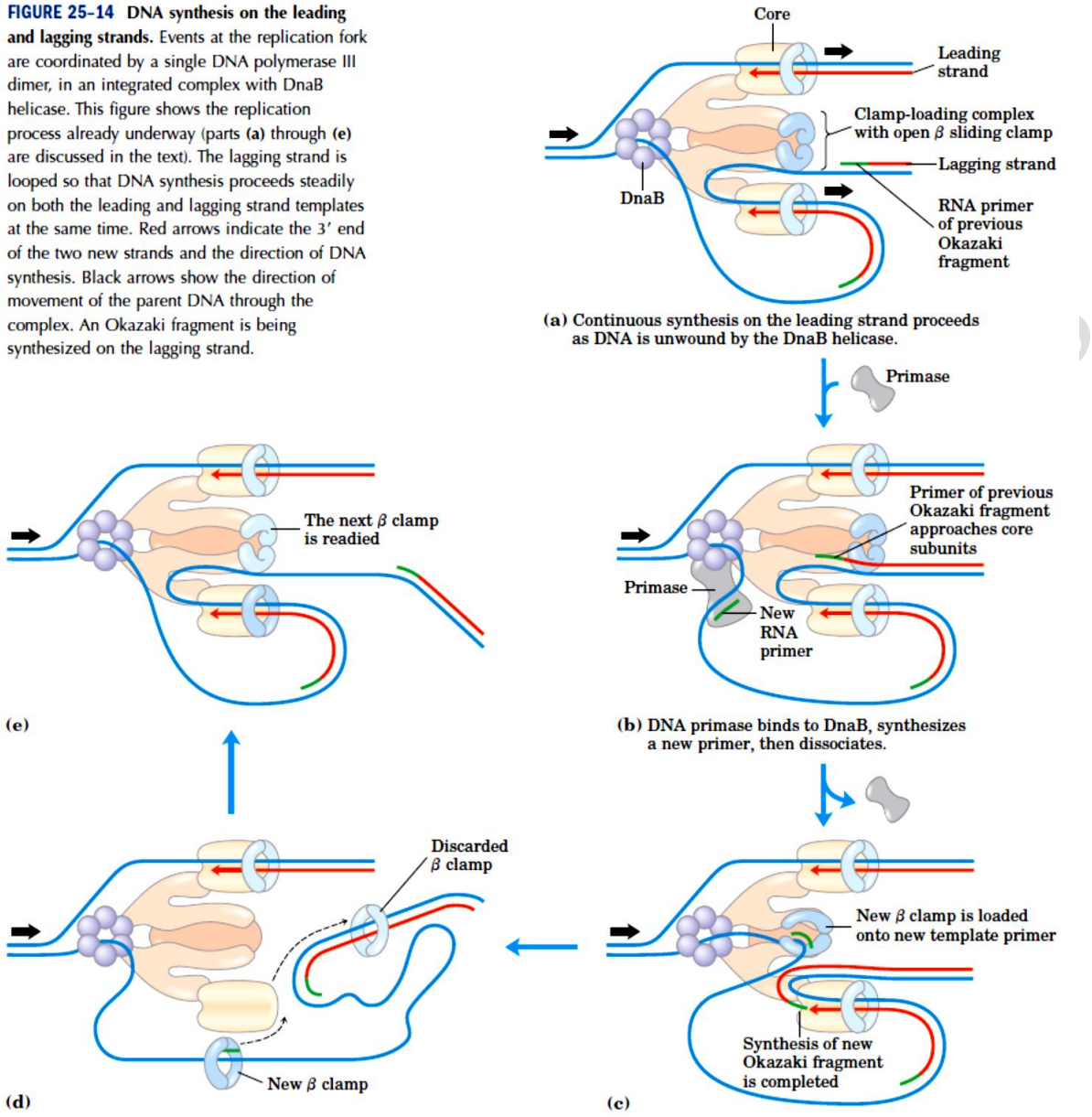


FIGURE 25–14 DNA synthesis on the leading and lagging strands. Events at the replication fork are coordinated by a single DNA polymerase III dimer, in an integrated complex with DnaB helicase. This figure shows the replication process already underway (parts **(a)** through **(e)** are discussed in the text). The lagging strand is looped so that DNA synthesis proceeds steadily on both the leading and lagging strand templates at the same time. Red arrows indicate the 3′ end of the two new strands and the direction of DNA synthesis. Black arrows show the direction of movement of the parent DNA through the complex. An Okazaki fragment is being synthesized on the lagging strand.

Proteins at the *E. coli* Replication Fork:

Protein	*M_r*	*Number of subunits*	*Function*
SSB	75,600	4	Binding to single-stranded DNA
DnaB protein (helicase)	300,000	6	DNA unwinding; primosome constituent
Primase (DnaG protein)	60,000	1	RNA primer synthesis; primosome constituent
DNA polymerase III	791,500	17	New strand elongation
DNA polymerase I	103,000	1	Filling of gaps; excision of primers
DNA ligase	74,000	1	Ligation
DNA gyrase (DNA topoisomerase II)	400,000	4	Supercoiling

Termination of DNA replication

- Eventually, the two replication forks of the circular *E. coli* chromosome meet at a terminus region containing multiple copies of a 20 bp sequence called Ter (for *ter*minus).
- The Ter sequences are arranged on the chromosome to create a sort of trap that a replication fork can enter but cannot leave.
- The Ter sequences function as binding sites for a protein called Tus (*t*erminus *u*tilization *s*ubstance).
- The Tus-Ter complex can arrest a replication fork from only one direction. Only one Tus-Ter complex functions per replication cycle—the complex first encountered by either replication fork.
- Given that opposing replication forks generally halt when they collide.
- Ter sequences do not seem essential, but they may prevent over-replication by one replication fork in the event that the other is delayed or halted by an encounter with DNA damage or some other obstacle.
- When either replication fork encounters a functional Tus-Ter complex, it halts; the other fork halts when it meets the first (arrested) fork.
- The final few hundred base pairs of DNA between these large protein complexes are then replicated (by an as yet unknown mechanism), completing two topologically interlinked (catenated) circular chromosomes. DNA circles linked in this way are known as **catenanes.** Separation of the catenated circles in *E. coli* requires Topoisomerase IV (a type II topoisomerase). The separated chromosomes then segregate into daughter cells at cell division.

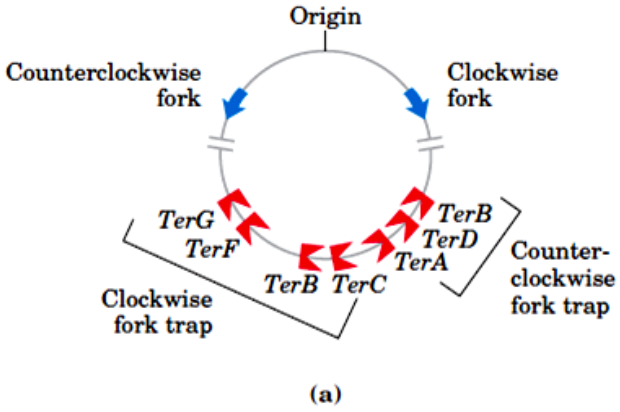
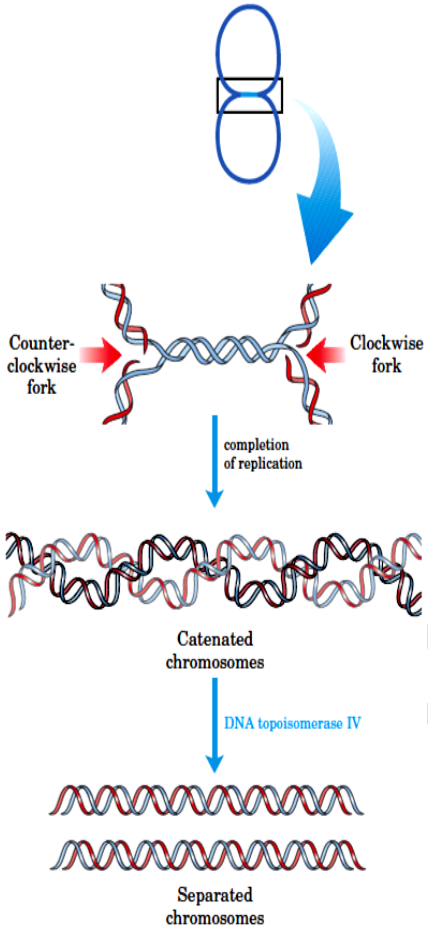


Fig: The Ter sequences are positioned on the chromosome in two clusters with opposite orientations.

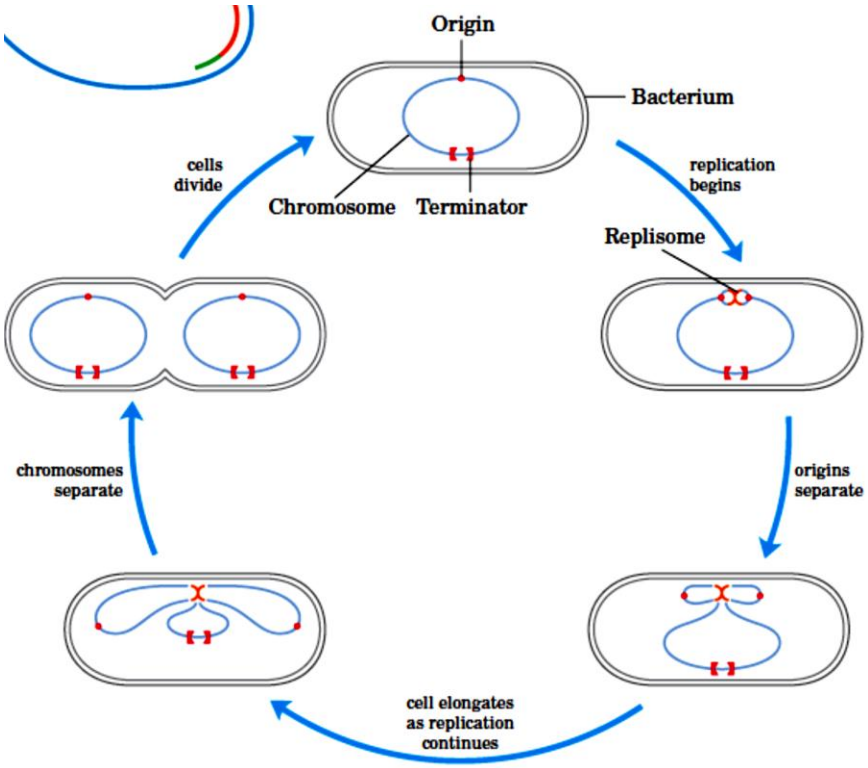
Career Avenues



Replication of the DNA separating the opposing replication forks leaves the completed chromosomes joined as catenanes, or topologically interlinked circles. The circles are not covalently linked, but because they are inter wound and each is covalently closed, they cannot be separated—except by the action of topoisomerases. In *E. coli, a type II topoisomerase known as DNA* topoisomerase IV plays the primary role in the separation of catenated chromosomes, transiently breaking both DNA strands of one chromosome and allowing the other chromosome to pass through the break.

The replication of a circular bacterial chromosome is highly organized. Once bidirectional replication is initiated at the origin, the two replisomes do not travel away from each other along the DNA. Instead, the replisomes are linked together and tethered to one point on the bacterial inner membrane, and the DNA substrate is fed through this "replication factory". The tethering point is at the center of the elongated bacterial cell. After initiation, each of the two newly synthesized replication origins is partitioned into one half of the cell, and continuing replication extrudes each new chromosome into that half. The elaborate spatial organization of the newly replicated chromosomes is orchestrated and maintained by many proteins, including bacterial homologs of the SMC proteins and topoisomerases.

Career Avenues



All replication is carried out at a central replication factory that includes two complete replication forks. The two replicated copies of the bacterial chromosome are extruded from the replication factory into the two halves of the cell, possibly with each newly synthesized origin bound separately to different points on the plasma membrane.
Sequestering the two chromosome copies in separate cell halves facilitates their proper segregation at cell division.

Replication in Eukaryotic Cells

The DNA molecules in eukaryotic cells are considerably larger than those in bacteria and are organized into complex nucleoprotein structures. The essential features of DNA replication are the same in eukaryotes and prokaryotes, and many of the protein complexes are functionally and structurally conserved. However, some interesting variations on the general principles discussed above promises new insights into the regulation of replication and its link with the cell cycle.

In eukaryotes, origins of replication, called *a*utonomously *r*eplicating *s*equences (ARS) or **replicators,** have been identified and best studied in yeast. Yeast replicators span ~150 bp and contain several essential conserved sequences. About 400 replicators are distributed among the 16 chromosomes in a haploid yeast genome. Initiation of replication in all eukaryotes requires a multisubunit protein, the *o*rigin *r*ecognition *c*omplex (ORC), which binds to several sequences within the replicator.

ORC interacts with and is regulated by a number of other proteins involved in control of the eukaryotic cell cycle. Two other proteins, CDC6 (discovered in a screen for genes affecting the *c*ell *d*ivision *c*ycle) and CDT1 (*C*dc10-*d*ependent *t*ranscript 1), bind to ORC and mediate the

loading of a heterohexamer of *m*ini*c*hromosome *m*aintenance proteins (MCM2 to MCM7). The MCM complex is a ring-shaped replicative helicase, analogous to the bacterial DnaB helicase. The CDC6 and CDT1 proteins have a role comparable to that of the bacterial DnaC protein, loading the MCM helicase onto the DNA near the replication origin. The rate of replication fork movement in eukaryotes (~50 nucleotides/sec) is only one-twentieth that observed in *E. coli.* At this rate, replication of an average human chromosome proceeding from a single origin would take more than 500 hours. Replication of human chromosomes in fact proceeds bi-directionally from many origins, spaced 30,000 to 300,000 bp apart. Eukaryotic chromosomes are almost always much larger than bacterial chromosomes, so multiple origins are probably a universal feature in eukaryotic cells.

Like bacteria, eukaryotes have several types of DNA polymerases. Some have been linked to particular functions, such as the replication of mitochondrial DNA. The replication of nuclear chromosomes involves DNA polymerase α, in association with DNA polymerase δ.

DNA polymerase α is typically a multisubunit enzyme with similar structure and properties in all eukaryotic cells. One subunit has a primase activity, and the largest subunit contains the polymerization activity. However, this polymerase has no proofreading 3'-5' exonuclease activity, making it unsuitable for high-fidelity DNA replication. DNA polymerase α is believed to function only in the synthesis of short primers (containing either RNA or DNA) for Okazaki fragments on the lagging strand. These primers are then extended by the multisubunit **DNA polymerase** δ. This enzyme is associated with and stimulated by a protein called *p*roliferating *c*ell *n*uclear *a*ntigen (PCNA). The three-dimensional structure of PCNA is remarkably similar to that of the β subunit of *E. coli* DNA polymerase III. PCNA has a function analogous to that of the β subunit, forming a circular clamp that greatly enhances the processivity of the polymerase. DNA polymerase δ has a 3'-5' proofreading exonuclease activity and appears to carry out both leading and lagging strand synthesis in a complex.

Another polymerase, **DNA polymerase** ε, replaces DNA polymerase δ in some situations, such as in DNA repair. DNA polymerase ε may also function at the replication fork, perhaps playing a role analogous to that of the bacterial DNA polymerase I, removing the primers of Okazaki fragments on the lagging strand.

Many DNA viruses encode their own DNA polymerases, and some of these have become targets for pharmaceuticals. For example, the DNA polymerase of the herpes simplex virus is inhibited by acyclovir. Acyclovir consists of guanine attached to an incomplete ribose ring. It is phosphorylated by a virally encoded thymidine kinase; acyclovir binds to this viral enzyme with an affinity 200-fold greater than its binding to the cellular thymidine kinase. This ensures that phosphorylation occurs mainly in virus-infected cells. Cellular kinases convert the resulting acyclo-GMP to acyclo-GTP, which is both an inhibitor and a substrate of DNA polymerases, and which competitively inhibits the herpes DNA polymerase more strongly than cellular DNA polymerases. Because it lacks a 3' hydroxyl, acyclo-GTP also acts as a chain terminator when incorporated into DNA. Thus viral replication is inhibited at several steps.

Two other protein complexes also function in eukaryotic DNA replication. RPA (*r*eplication *p*rotein *A*) is a eukaryotic single-stranded DNA–binding protein, equivalent in

function to the *E. coli* SSB protein. RFC (*r*eplication *f*actor *C*) is a clamp loader for PCNA and facilitates the assembly of active replication complexes. The subunits of the RFC complex have significant sequence similarity to the subunits of the bacterial clamploading (γ) complex. The termination of replication on linear eukaryotic chromosomes involves the synthesis of special structures called **telomeres** at the ends of each chromosome.

DNA Replication Control

DNA Replication Control by DnaA proteins

The regulation of DNA replication is a vital cellular process. In a general view, DNA replication is controlled by a series of mechanisms that are centered on the control of cellular DnaA levels, its availability as a free protein and modulation of its activity by binding the small-molecule ligand ATP the other point of control is by modulating the accessibility of replisome components to the oriC region on the DNA.

One of the main mechanisms associated with DNA replication is the so-called RIDA system (Regulatory Inactivation of DnaA). The elemen ts of this system are the sliding-clamp of DNA polymerase III and Hda (Homologous to DnaA). This mechanism takes place when DnaA is activated by its binding to ATP. The accumulation of DnaA in this active form leads to the initiation of chromosomal replication n since it facilitates its binding to the oriC on the DNA. DnaA reverts to its inactive form DnaA-ADP by hydrolys is of ATP. Hda-ADP is the monomeric active form for promoting the hydrolysis of DNA-ATP, a process which is mediated by the slider-loader clamp. This inactivating regulation of DnaA is key for preventing the over-initiation of replicative events during the cell cycle

Another mechanism that regulates the initiate on of DNA replication is by controlling the availability of free DnaA to bind to DnaA boxes on the oriC. Here the role of the 1kb datA locus, which is localized near (downstream) from the oriC is important. The datAlocus shows high affinity for DnaA, even more than the DnaA boxes on the oriC. Thus the datA region is able to bind over 300 DnaA molecules whereas oriC binds to 45 DnaA monomers. The operability of this mechanism is facilitated by the fact that the oriC had only few DnaA boxes compared to the datA locus and by the close proximity of data in respect to oriC on the DNA molecule. One related control system depends on the property of DnaA to act as a transcription factor and to the presence of DnaA boxes in the promoter regions of several genes. In most cases DnaA represses the expression of the associated gene but in some cases it can activates certain genes. DnaA regulates around 10 genes in E. coli as documented in RegulonDB. The transcription of DnaA is one of the most important regulatory mechanisms that directly affect the replication of DNA and one of the proteins that negatively regulate the expression of dnaA is DnaA itself. At high levels DnaA binds to the DnaA boxes in the promoter region and impedes transcription. This auto-repressive process directly affects the amount of DnaA-ATP available and controls the efficiency of initiation of DNA replication

Career Avenues

DNA Replication Control by DNA Methylation

A requirement for initiation of DNA replication is that both DNA strands are methylated, principally the adenine nucleotide in the GATC motifs, this process is mediated by Dam (DNA adenine methyltranferase). Dam binds to the DNA nonspecifically, and methylates the GATC motifs. On DNA strands recently synthesized these motifs are rapidly methylated and exist in the hemimethylated state only during a fraction of the time needed for the replication of the entire DNA.

The methylation process occurs asynchronously on the newly synthesized strands; i. e. methylation on the lagging arm occurs only after the ligation of the Okazaki fragments. It is postulated that Dam is always present in a complex bound near the replication origin, thus the methylation of nascent DNA strands occurs as soon as polymerization begins. In summary, the presence of hemimethylated GATC sites provides a cue to indicate that DNA replication has just occurred.

Another way to repress the transcription of dnaA is that which occurs immediately after the initiation of DNA replication. Here, SeqA binds to the hemimethylated GATC sequences in the regulatory regions of the dnaA gene. Similarly, SeqA also represses the replication of DNA by binding to the hemimethylated GATC sequence at the oriC, this is possible because SeqA DNA-binding sites overlap with those of low affinity for DnaA (DnaA boxes) on the oriC. This overlap impedes the complete access of DnaA-ATP to the oriC. This prevention of replication, dependant of DNA methylation, has been considered as an epigenetic regulatory mechanism because it depends on the chemical modification of the nucleotide residues of the DNA and not in its sequence.

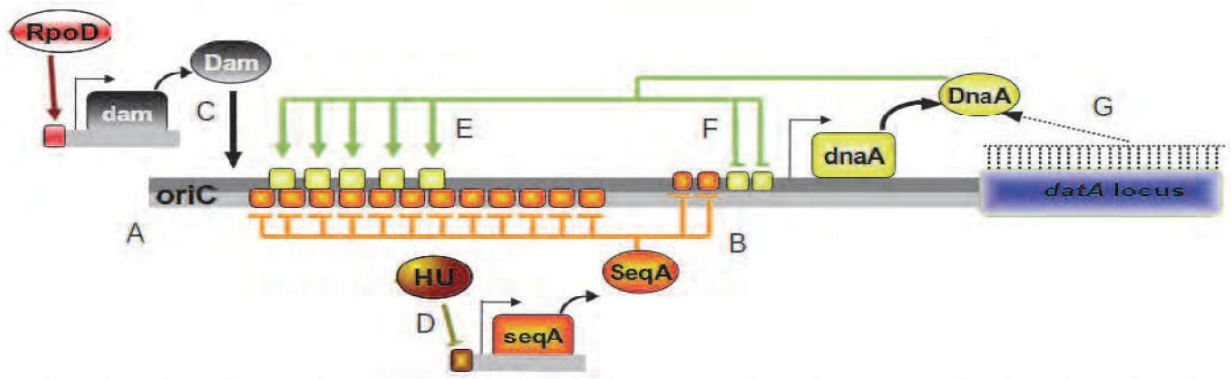


Fig. **Mechanisms that regulate DNA replication in *E. coli***. A) The newly replicated DNA duplex is in a hemimethylated state. B) SeqA binds to the hemimethylated GATC sites immediately after they are replicated. C) RpoD activates the transcription of dam and Dam methylates GATC sites of the newly synthesized strand. D) HU represses the transcription of SeqA. E) DnaA binds to the DnaA boxes on the oriC region. F) When there are many DnaA molecules they repress the transcription of the dnaA gene. G) data locus binds many DnaA molecules.

Special notes:

Unidirectional Replication:

Not all genetic systems follow bidirectional replication For example, replication of a plasmid called **colE1.** Replication of colE1 DNA is unidirectional.

Rolling Circle Replication:

Certain circular DNAs replicate, not by the common replication mechanism, but by a mechanism called **rolling circle** replication. For example, *E. coli* phages with single-stranded circular DNA genomes, such as ϕX174.

This mechanism the rolling circle name because the double-stranded part of the replicating DNA can be considered to be rolling counterclockwise and trailing out the progeny single-stranded DNA, rather like a roll of toilet paper unrolling as it speeds across the floor. This intermediate also somewhat resembles the lowercase Greek letter σ (sigma), so this mechanism is sometimes called the σ mode, to distinguish it from the θ mode.

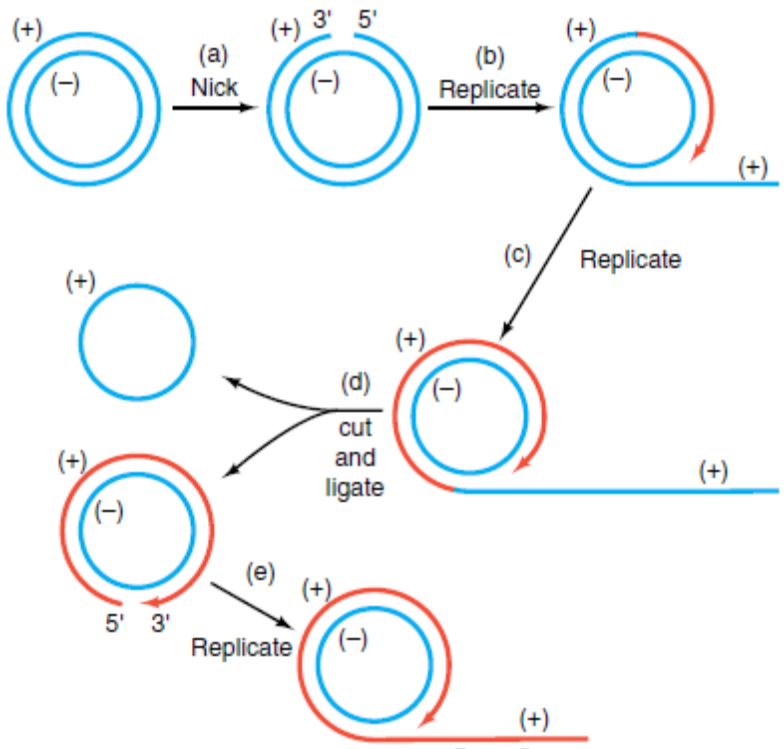
The rolling circle mechanism is not confined to production of single-stranded DNA. Some phages (e.g., λ) use this mechanism to replicate double-stranded DNA.

Rolling circle DNA replication mechanism:

Schematic representation (given below) of rolling circle replication that produces single-stranded circular progeny DNAs.

(**a**) An endonuclease creates a nick in the positive strand of the double-stranded replicative form.
(**b**) The free 3′-end created by the nick serves as the primer for positive strand elongation, as the other end of the positive strand is displaced. The negative strand is the template.
(**c**) Further replication occurs, as the positive strand approaches double length. The circle can be considered to be rolling counterclockwise.
(**d**) The unit length of positive-strand DNA that has been displaced is cleaved off by an endonuclease.
(**e**) Replication continues, producing another new positive strand, using the negative strand as template. This process repeats over and over to yield many copies of the circular positive strand.

Career Avenues



DNA REPAIRS:

A cell generally has only one or two sets of genomic DNA. Damaged proteins and RNA molecules can be quickly replaced by using information encoded in the DNA, but DNA molecules themselves are irreplaceable. Maintaining the integrity of the information in DNA is a cellular imperative, supported by an elaborate set of DNA repair systems. DNA can become damaged by a variety of processes, some spontaneous, others catalyzed by environmental agents. Replication itself can very occasionally damage the information content in DNA when errors introduce mismatched base pairs (such as G paired with T).

DNA can be damaged in many different ways, and this damage, if left unrepaired, can lead to mutations: changes in the base sequence of a DNA. This distinction is worth emphasizing at the outset: *DNA damage is not the same as mutation.* DNA damage is simply a chemical alteration to DNA. A mutation is a change in a base pair. For example, the change from a G–C pair to a methyl-G–C pair is DNA damage; the change from a G–C pair to any other natural base pair (A–T or T–A or C–G) is a mutation.

Damage Caused by Alkylation of Bases:

Alkylating agents like ethylmethane sulfonate add alkyl groups to bases. Some of these alkylations do not change base-pairing, so they are innocuous. Others cause DNA replication to stall, so they are cytotoxic, and can lead to mutations if the cell attempts to replicate its DNA without repairing the damage. Other alkylations change the base pairing properties of a base, so they are mutagenic. Alkylating agents like ethylmethane sulfonate add alkyl groups to bases. Some of these alkylations do not change base-pairing, so they are innocuous. Others cause DNA

Career Avenues

replication to stall, so they are cytotoxic, and can lead to mutations if the cell attempts to replicate its DNA without repairing the damage. Other alkylations change the base pairing properties of a base, so they are mutagenic.

DNA repair systems

The number and diversity of repair systems reflect both the importance of DNA repair to cell survival and the diverse sources of DNA damage. In mammals there is a strong correlation between the accumulation of mutations and cancer. A simple test developed by Bruce Ames measures the potential of a given chemical compound to promote certain easily detected mutations in a specialized bacterial strain.

A)- Mismatch Repair system

Correction of the rare mismatches left after replication in *E. coli* improves the overall fidelity of replication. The mismatches are nearly always corrected to reflect the information in the old (template) strand, so the repair system must somehow discriminate between the template and the newly synthesized strand. The cell accomplishes this by tagging the template DNA with methyl groups to distinguish it from newly synthesized strands.

Strand discrimination is based on the action of Dam methylase methylates DNA at the *N*6 position of all adenines within (5')GATC sequences. Immediately after passage of the replication fork, there is a short period (a few seconds or minutes) during which the template strand is methylated but the newly synthesized strand is not. The transient unmethylated state of GATC sequences in the newly synthesized strand permits the new strand to be distinguished from the template strand. Replication mismatches in the vicinity of a hemimethylated GATC sequence are then repaired according to the information in the methylated parent (template) strand. Tests in vitro show that if both strands are methylated at a GATC sequence, few mismatches are repaired; if neither strand is methylated, repair occurs but does not favor either strand. methyl-directed mismatch repair system efficiently repairs mismatches up to 1,000 bp from a hemimethylated GATC sequence.

MutL protein forms a complex with MutS protein, and the complex binds to all mismatched base pairs (except C–C). MutH protein binds to MutL and to GATC sequences encountered by the MutL-MutS complex. DNA on both sides of the mismatch is threaded through the MutL-MutS complex, creating a DNA loop; simultaneous movement of both legs of the loop through the complex is equivalent to the complex moving in both directions at once along the DNA. MutH has a site-specific endonuclease activity that is inactive until the complex encounters a hemimethylated GATC sequence. At this site, MutH catalyzes cleavage of the unmethylated strand on the 5' side of the G in GATC, which marks the strand for repair. Further steps in the pathway depend on where the mismatch is located relative to this cleavage site.

When the mismatch is on the 5' side of the cleavage site, the unmethylated strand is unwound and degraded in the 3'-5' direction from the cleavage site through the mismatch, and this segment is replaced with new DNA. This process requires the combined action of DNA helicase II, SSB, exonuclease I or exonuclease X (both of which degrade strands of DNA in the 3'-5' direction), DNA polymerase III, and DNA ligase. The pathway for repair of mismatches on the 3' side of the cleavage site is similar, except that the exonuclease is either exonuclease VII

Career Avenues

(which degrades single-stranded DNA in the 5'-3' or 3'-5' direction) or RecJ nuclease (which degrades single-stranded DNA in the 5'-3' direction).

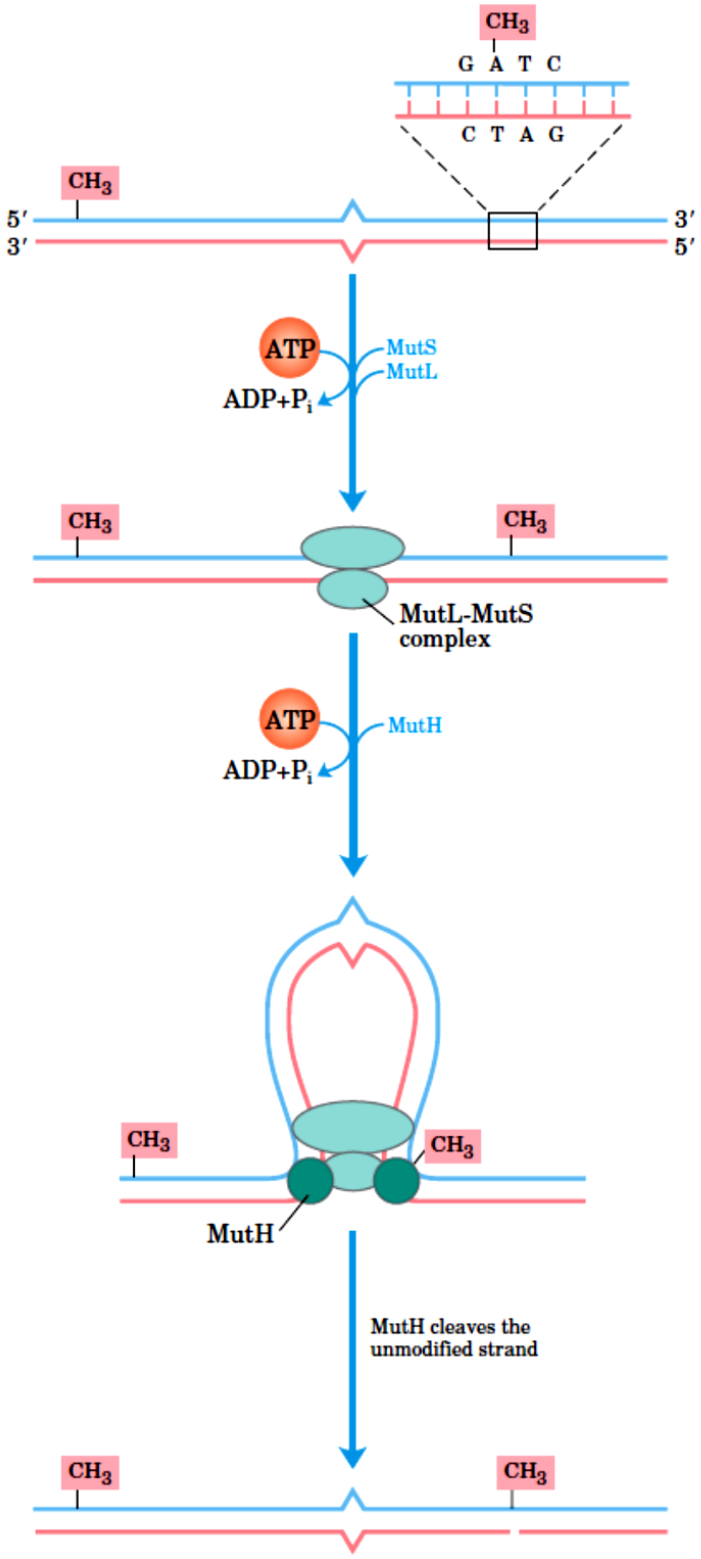


Figure: A model for the early steps of methyl-directed mismatch repair.

B)- Base-Excision Repair

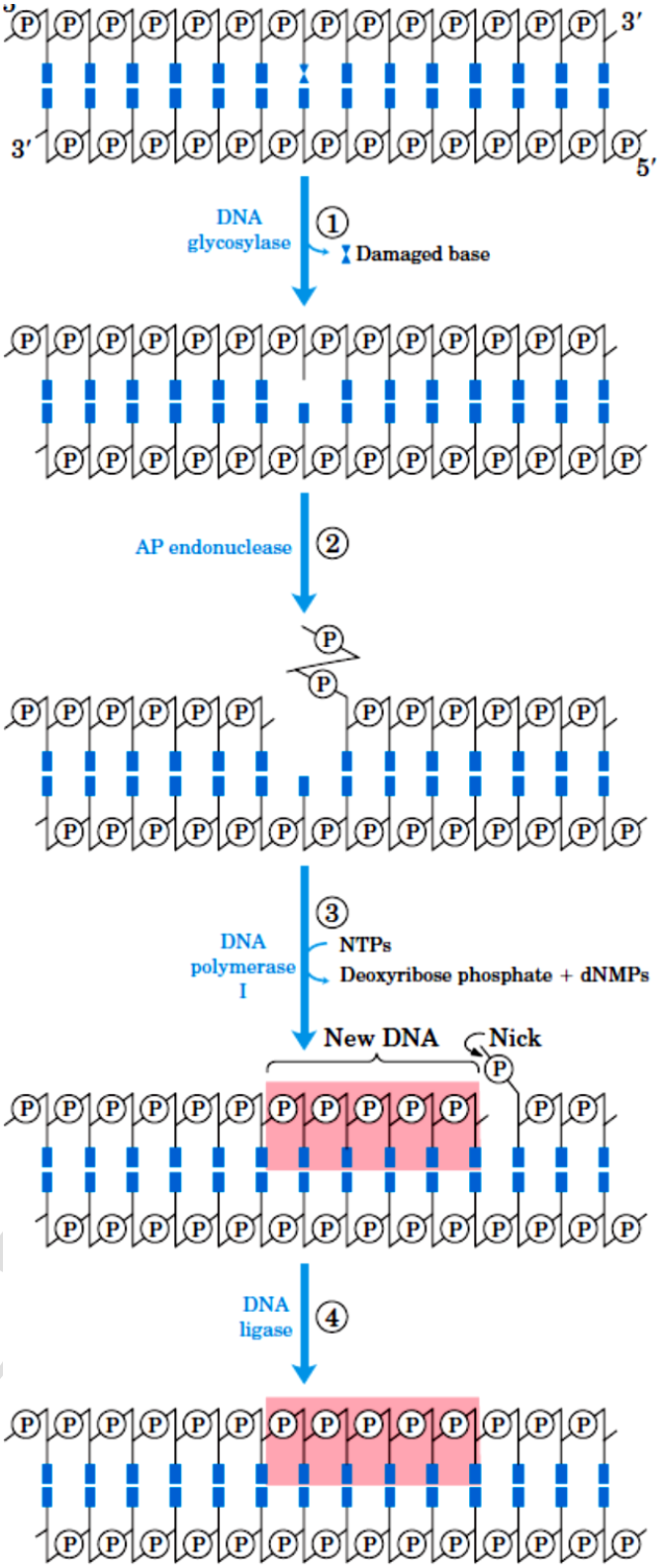
Every cell has a class of enzymes called **DNA glycosylases** that recognize particularly common DNA lesions (such as the products of cytosine and adenine deamination and remove the affected base by cleaving the *N*-glycosyl bond. This cleavage creates an apurinic or apyrimidinic site in the DNA, commonly referred to as an **AP site** or **abasic site.** Each DNA glycosylase is generally specific for one type of lesion.

Uracil DNA glycosylases, for example, found in most cells, specifically remove from DNA the uracil that results from spontaneous deamination of cytosine. Mutant cells that lack this enzyme have a high rate of G-C to A-T mutations. This glycosylase does not remove uracil residues from RNA or thymine residues from DNA. The capacity to distinguish thymine from uracil, the product of cytosine deamination—necessary for the selective repair of the latter—may be one reason why DNA evolved to contain thymine instead of uracil.

Bacteria generally have just one type of uracil DNA glycosylase, whereas humans have at least four types, with different specificities—an indicator of the importance of uracil removal from DNA. The most abundant human uracil glycosylase, UNG, is associated with the human replisome, where it eliminates the occasional U residue inserted in place of a T during replication. The deamination of C residues is 100-fold faster in singlestranded DNA than in double-stranded DNA, and humans have the enzyme hSMUG1, which removes any U residues that occur in single-stranded DNA during replication or transcription. Two other human DNA glycosylases, TDG and MBD4, remove either U or T residues paired with G, generated by deamination of cytosine or 5-methylcytosine, respectively.

Once an AP site has formed, another group of enzymes must repair it. The repair is *not* made by simply inserting a new base and re-forming the *N*-glycosyl bond. Instead, the deoxyribose 5'-phosphate left behind is removed and replaced with a new nucleotide. This process begins with **AP endonucleases,** enzymes that cut the DNA strand containing the AP site. The position of the incision relative to the AP site (5' or 3' to the site) varies with the type of AP endonuclease. A segment of DNA including the AP site is then removed, DNA polymerase I replaces the DNA, and DNA ligase seals the remaining nick.

Career Avenues



Career Avenues

DNA repair by the base-excision repair pathway.

1. A DNA glycosylase recognizes a damaged base and cleaves between the base and deoxyribose in the backbone.
2. An AP endonuclease cleaves the phosphodiester backbone near the AP site.
3. DNA polymerase I initiates repair synthesis from the free 3' hydroxyl at the nick, removing (with its 5'-3' exonuclease activity) a portion of the damaged strand and replacing it with undamaged DNA.
4. The nick remaining after DNA polymerase I has dissociated is sealed by DNA ligase.

C)- Nucleotide-Excision Repair

DNA lesions that cause large distortions in the helical structure of DNA generally are repaired by the nucleotide-excision system, a repair pathway critical to the survival of all free-living organisms. In nucleotide-excision repair, a multisubunit enzyme hydrolyzes two phosphodiester bonds, one on either side of the distortion caused by the lesion.

In *E. coli* and other prokaryotes, the enzyme system hydrolyzes the fifth phosphodiester bond on the 3' side and the eighth phosphodiester bond on the 5' side to generate a fragment of 12 to 13 nucleotides (depending on whether the lesion involves one or two bases).

In humans and other eukaryotes, the enzyme system hydrolyzes the sixth phosphodiester bond on the 3' side and the twenty-second phosphodiester bond on the 5' side, producing a fragment of 27 to 29 nucleotides. Following the dual incision, the excised oligonucleotides are released from the duplex and the resulting gap is filled—by DNA polymerase I in *E. coli* and DNA polymerase ε in humans. DNA ligase seals the nick.

In *E. coli*, the key enzymatic complex is the ABC excinuclease, which has three subunits, UvrA, UvrB, and UvrC. The term "excinuclease" is used to describe the unique capacity of this enzyme complex to catalyze two specific endonucleolytic cleavages, distinguishing this activity from that of standard endonucleases. A complex of the UvrA and UvrB proteins (A_2B) scans the DNA and binds to the site of a lesion. The UvrA dimer then dissociates, leaving a tight UvrB-DNA complex. UvrC protein then binds to UvrB, and UvrB makes an incision at the fifth phosphodiester bond on the 3' side of the lesion. This is followed by a UvrC-mediated incision at the eighth phosphodiester bond on the 5' side. The resulting 12 to 13 nucleotide fragment is removed by UvrD helicase. The short gap thus created is filled in by DNA polymerase I and DNA ligase.

This pathway is a primary repair route for many types of lesions, including cyclobutane pyrimidine dimers, 6-4 photoproducts, and several other types of base adducts including benzo[*a*]pyrene-guanine, which is formed in DNA by exposure to cigarette smoke. The nucleolytic activity of the ABC excinuclease is novel in the sense that two cuts are made in the DNA. The mechanism of eukaryotic excinucleases is quite similar to that of the bacterial enzyme, although 16 polypeptides with no similarity to the *E. coli* excinuclease subunits are required for the dual incision. Genetic deficiencies in nucleotide excision repair in humans give rise to a variety of serious diseases

Career Avenues

D)- Direct Repair

Several types of damage are repaired without removing a base or nucleotide. The best-characterized example is direct photoreactivation of cyclobutane pyrimidine dimers, a reaction promoted by **DNA photolyases.**

Pyrimidine dimers result from an ultraviolet light–induced reaction, and photolyases use energy derived from absorbed light to reverse the damage. Photolyases generally contain two cofactors that serve as light-absorbing agents, or chromophores. One of the chromophores is always $FADH^+$. In *E. coli* and yeast, the other chromophore is a folate. The reaction mechanism entails the generation of free radicals. DNA photolyases are not present in the cells of placental mammals (which include humans). Additional examples can be seen in the repair of nucleotides with alkylation damage. The modified nucleotide O^6-methylguanine forms in the presence of alkylating agents and is a common and highly mutagenic lesion. It tends to pair with thymine rather than cytosine during replication, and therefore causes GmC to AUT mutations. Direct repair of O^6- methylguanine is carried out by O^6-methylguanine-DNA methyltransferase, a protein that catalyzes transfer of the methyl group of *O*6-methylguanine to one of its own Cys residues. This methyltransferase is not strictly an enzyme, because a single methyl transfer event permanently methylates the protein, making it inactive in this pathway. The consumption of an entire protein molecule to correct a single damaged base is another vivid illustration of the priority given to maintaining the integrity of cellular DNA.

The repair pathways considered to this point generally work only for lesions in double-stranded DNA, the undamaged strand providing the correct genetic information to restore the damaged strand to its original state. However, in certain types of lesions, such as doublestrand breaks, double-strand cross-links, or lesions in a single-stranded DNA, the complementary strand is itself damaged or is absent. Double-strand breaks and lesions in single-stranded DNA most often arise when a replication fork encounters an unrepaired DNA lesion. Such lesions and DNA cross-links can also result from ionizing radiation and oxidative reactions. At a stalled bacterial replication fork, there are two avenues for repair. In the absence of a second strand, the information required for accurate repair must come from a separate, homologous chromosome. The repair system thus involves homologous genetic recombination.

A second repair pathway, **error-prone translesion DNA synthesis** (often abbreviated TLS), becomes available. When this pathway is active, DNA repair becomes significantly less accurate and a high mutation rate can result. In bacteria, error-prone translesion DNA synthesis is part of a cellular stress response to extensive DNA damage known, appropriately enough, as the **SOS response.** Some SOS proteins, such as the UvrA and UvrB proteins are normally present in the cell but are induced to higher levels as part of the SOS response. Additional SOS proteins participate in the pathway for error-prone repair; these include the UmuC and UmuD proteins ("Umu" from *un**mu**t*able; lack of the *umu* gene function eliminates error-prone repair). The UmuD protein is cleaved in an SOS-regulated process to a shorter form called UmuD, which forms a complex with UmuC to create a specialized DNA polymerase (DNA polymerase V) that can replicate past many of the DNA lesions that would normally block replication. Proper base pairing is often impossible at the site of such a lesion, so this translesion replication is error-prone. The bacterial DNA polymerases IV and V are part of a family of TLS polymerases found in all organisms. These enzymes lack a proofreading exonuclease activity, and the fidelity of replicative base selection can be enduced.

Career Avenues

Key Notes:

■ Cells have many systems for DNA repair. Mismatch repair in *E. coli* is directed by transient nonmethylation of (5') GATC sequences on the newly synthesized strand.

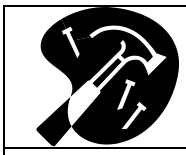
■ Base-excision repair systems recognize and repair damage caused by environmental agents (such as radiation and alkylating agents) and spontaneous reactions of nucleotides. Some repair systems recognize and excise only damaged or incorrect bases, leaving an AP (abasic) site in the DNA. This is repaired by excision and replacement of the DNA segment containing the AP site.

■ Nucleotide-excision repair systems recognize and remove a variety of bulky lesions and pyrimidine dimers. They excise a segment of the DNA strand including the lesion, leaving a gap that is filled in by DNA polymerase and ligase activities.

■ Some DNA damage is repaired by direct reversal of the reaction causing the damage: pyrimidine dimers are directly converted to monomeric pyrimidines by a photolyase, and the methyl group of *O*6-methylguanine is removed by a methyltransferase.

■ In bacteria, error-prone translesion DNA synthesis, involving TLS DNA polymerases, occurs in response to very heavy DNA damage. In eukaryotes, similar polymerases have specialized roles in DNA repair that minimize the introduction of mutations.

Career Avenues



GATE Questions

GATE 2007

Q.1 Which of the DNA polymerases listed below is primarily responsible for the *de novo* synthesis of new DNA strands ?

(A) DNA polymerase I
(B) DNA polymerase II
(C) DNA polymerase III
(D) DNA polymerase IV

GATE 2009

Q.2 Human DNA (3×10^6 kb) is replicated in 5 hrs at a rate of 1 kb/min. The number of origins of replication utilized are

(A)1 (B) 3 (C) 300 (D) 10,000

Answers	
Q.1 – C	**Q.2 – A.**

Career Avenues

TRANSCRIPTION AND ITS CONTROL

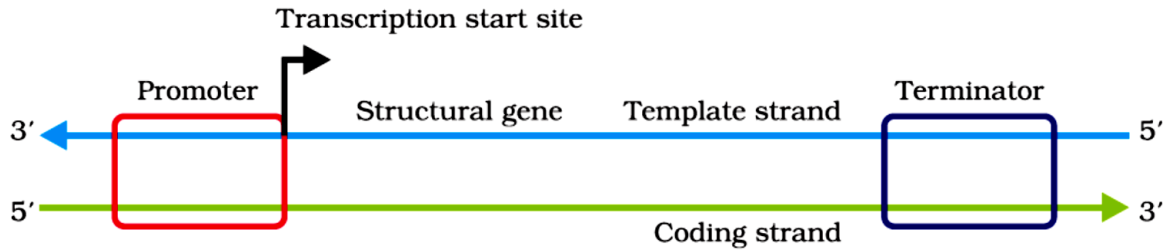
The process of copying genetic information from one strand of the DNA into RNA is termed as transcription. Here also, the principle of complementarity governs the process of transcription, except the adenosine now forms base pair with uracil instead of thymine. However, unlike in the process of replication, which once set in, the total DNA of an organism gets duplicated, in transcription only a segment of DNA and only one of the strands is copied into RNA. both the strands are not copied during transcription. There are two reasons. First, if both strands act a template, they would code for RNA molecule with different sequences (Remember complementarity does not mean identical), and in turn, if they code for proteins, the sequence of amino acids in the proteins would be different. Hence, one segment of the DNA would be coding for two different proteins, and this would complicate the genetic information transfer machinery. Second, the two RNA molecules if produced simultaneously would be complementary to each other, hence would form a double stranded RNA. This would prevent RNA from being translated into protein and the exercise of transcription would become a futile one.

TRANSCRIPTION UNIT

A transcription unit in DNA is defined primarily by the three regions in the DNA:
(i) A Promoter
(ii) The Structural gene
(iii) A Terminator

There is a convention in defining the two strands of the DNA in the structural gene of a transcription unit. Since the two strands have opposite polarity and the **DNA-dependent RNA polymerase** also catalyse the polymerisation in only one direction, that is, 5′→3′ , the strand that has the polarity 3′→5′ acts as a template, and is also referred to as **template strand**. The other strand which has the polarity (5′→3′) and the sequence same as RNA (except thymine at the place of uracil), is displaced during transcription. Strangely, this strand (which does not code for anything) is referred to as **coding strand**. All the reference point while defining a transcription unit is made with coding strand. To explain the point, a hypothetical sequence from a transcription unit is represented below:

3′-ATGCATGCATGCATGCATGCATGC-5′ Template Strand
5′-TACGTACGTACGTACGTACGTACG-3′ Coding Strand



Career Avenues

The **promoter** and **terminator** flank the **structural gene** in a transcription unit. The promoter is said to be located towards 5′-end (upstream) of the structural gene (the reference is made with respect to the polarity of coding strand). The terminator is located towards 3′-end (downstream) of the coding strand and it usually defines the end of the process of transcription.

TRANSCRIPTION UNIT AND THE GENE

A gene is defined as the functional unit of inheritance. Genes are considered as a unit of function *(cistron),* a unit of mutation *(mutan)* and a unit of recombination *(recon).*

Cistron is defined as a segment of DNA coding for a polypeptide. The structural gene in a transcription unit could be said as **monocistronic** (mostly in eukaryotes) or **polycistronic** (mostly in bacteria or prokaryotes). In eukaryotes, the monocistronic structural genes have interrupted coding sequences – the genes in eukaryotes are split. The coding sequences or expressed sequences are defined as **exons**. Exons are said to be those sequence that appear in mature or processed RNA. The exons are interrupted by **introns**. Introns or intervening sequences do not appear in mature or processed RNA. The split-gene arrangement further complicates the definition of a gene in terms of a DNA segment.

TYPES OF RNA AND THE PROCESS OF TRANSCRIPTION

In bacteria, there are three major types of RNAs: mRNA (messenger RNA), tRNA (transfer RNA), and rRNA (ribosomal RNA). All three RNAs are needed to synthesise a protein in a cell. The mRNA provides the template, tRNA brings aminoacids and reads the genetic code, and rRNAs play structural and catalytic role during translation. There is single DNA-dependent RNA polymerase that catalyses transcription of all types of RNA in bacteria.

RNA polymerase

The DNA-dependent RNA polymerase of E. coli is a large, complex enzyme with five core subunits and a sixth subunit, one of a group designated σ , with variants designated by size (molecular weight). The σ subunit binds transiently to the core and directs the enzyme to specific binding sites on the DNA. These six subunits constitute the RNA polymerase holoenzyme (α_2 β β'ω σ). RNA polymerase core enzyme is (α_2 β β'ω). RNA polymerases lack a proofreading 3' – 5' exonuclease active site. So the error rate for transcription is higher than that for chromosomal DNA replication— approximately one error for every 10^4 to 10^5 ribonucleotides incorporated into RNA.

The RNA polymerase moves stepwise along the DNA, unwinding the DNA helix just ahead of the active site for polymerization to expose a new region of the template strand for complementary base-pairing. In this way, the growing RNA chain is extended by one nucleotide at a time in the 5¢-to-3¢ direction.

Although RNA polymerase catalyzes essentially the same chemical reaction as DNA polymerase, there are some important differences between the activities of the two enzymes. First, and most obviously, RNA polymerase catalyzes the linkage of ribonucleotides, not deoxyribonucleotides. Second, unlike the DNA polymerases involved in DNA replication, RNA

Career Avenues

polymerases can start an RNA chain without a primer. This difference may exist because transcription need not be as accurate as DNA replication. Unlike DNA, RNA does not permanently store genetic information in cells. RNA polymerases make about one mistake for every 104 nucleotides copied into RNA (compared with an error rate for direct copying by DNA polymerase of about one in 107 nucleotides), and the consequences of an error in RNA transcription are much less significant than that in DNA replication.

TYPE OF RNA	FUNCTION
mRNAs	messenger RNAs, code for proteins
rRNAs	ribosomal RNAs, form the basic structure of the ribosome and catalyze protein synthesis
tRNAs	transfer RNAs, central to protein synthesis as adaptors between mRNA and amino acids
snRNAs	small nuclear RNAs, function in a variety of nuclear processes, including the splicing of pre-mRNA
snoRNAs	small nucleolar RNAs, used to process and chemically modify rRNAs
scaRNAs	small cajal RNAs, used to modify snoRNAs and snRNAs
miRNAs	microRNAs, regulate gene expression typically by blocking translation of selective mRNAs
siRNAs	small interfering RNAs, turn off gene expression by directing degradation of selective mRNAs and the establishment of compact chromatin structures
Other noncoding RNAs	function in diverse cell processes, including telomere synthesis, X-chromosome inactivation, and the transport of proteins into the ER

Career Avenues

THE TRANSCRIPTION BY BACTERIAL RNA POLYMERASE

In *E. coli*, RNA polymerase binding occurs within a region stretching from about 70 bp before the transcription start site to about 30 bp beyond it. By convention, the DNA base pairs that correspond to the beginning of an RNA molecule are given positive numbers, and those preceding the RNA start site are given negative numbers. The promoter region thus extends between positions -70 and +30. Analyses and comparisons of the most common class of bacterial promoters (those recognized by an RNA polymerase holoenzyme containing -70) have revealed similarities in two short sequences centered about positions -10 and -35. These sequences are important interaction sites for the σ_{70} subunit. Although the sequences are not identical for all bacterial promoters in this class, certain nucleotides that are particularly common at each position form a **consensus sequence.**

The consensus sequence at the -10 region is (5')TATAAT(3'); the consensus sequence at the -35 region is (5-)TTGACA(3'). A third AT-rich recognition element, called the UP (upstream promoter) element, occurs between positions -40 and -60 in the promoters of certain highly expressed genes. The UP element is bound by the α subunit of RNA polymerase.

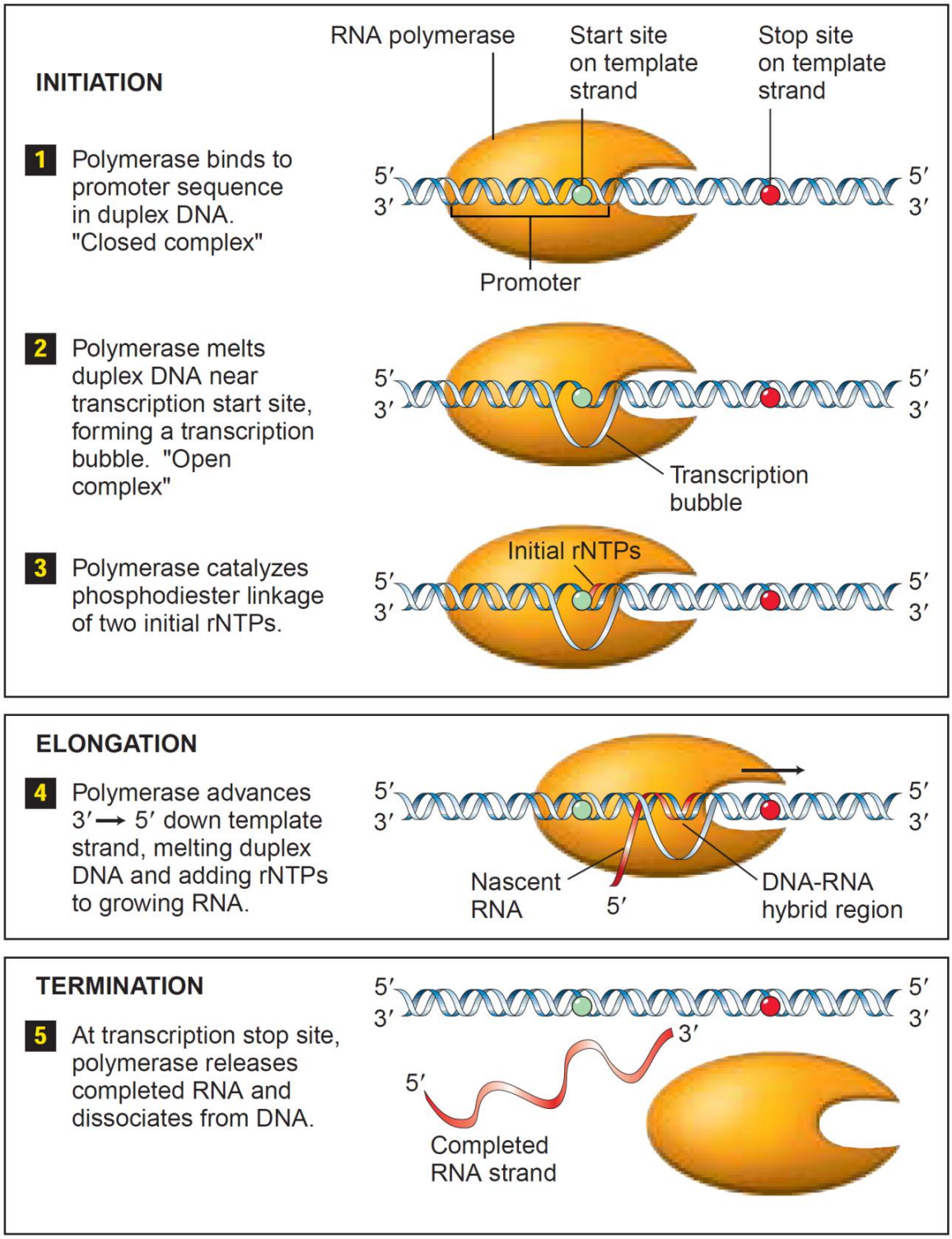
The efficiency with which an RNA polymerase binds to a promoter and initiates transcription is determined in large measure by these sequences, the spacing between them, and their distance from the transcription start site. The pathway of transcription initiation is becoming much better defined. It consists of two major parts, binding and initiation, each with multiple steps. First, the polymerase binds to the promoter, forming, in succession, a closed complex (in which the bound DNA is intact) and an open complex (in which the bound DNA is intact and partially unwound near the -10 sequence). Second, transcription is initiated within the complex, leading to a conformational change that converts the complex to the elongation form, followed by movement of the transcription complex away from the promoter (promoter clearance). The σ subunit dissociates as the polymerase enters the elongation phase of transcription.

E. coli has at least two classes of termination signals: one class relies on a protein factor called ρ (rho) and the other is ρ -independent. Most ρ -independent terminators have two distinguishing features. The first is a region that produces an RNA transcript with self-complementary sequences, permitting the formation of a hairpin structure centered 15 to 20 nucleotides before the projected end of the RNA strand. The second feature is a highly conserved string of three A residues in the template strand that are transcribed into U residues near the 3' end of the hairpin. When a polymerase arrives at a termination site with this structure, it pauses. Formation of the hairpin structure in the RNA disrupts several AUU base pairs in the RNA-DNA hybrid segment and may disrupt important interactions between RNA and the RNA polymerase, facilitating dissociation of the transcript.

The ρ -dependent terminators lack the sequence of repeated A residues in the template strand but usually include a CA-rich sequence called a *rut* (*r*ho *ut*ilization) element. The ρ protein associates with the RNA at specific binding sites and migrates in the 5'-3' direction until it reaches the transcription complex that is paused at a termination site. Here it contributes to release of the RNA transcript. The ρ protein has an ATP-dependent RNA-DNA helicase activity

Career Avenues

that promotes translocation of the protein along the RNA, and ATP is hydrolyzed by ρ protein during the termination process.

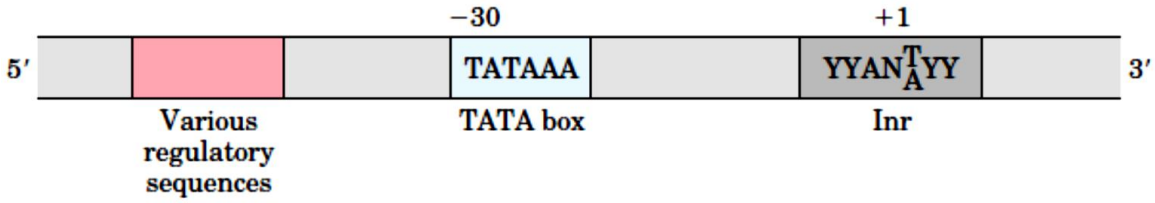


Eukaryotic transcription

The transcriptional machinery in the nucleus of a eukaryotic cell is much more complex than that in bacteria. Eukaryotes have three RNA polymerases, designated I, II, and III, which are distinct complexes but have certain subunits in common. Each polymerase has a specific function and is recruited to a specific promoter sequence.

RNA polymerase I (Pol I) is responsible for the synthesis of only one type of RNA, a transcript called preribosomal RNA (or pre-rRNA), which contains the precursor for the 18S, 5.8S, and 28S rRNAs. Pol I promoters vary greatly in sequence from one species to another. The principal function of RNA polymerase II (Pol II) is synthesis of mRNAs and some specialized RNAs. This enzyme can recognize thousands of promoters that vary greatly in sequence. Many Pol II promoters have a few sequence features in common, including a TATA box (eukaryotic consensus sequence TATAAA) near base pair -30 and an Inr sequence (initiator) near the RNA start site at +1. RNA polymerase III (Pol III) makes tRNAs, the 5S rRNA, and some other small specialized RNAs. The promoters recognized by Pol III are well characterized.

Common sequences in promoters recognized by eukaryotic RNA polymerase II.

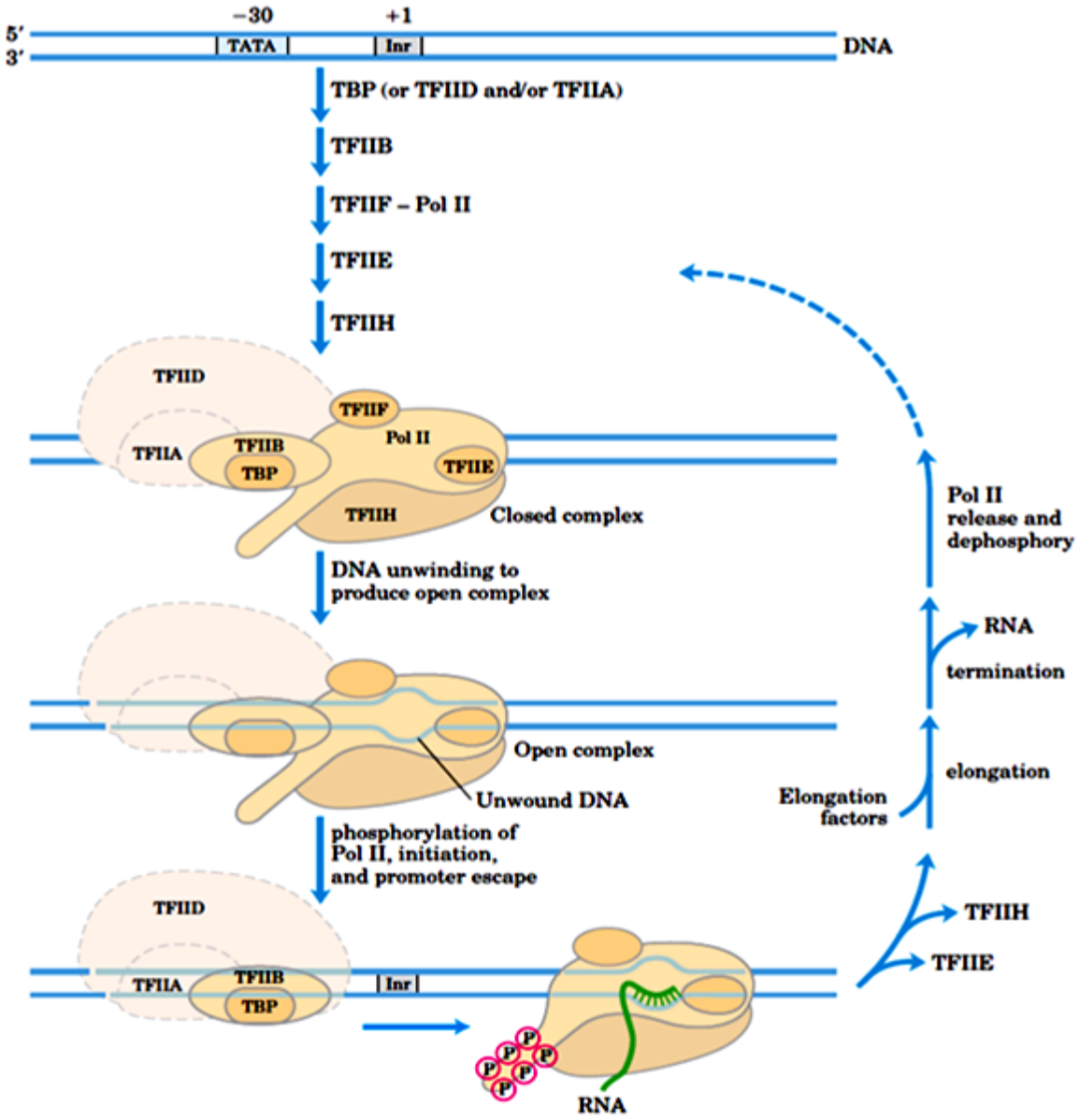


The TATA box is the major assembly point for the proteins of the preinitiation complexes of Pol II. The DNA is unwound at the initiator sequence (Inr), and the transcription start site is usually within or very near this sequence.

EUKARYOTIC RNA POLYMERASE

RNA polymerase II is central to eukaryotic gene expression. Pol II is a huge enzyme with 12 subunits. The largest subunit (RBP1) exhibits, a high degree of homology to the *β'* subunit of bacterial RNA polymerase. Another subunit (RBP2) is structurally similar to the bacterial *β* subunit, and two others (RBP3 and RBP11) show some structural homology to the two bacterial *α* subunits. The largest subunit of Pol II also has a long carboxyl-terminal tail.

RNA polymerase II requires an array of other proteins, called **transcription factors,** in order to form the active transcription complex. The **general transcription factors** required at every Pol II promoter (factors usually designated TFII with an additional identifier) are highly conserved in all eukaryotes. The process of transcription by Pol II can be described in terms of several phases—assembly, initiation, elongation, termination—each associated with characteristic proteins.



TRANSCRIPTION AT RNA POLYMERASE II PROMOTERS:

The sequential assembly of TBP (often with TFIIA), TFIIB, TFIIF plus Pol II, TFIIE, and TFIIH results in a closed complex. TBP often binds as part of a larger complex, TFIID. Some of the TFIID subunits play a role in transcription regulation. Within the complex, the DNA is unwound at the Inr region by the helicase activity of TFIIH and perhaps of TFIIE, creating an open complex. The carboxyl-terminal domain of the largest Pol II subunit is phosphorylated by TFIIH, and the polymerase then escapes the promoter and begins transcription. Elongation is accompanied by the release of many transcription factors and is also enhanced by elongation factors. After termination, Pol II is released, dephosphorylated, and recycled.

TABLE **Proteins Required for Initiation of Transcription at the RNA Polymerase II (Pol II) Promoters of Eukaryotes**

Transcription protein	*Number of subunits*	*Subunit(s) M,*	*Function(s)*
Initiation			
Pol II	12	10,000–220,000	Catalyzes RNA synthesis
TBP (TATA-binding protein)	1	38,000	Specifically recognizes the TATA box
TFIIA	3	12,000, 19,000, 35,000	Stabilizes binding of TFIIB and TBP to the promoter
TFIIB	1	35,000	Binds to TBP; recruits Pol II–TFIIF complex
TFIIE	2	34,000, 57,000	Recruits TFIIH; has ATPase and helicase activities
TFIIF	2	30,000, 74,000	Binds tightly to Pol II; binds to TFIIB and prevents binding of Pol II to nonspecific DNA sequences
TFIIH	12	35,000–89,000	Unwinds DNA at promoter (helicase activity); phosphorylates Pol II (within the CTD); recruits nucleotide-excision repair proteins
Elongation*			
ELL†	1	80,000	
p-TEFb	2	43,000, 124,000	Phosphorylates Pol II (within the CTD)
SII (TFIIS)	1	38,000	
Elongin (SIII)	3	15,000, 18,000, 110,000	

ASSEMBLY OF RNA POLYMERASE AND TRANSCRIPTION FACTORS AT A PROMOTER:

The formation of a closed complex begins when the TATA-binding protein (TBP) binds to the TATA box. TBP is bound in turn by the transcription factor TFIIB, which also binds to DNA on either side of TBP. TFIIA binding, although not always essential, can stabilize the TFIIB-TBP complex on the DNA and can be important at non-consensus promoters where TBP binding is relatively weak. The TFIIB-TBP complex is next bound by another complex consisting of TFIIF and Pol II. TFIIF helps target Pol II to its promoters, both by interacting with TFIIB and by reducing the binding of the polymerase to nonspecific sites on the DNA. Finally, TFIIE and TFIIH bind to create the closed complex. TFIIH has DNA helicase activity that promotes the unwinding of DNA near the RNA start site (a process requiring the hydrolysis of ATP), thereby creating an open complex. Counting all the subunits of the various essential factors (excluding TFIIA), this minimal active assembly has more than 30 polypeptides.

RNA Strand Initiation and Promoter Clearance:

TFIIH has an additional function during the initiation phase. A kinase activity in one of its subunits phosphorylates Pol II at many places in the CTD. Several other protein kinases, including CDK9 (cyclin-dependent kinase 9), which is part of the complex pTEFb (*p*ositive *t*ranscription *e*longation *f*actor *b*), also phosphorylate the CTD. This causes a conformational change in the overall complex, initiating transcription. Phosphorylation of the CTD is also important during the subsequent elongation phase, and it affects the interactions between the transcription complex and other enzymes involved in processing the transcript (as described below). During synthesis of the initial 60 to 70 nucleotides of RNA, first TFIIE and then TFIIH is released, and Pol II enters the elongation phase of transcription.

Career Avenues

Elongation, Termination, and Release:

TFIIF remains associated with Pol II throughout elongation. During this stage, the activity of the polymerase is greatly enhanced by proteins called elongation factors. The elongation factors suppress pausing during transcription and also coordinate interactions between proteins complexes involved in the posttranscriptional processing of mRNAs. Once the RNA transcript is completed, transcription is terminated. Pol II is dephosphorylated and recycled, ready to initiate another transcript.

Regulation of RNA Polymerase II Activity:
Regulation of transcription at Pol II promoters is quite elaborate. It involves the interaction of a wide variety of other proteins with the pre-initiation complex. Some of these regulatory proteins interact with transcription factors, others with Pol II itself. Many interact through TFIID, a complex of about 12 proteins, including TBP and certain *T*BP-*a*ssociated *f*actors, or TAFs.

Diverse Functions of TFII:
In eukaryotes, the repair of damaged DNA is more efficient within genes that are actively being transcribed than for other damaged DNA, and the template strand is repaired somewhat more efficiently than the non-template strand. These remarkable observations are explained by the alternative roles of the TFIIH subunits. Not only does TFIIH participate in the formation of the closed complex during assembly of a transcription complex (as described above), but some of its subunits are also essential components of the separate nucleotide-excision repair complex. When Pol II transcription halts at the site of a DNA lesion, TFIIH can interact with the lesion and recruit the entire nucleotide-excision repair complex. Genetic loss of certain TFIIH subunits can produce human diseases. Some examples are xeroderma pigmentosum and Cockayne's syndrome, which is characterized by arrested growth, photosensitivity, and neurological disorders.

DNA-DEPENDENT RNA POLYMERASE UNDERGOES SELECTIVE INHIBITION

1. The elongation of RNA strands by RNA polymerase in both bacteria and eukaryotes is inhibited by the antibiotic **actinomycin D.** The planar portion of this molecule inserts (intercalates) into the double helical DNA between successive GqC base pairs, deforming the DNA. This prevents movement of the polymerase along the template.

2. **Acridine** inhibits RNA synthesis in a similar fashion.

3. **Rifampicin** inhibits bacterial RNA synthesis by binding to the subunit of bacterial RNA polymerases.

4. **α-amanitin:** disrupts mRNA formation in animal cells by blocking Pol II and, at higher concentrations, Pol III. Neither Pol I nor bacterial RNA polymerase is sensitive to -amanitin—nor is the RNA polymerase II of *A. phalloides* itself!

Career Avenues

RNA PROCESSING

Many of the RNA molecules in bacteria and virtually all RNA molecules in eukaryotes are processed to some degree after synthesis. A newly synthesized RNA molecule is called a **primary transcript (or hnRNA).** The primary transcript for a eukaryotic mRNA contain both introns (non coding region) and exons (coding region). In a process where the introns are removed from the primary transcript and the exons are joined to form a continuous sequence that specifies a functional polypeptide called **splicing**.

Processing of mRNA in eukaryotes

- 5' capping
- 3' tailing
- Splicing

5' CAPPING

Most eukaryotic mRNAs have a **5' cap,** a residue of 7- methylguanosine linked to the 5'-terminal residue of the mRNA through an unusual 5',5'-triphosphate linkage. The 5' cap helps protect mRNA from ribonucleases. The cap also binds to a specific cap binding complex of proteins and participates in binding of the mRNA to the ribosome to initiate translation. The 5' cap is formed by condensation of a molecule of GTP with the triphosphate at the 5' end of the transcript. The guanine is subsequently methylated at N-7, and additional methyl groups are often added at the 2' hydroxyls of the first and second nucleotides adjacent to the cap. The methyl groups are derived from *S*-adenosylmethionine. All these reactions occur very early in transcription, after the first 20 to 30 nucleotides of the transcript have been added. All three of the capping enzymes, and through them the 5' end of the transcript itself, are associated with the RNA polymerase II CTD until the cap is synthesized. The capped 5' end is then released from the capping enzymes and bound by the cap-binding complex.

Career Avenues

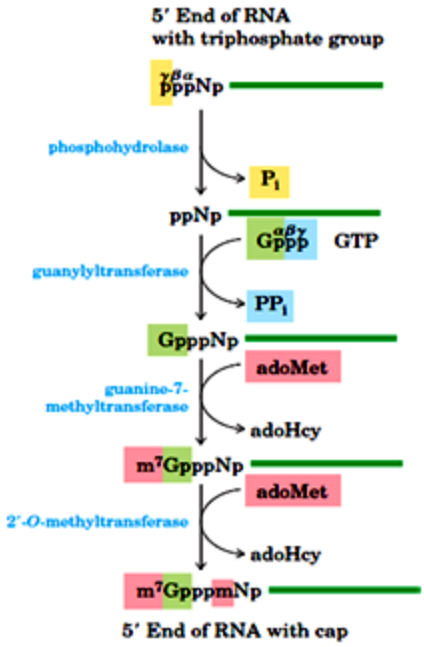


Fig: Generation of the 5 cap involves four to five separate steps

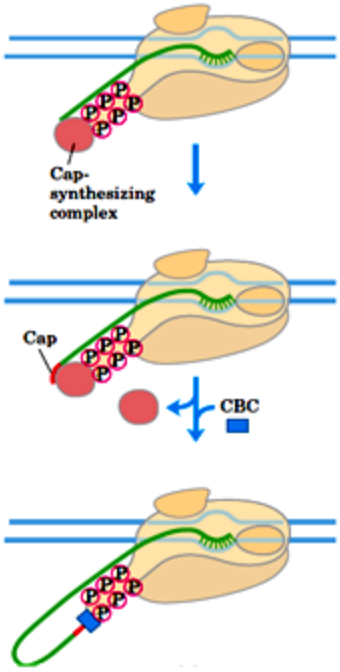


Fig: Synthesis of the cap is carried out by enzymes tethered to the CTD of Pol II.

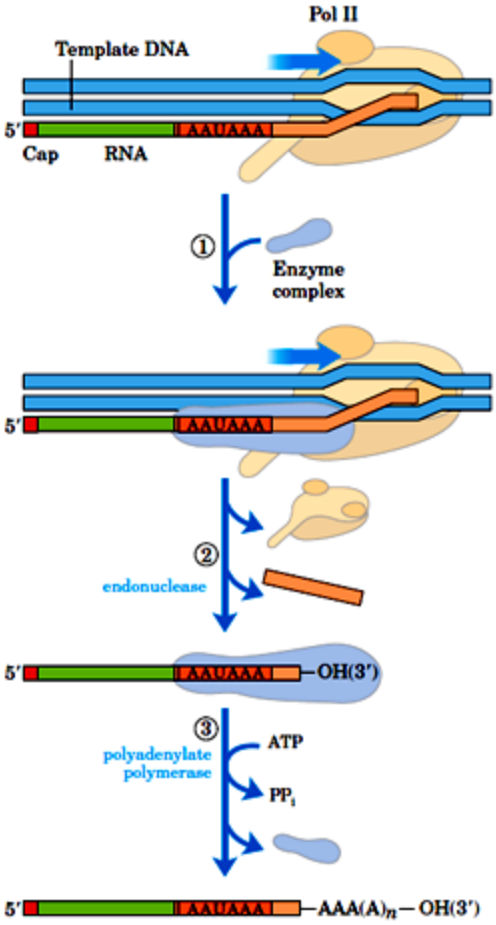
3'TAILING:

At their 3' end, most eukaryotic mRNAs have a string of 80 to 250 A residues, making up the **poly(A) tail.** This tail serves as a binding site for one or more specific proteins. The poly(A) tail and its associated proteins probably help protect mRNA from enzymatic destruction. Many prokaryotic mRNAs also acquire poly(A) tails, but these tails stimulate decay of mRNA rather than protecting it from degradation. The poly(A) tail is added in a multistep process. The transcript is extended beyond the site where the poly(A) tail is to be added, then is cleaved at the poly(A) addition site by an endonuclease component of a large enzyme complex, again associated with the CTD of RNA polymerase II. The mRNA site where cleavage occurs is marked by two sequence elements: the highly conserved sequence (5')AAUAAA(3'), 10 to 30 nucleotides on the 5' side (upstream) of the cleavage site, and a less well-defined sequence rich in G and U residues, 20 to 40 nucleotides downstream of the cleavage site. Cleavage generates the free 3'-hydroxyl group that defines the end of the mRNA, to which A residues are immediately added by **polyadenylate polymerase,** which catalyzes the reaction

Career A

Addition of the poly(A) tail to the primary RNA transcript of eukaryotes.
Pol II synthesizes RNA beyond the segment of the transcript containing the cleavage signal sequences, including the highly conserved upstream sequence (5')AAUAAA.

1- The cleavage signal sequence is bound by an enzyme complex that includes an endonuclease, a polyadenylate polymerase, and several other multisubunit proteins involved in sequence recognition, stimulation of cleavage, and regulation of the length of the poly(A) tail.
2- The RNA is cleaved by the endonuclease at a point 10 to 30 nucleotides 3' to (downstream of) the sequence AAUAAA.
3- The polyadenylate polymerase synthesizes a poly(A) tail 80 to 250 nucleotides long, beginning at the cleavage site.



SPLICING

Many genes for polypeptides in eukaryotes are interrupted by non-coding sequences (introns). The vast majority of genes in vertebrates contain introns; among the few exceptions are those that encode histones.

There are 4 types of introns:

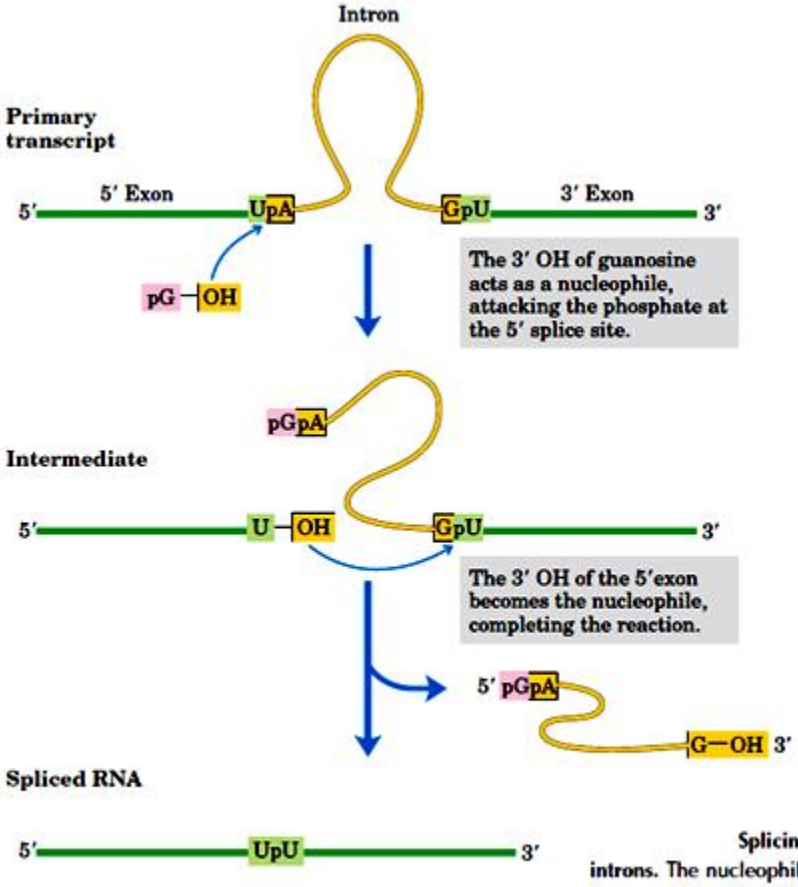
- Group I
- Group II
- Group III
- Group IV

Career

Group I intron and their splicing

Group I Introns found in some nuclear, mitochondria, chloroplast genes (which are coding for mRNA, rRNA and tRNA). There splicing does not utilize any enzyme and ATP (as energy). Group I requires guanine nucleoside or guanine nucleotide as cofactor from outside. In group I intron splicing, lariat does not form.

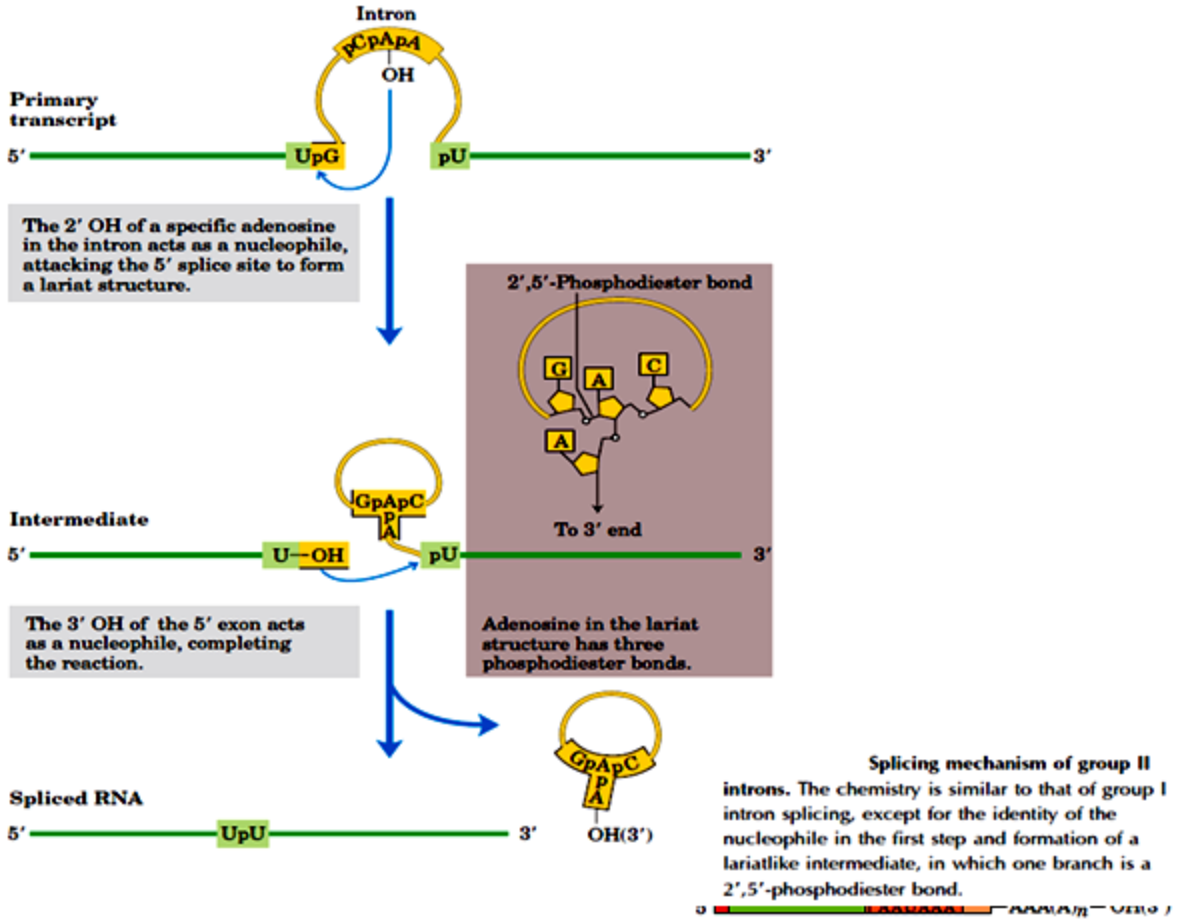
The group I splicing reaction requires a guanine nucleoside or nucleotide cofactor, but the cofactor is not used as a source of energy; instead, the 3'-hydroxyl group of guanosine is used as a nucleophile in the first step of the splicing pathway. The guanosine 3'-hydroxyl group forms a normal 3',5'-phosphodiester bond with the 5' end of the intron. The 3' hydroxyl of the exon that is displaced in this step then acts as a nucleophile in a similar reaction at the 3' end of the intron. The result is precise excision of the intron and ligation of the exons.



Splicing mechanism of group I introns. The nucleophile in the first step may be guanosine, GMP, GDP, or GTP. The spliced intron is eventually degraded.

Group II intron and their splicing

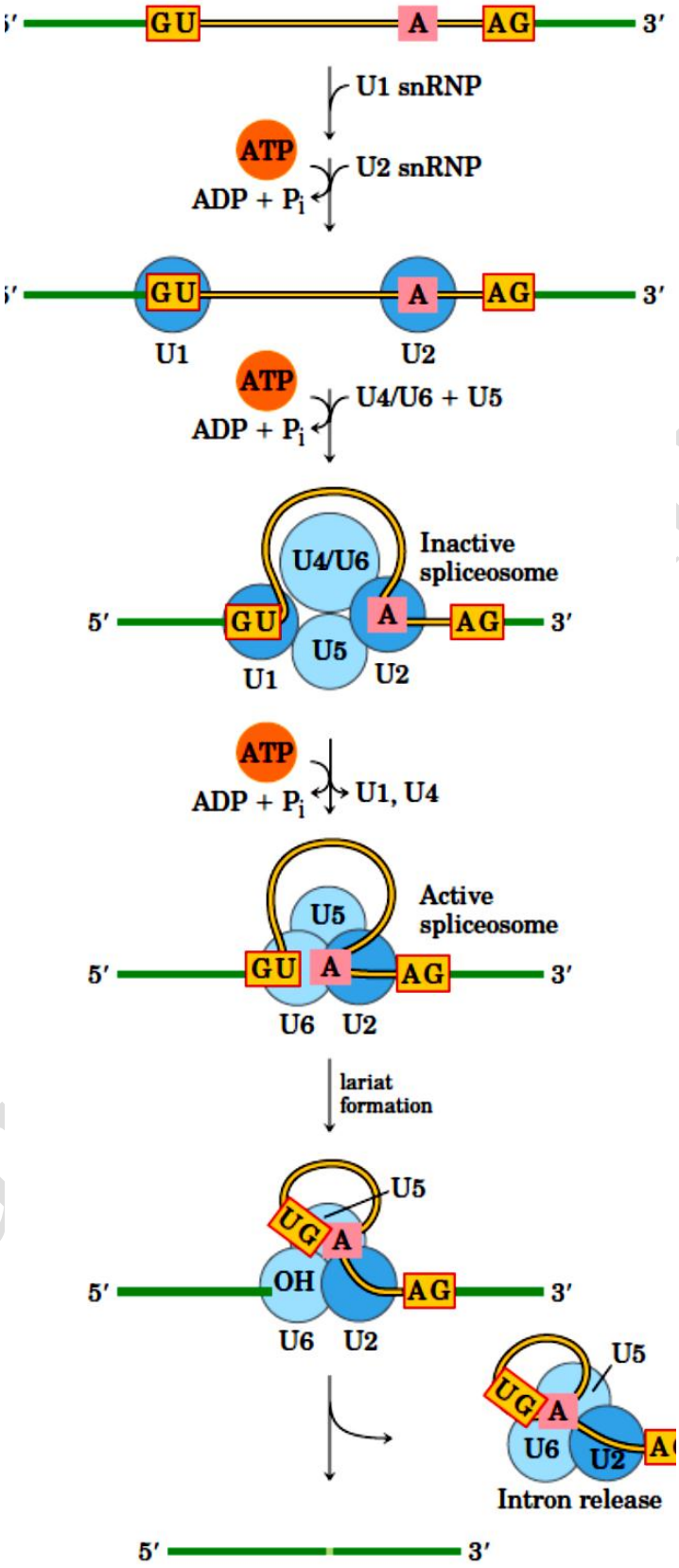
Group II introns Found in primary transcript of mitochondrial and chloroplast mRNA of eukaryotic plants (like algae, fungi and higher plants). It requires adenine nucleotide within the intron. In group II intron splicing lariat will form. Some of the group II intron codes reverse transcriptase.



Splicing mechanism of group II introns. The chemistry is similar to that of group I intron splicing, except for the identity of the nucleophile in the first step and formation of a lariatlike intermediate, in which one branch is a 2′,5′-phosphodiester bond.

Group III intron and their splicing

Most introns are *not* self-splicing, and these types are not designated with a group number. The third and largest class of introns includes those found in nuclear mRNA primary transcripts. These are called **Spliceosomal introns,** because their removal occurs within and is catalyzed by a large protein complex called a **spliceosome.** Within the spliceosome, the introns undergo splicing by the same lariat-forming mechanism as the group II introns. The spliceosome is made up of specialized RNA-protein complexes, *s*mall *n*uclear *r*ibo*n*ucleo*p*roteins (snRNPs, often pronounced "snurps"). Each snRNP contains one of a class of eukaryotic RNAs, 100 to 200 nucleotides long, known as **small nuclear RNAs (snRNAs).** Five snRNAs (U1, U2, U4, U5, and U6) involved in splicing reactions are generally found in abundance in eukaryotic nuclei. The RNAs and proteins in snRNPs are highly conserved in eukaryotes from yeasts to humans.


GU
A
AG
3′
U1 snRNP
ATP
U2 snRNP
ADP + Pi
U1
U2
U4/U6 + U5
Inactive
spliceosome
U4/U6
5′
U5
U1, U4
Active
spliceosome
U6
lariat
formation
UG
OH
Intron release

Ca ues

Assembly of Spliceosomes: The U1 and U2 snRNPs bind, then the remaining snRNPs (the U4/U6 complex and U5) bind to form an inactive spliceosome. Internal rearrangements convert this species to an active spliceosome in which U1 and U4 have been expelled and U6 is paired with both the 5' splice site and U2. This is followed by the catalytic steps, which parallel those of the splicing of group II introns.

Group IV intron and their splicing

The fourth class of introns, found in certain tRNAs, is distinguished from the group I and II introns in that the splicing reaction requires ATP and an endonuclease. The splicing endonuclease cleaves the phosphodiester bonds at both ends of the intron, and the two exons are joined by a mechanism similar to the DNA ligase reaction.

Transcription control

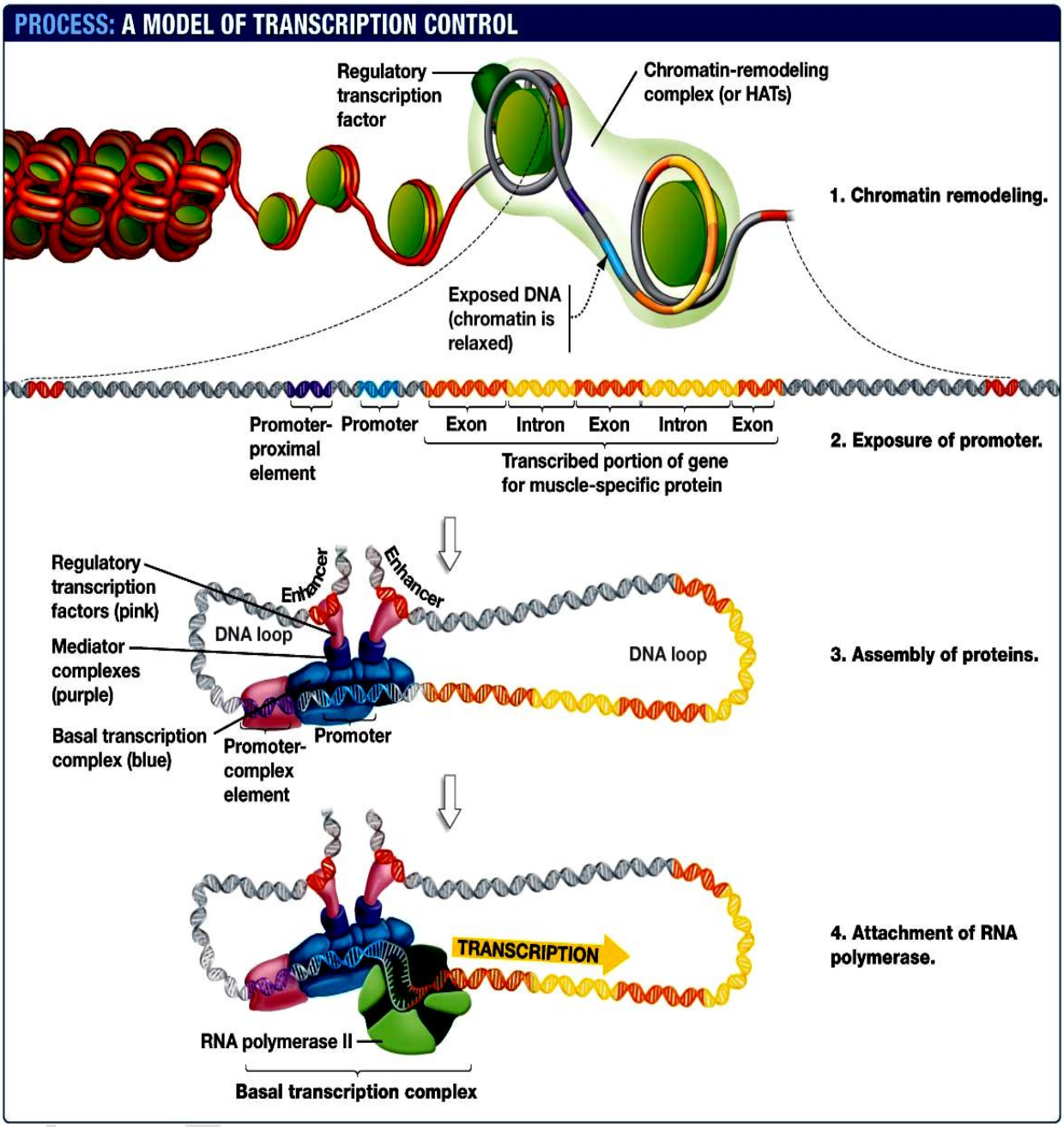
The lactose operon provides a good model system of several concepts of prokaryotic gene regulation. It consists of three structural genes (lacZ, lacY, and lacA) as well as three controlling sites (lacCRP, lacP1, and lacO). The structural genes lacZ, lacY, and lacA encode the enzymes b-galactosidase, permease, and transacetylase, respectively. Catabolism of lactose is dependent upon these proteins. The controlling sites lacCRP, lacP1, and lacO are binding sites for the cAMP receptor protein, RNA polymerase, and the lactose repressor, respectively.

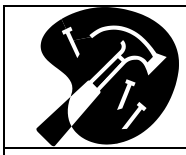
In eukaryotes, just as in bacteria, genes are regulated so that they are expressed at the right time and at the correct levels to maintain the cell or promote growth and proliferation. Cells in multicellular organisms must not only respond to the chemicals in their environment but must also work together through complex signaling that often affects gene activity.

The controlling sites in eukaryotes are similar to those found in bacteria, however there are many more controlling sites and proteins affecting each eukaryotic gene. Transcription factors (TFs) attach to binding sites in promoter regions and stimulate RNA polymerase binding to the promoter site. Transcription factors are produced constitutively since great numbers of genes depend on these for expression.

Enhancers are bound by transcription activators (TAs) that are synthesized in response to specific signals. Most enhancers, such as the one that binds Gal4, are located hundreds or thousands of base pairs from the promoter sites. However, some induced activators, such as Fos and Jun, bind very near promoter sites. TAs cause the DNA to loop back on itself when they interact with the TFs near the promoter. This interaction between enhancer sites and initiator sites is usually necessary for transcription above a basal level. Coactivators are activator proteins that often connect TFs and/or TAs with the RNA polymerase and may be essential for gene expression

Career Avenues


PROCESS: A MODEL OF TRANSCRIPTION CONTROL
Regulatory transcription factor
Chromatin-remodeling complex (or HATs)
1. Chromatin remodeling.
Exposed DNA (chromatin is relaxed)
Promoter-proximal element
Promoter
Exon
Intron
Exon
Intron
Exon
Transcribed portion of gene for muscle-specific protein
2. Exposure of promoter.
Regulatory transcription factors (pink)
Enhancer
Enhancer
DNA loop
Mediator complexes (purple)
DNA loop
3. Assembly of proteins.
Basal transcription complex (blue)
Promoter-complex element
Promoter
TRANSCRIPTION
4. Attachment of RNA polymerase.
RNA polymerase II
Basal transcription complex



GATE Questions

GATE 2008

Q.3 Match the items on the left with the inhibitors on the right

(P) DNA Polymerase α	(1) Phenyl methyl sulphonyl fluoride (PMSF)
(Q) RNA Polymerase II	(2) Aphidicolin
(R) Serine protease	(3) α amanitin
	(4) Actinomycin

(A)	(B)	(C)	(D)
P-2	P-3	P-2	P-1
Q-3	Q-1	Q-1	Q-2
R-1	R-2	R-2	R-4

GATE 2010

Q.4 Nucleolus is involved in the synthesis of

(A) rRNA (B) tRNA (C) DNA (D) mRNA

Q.5 Multiple RNA polymerase transcribes a DNA template, unwinding about 1.5 turns of DNA template per transcription bubble. From the structural information of classical B-DNA, how many transcription bubbles are possible for a 180 base pair DNA molecule?

(A) 12 (B) 27 (C) 6 (D) 270

Answers		
Q.3 – A	**Q.4 – A**	**Q.5 – A**

Career Avenues

TRANSLATIONATIONAL PROCESS

PAIRING RELATIONSHIP OF CODON AND ANTICODON:

When several different codons specify one amino acid, the difference between them usually lies at the third base position (at the 3_ end). For example, alanine is coded by the triplets GCU, GCC, GCA, and GCG.

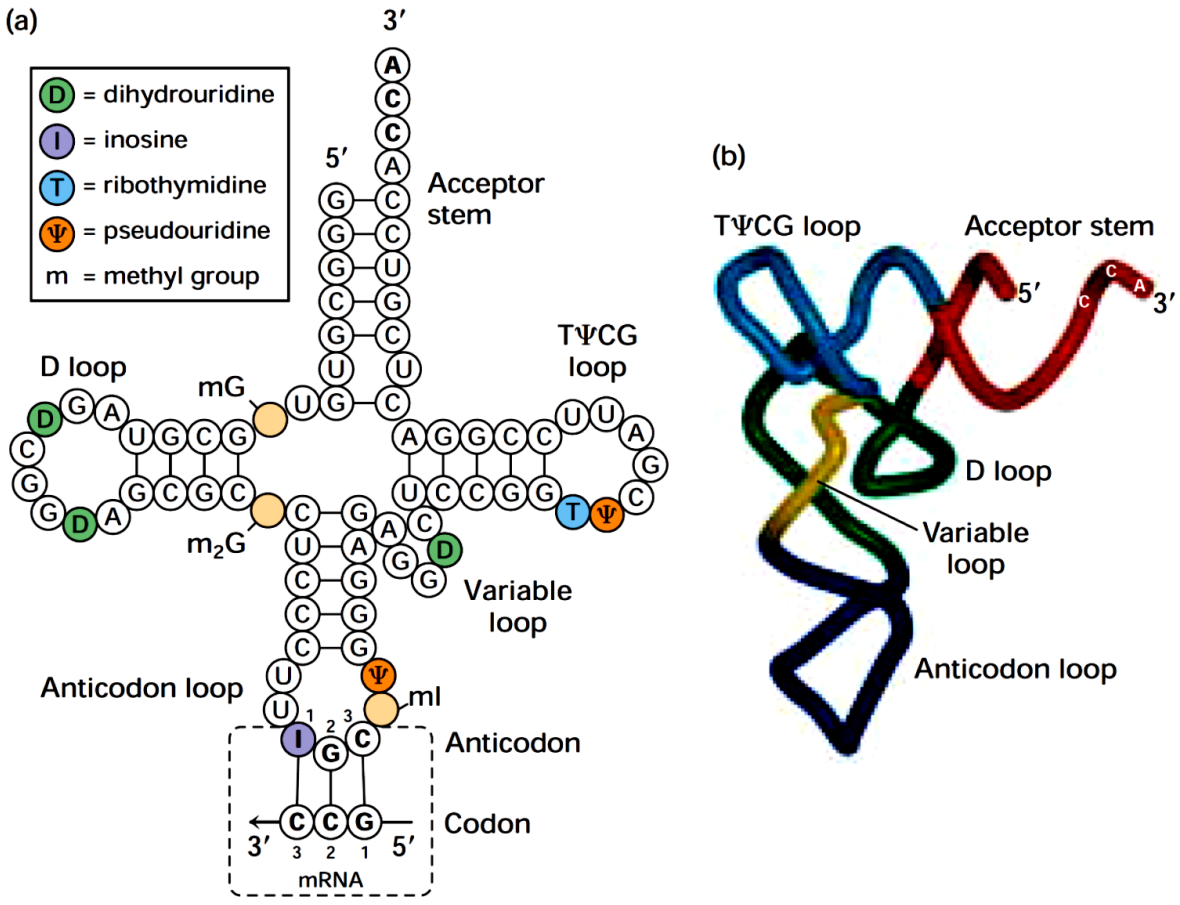


Figure: Pairing relationship of codon and anticodon. **(a)** Alignment of the two RNAs is antiparallel. The tRNA is shown in the traditional cloverleaf configuration. **(b)** Three different codon pairing relationships are possible when the tRNA anticodon contains inosinate.

WOBBLE HYPOTHESIS:

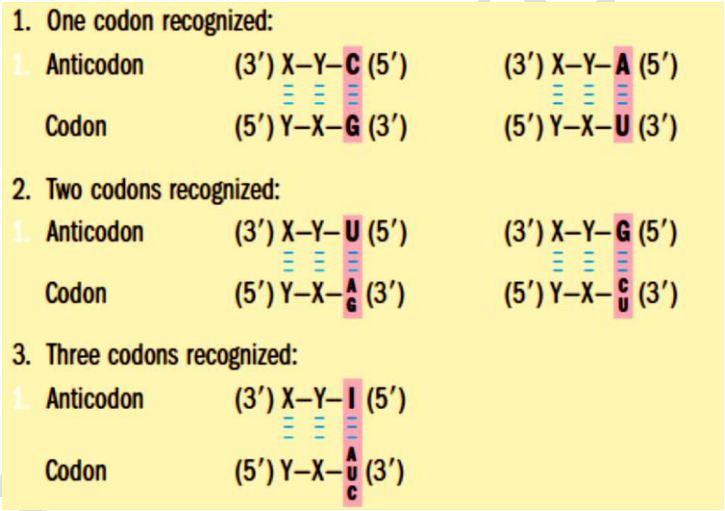
1. The first two bases of an mRNA codon always form strong Watson-Crick base pairs with the corresponding bases of the tRNA anticodon and confer most of the coding specificity.
2. The first base of the anticodon (reading in the 5'-3' direction; this pairs with the third base of the codon) determines the number of codons recognized by the tRNA. When the first

base of the anticodon is C or A, base pairing is specific and only one codon is recognized by that tRNA. When the first base is U or G, binding is less specific and two different codons may be read. When inosine(I) is the first (wobble) nucleotide of an anticodon, three different codons can be recognized—the maximum number for any tRNA.

3. When an amino acid is specified by several different codons, the codons that differ in either of the first two bases require different tRNAs.
4. A minimum of 32 tRNAs are required to translate all 61 codons (31 to encode the amino acids and 1 for initiation).

The wobble (or third) base of the codon contributes to specificity, but, because it pairs only loosely with its corresponding base in the anticodon, it permits rapid dissociation of the tRNA from its codon during protein synthesis. If all three bases of a codon engaged in strong Watson-Crick pairing with the three bases of the anticodon, tRNAs would dissociate too slowly and this would severely limit the rate of protein synthesis. Codon-anticodon interactions balance the requirements for accuracy and speed.

How the Wobble Base of the Anticodon Determines the Number of Codons a tRNA Can Recognize:



KEY POINTS:

■ The particular amino acid sequence of a protein is constructed through the translation of information encoded in mRNA. This process is carried out by ribosomes.

■ Amino acids are specified by mRNA codons consisting of nucleotide triplets. Translation requires adaptor molecules, the tRNAs, that recognize codons and insert amino acids into their appropriate sequential positions in the polypeptide.

■ The base sequences of the codons were deduced from experiments using synthetic mRNAs of known composition and sequence.

Career Avenues

■ The codon AUG signals initiation of translation. The triplets UAA, UAG, and UGA are signals for termination.

■ The genetic code is degenerate: it has multiple code words for almost every amino acid.

■ The standard genetic code words are universal in all species, with some minor deviations in mitochondria and a few single-celled organisms.

■ The third position in each codon is much less specific than the first and second and is said to wobble.

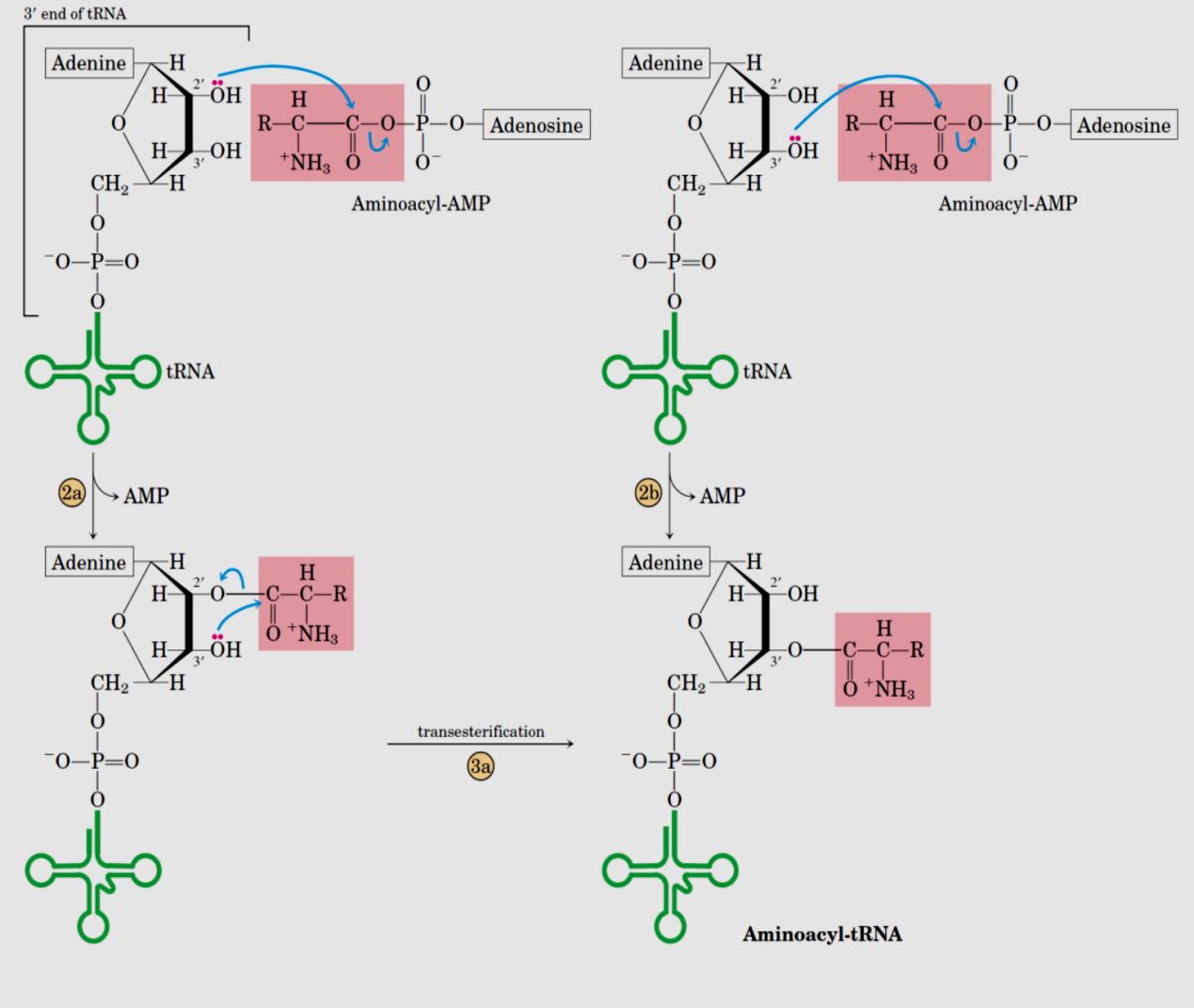
PROTEIN SYNTHESIS:

Protein Biosynthesis Takes Place in Five Stages:

Stage 1: Activation of Amino Acids

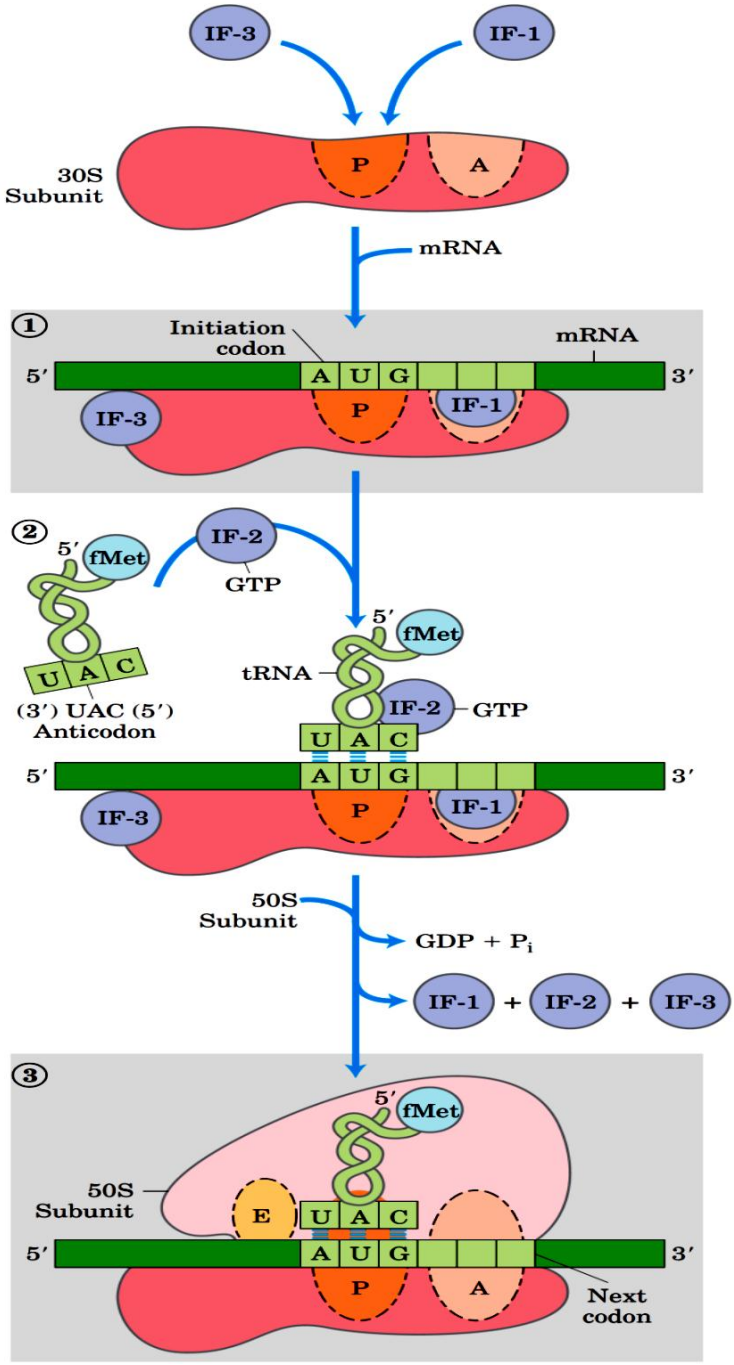
For the synthesis of a polypeptide with a defined sequence, two fundamental chemical requirements must be met: (1) the carboxyl group of each amino acid must be activated to facilitate formation of a peptide bond, and (2) a link must be established between each new amino acid and the information in the mRNA that encodes it. Both these requirements are met by attaching the amino acid to a tRNA in the first stage of protein synthesis. Attaching the right amino acid to the right tRNA is critical. This reaction takes place in the cytosol, not on the ribosome. Each of the 20 amino acids is covalently attached to a specific tRNA at the expense of ATP energy, using Mg^{2+}- dependent activating enzymes known as aminoacyltRNA synthetases. When attached to their amino acid (aminoacylated) the tRNAs are said to be "charged." Below in the image you can see the charging of amino acid to it's corresponding tRNA.

Career Avenues



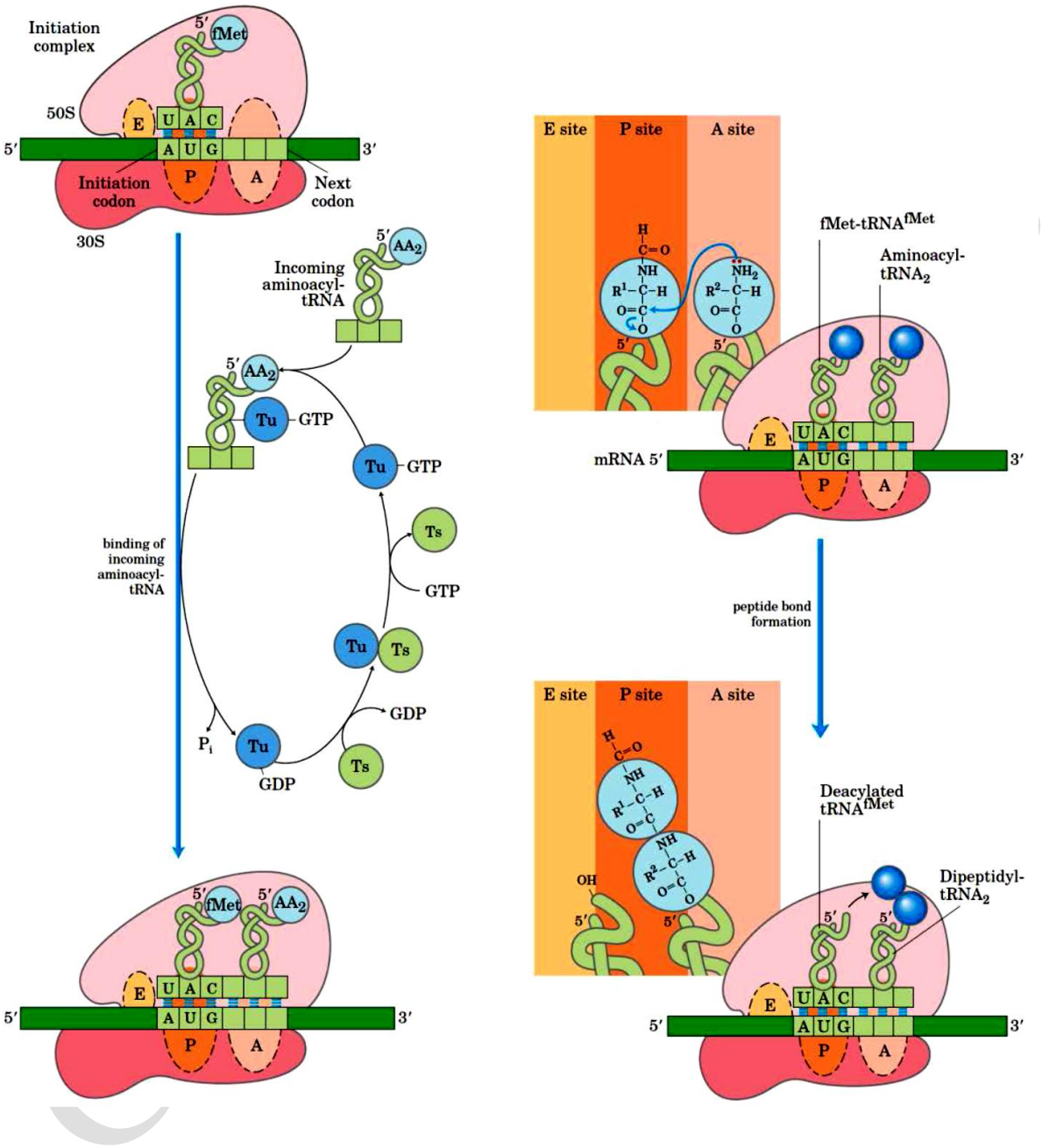
Stage 2: Initiation

The mRNA bearing the code for the polypeptide to be made binds to the smaller of two ribosomal subunits and to the initiating aminoacyl-tRNA. the large ribosomal subunit then binds to form an initiation complex. The initiating aminoacyl-tRNA base pairs with the mRNA codon AUG, that signals the beginning of the polypeptide. This process, which requires GTP, is promoted by cytosolic proteins called initiation factors.

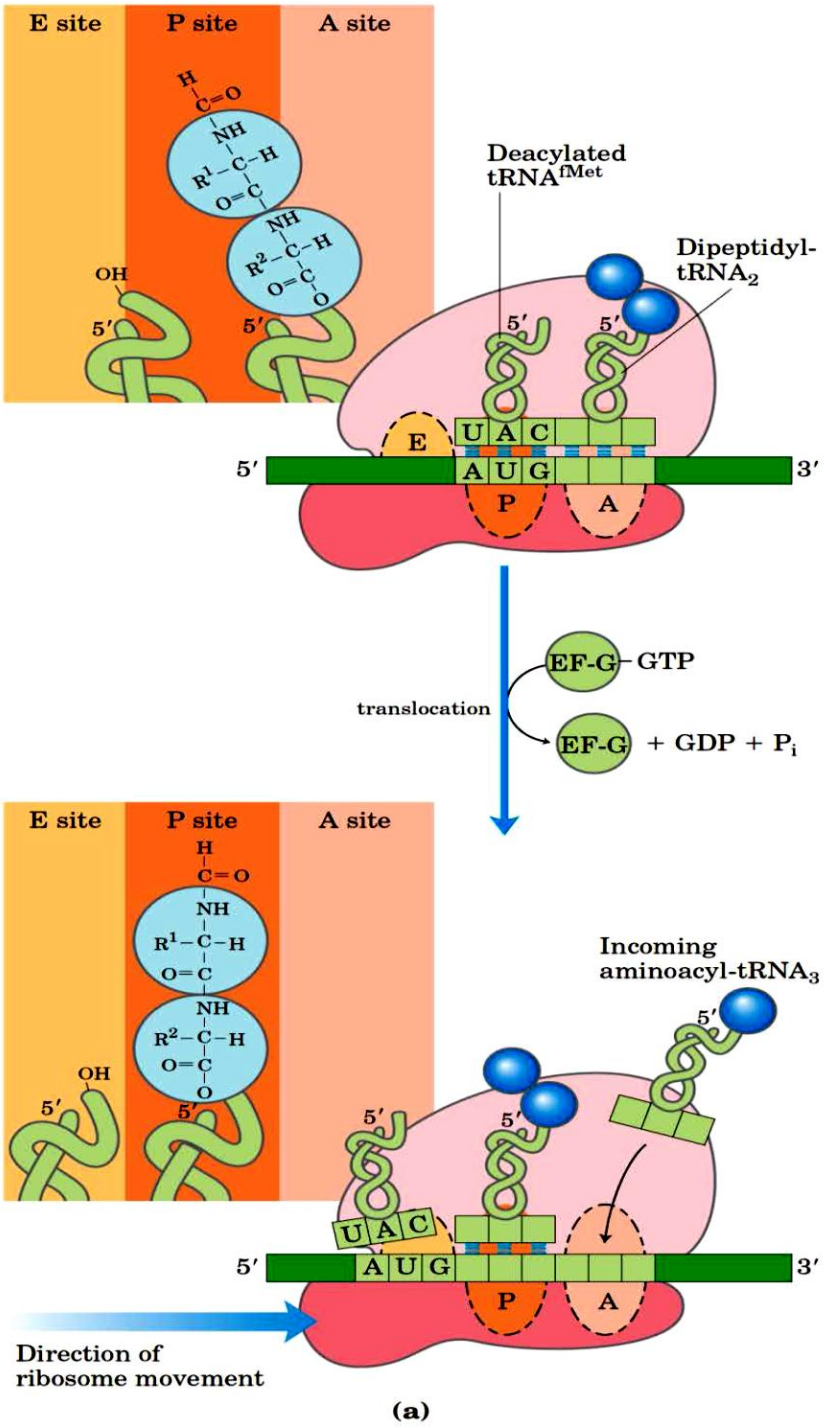


Stage 3: Elongation

The nascent polypeptide is lengthened by covalent attachment of successive amino acid units, each carried to the ribosome and correctly positioned by its tRNA, which base-pairs to its corresponding codon in the mRNA. Elongation requires cytosolic proteins known as elongation factors. The binding of each incoming aminoacyl-tRNA and the movement of the ribosome along the mRNA are facilitated by the hydrolysis of GTP as each residue is added to the growing polypeptide.

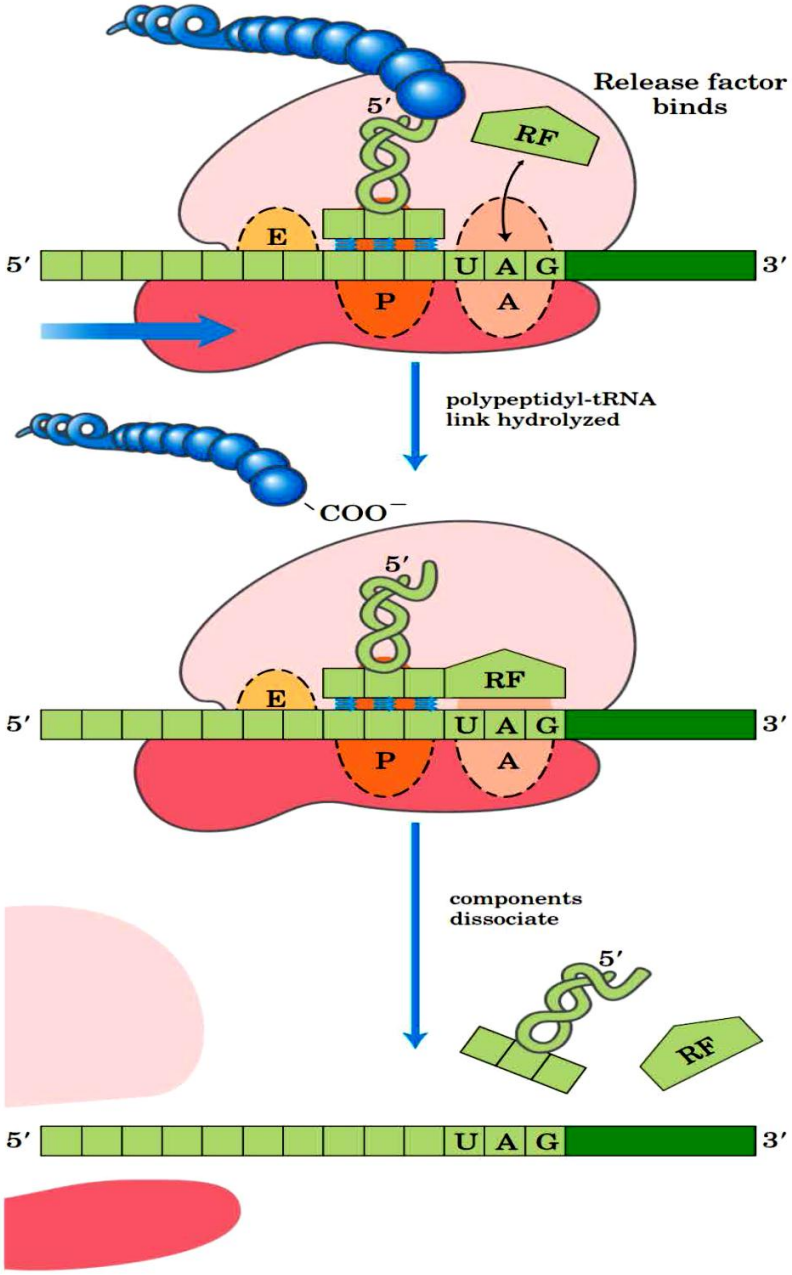

Initiation complex
5′
fMet
50S
E
U A C
A U G
5′
3′
Initiation codon
P
A
Next codon
30S
5′
AA2
Incoming aminoacyl-tRNA
5′
AA2
Tu
GTP
Tu
GTP
Ts
GTP
binding of incoming aminoacyl-tRNA
Tu
Ts
GDP
Pi
Tu
Ts
GDP
5′
fMet
5′
AA2
E
U A C
A U G
5′
3′
P
A
E site
P site
A site
H
C=O
NH
R1–C–H
O=C
O
NH2
R2–C–H
O=C
O
5′
5′
fMet-tRNAfMet
Aminoacyl-tRNA2
E
U A C
A U G
mRNA 5′
3′
P
A
peptide bond formation
E site
P site
A site
H
C=O
NH
R1–C–H
O=C
NH
R2–C–H
O=C
O
OH
5′
5′
Deacylated tRNAfMet
Dipeptidyl-tRNA2
5′
5′
E
U A C
A U G
5′
3′
P
A

Stage 4: Translocation



Stage 5: Termination and Release

Completion of the polypeptide chain is signaled by a termination codon in the mRNA. The new polypeptide is released from the ribosome, aided by proteins called release factors.



Stage 6: Folding and Posttranslational Processing

In order to achieve its biologically active form, the new polypeptide must fold into its proper three-dimensional conformation. Before or after folding, the new polypeptide may undergo enzymatic processing, including removal of one or more amino acids (usually from the amino terminus); addition of acetyl, phosphoryl, methyl, carboxyl, or other groups to certain amino acid residues; proteolytic cleavage; and/or attachment of oligosaccharides or prosthetic groups.

COMPONENTS REQUIRED FOR THE FIVE MAJOR STAGES OF PROTEIN:

Synthesis in *E. coli:*

Stage	*Essential components*
1. Activation of amino acids	20 amino acids 20 aminoacyl-tRNA synthetases 32 or more tRNAs ATP Mg^{2+}
2. Initiation	mRNA *N*-Formylmethionyl-$tRNA^{fmet}$ Initiation codon in mRNA (AUG) 30S ribosomal subunit 50S ribosomal subunit Initiation factors (IF-1, IF-2, IF-3) GTP Mg^{2+}
3. Elongation	Functional 70S ribosome (initiation complex) Aminoacyl-tRNAs specified by codons Elongation factors (EF-Tu, EF-Ts, EF-G) GTP Mg^{2+}
4. Termination and release	Termination codon in mRNA Release factors (RF-1, RF-2, RF-3)
5. Folding and posttranslational processing	Specific enzymes, cofactors, and other components for removal of initiating residues and signal sequences, additional proteolytic processing, modification of terminal residues, and attachment of phosphate, methyl, carboxyl, carbohydrate, or prosthetic groups

PROTEIN SYNTHESIS IS INHIBITED BY MANY ANTIBIOTICS AND TOXINS:

In general, protein synthesis inhibitors work at different stages of prokaryotic mRNA translation into proteins, like initiation, elongation (including aminoacyl tRNA entry, proofreading, peptidyl transfer, and ribosomal translocation) and termination:

Earlier stages

• Rifamycin inhibits prokaryotic DNA transcription into mRNA by inhibiting DNA-dependent RNA polymerase by binding its beta-subunit.

Initiation

• Linezolid acts at the initiation stage, probably by preventing the formation of the initiation complex.

Aminoacyl tRNA entry

• Tetracyclines and Tigecycline (a glycylcycline related to tetracyclines) block the A site on the ribosome, preventing the binding of aminoacyl tRNAs.

Proofreading

• Aminoglycosides, among other potential mechanisms of action, interfere with the proofreading process, causing increased rate of error in synthesis with premature termination.

Peptidyl transfer

• Chloramphenicol blocks the peptidyl transfer step of elongation on the 50S ribosomal subunit in both bacteria and mitochondria.
• Macrolides (as well as inhibiting *ribosomal translocation* and other potential mechanisms) bind to the 50s ribosomal subunits, inhibiting *peptidyl transfer*.
• Quinupristin/dalfopristin act synergistically, with dalfopristin, enhancing the binding of quinupristin, as well as inhibiting *peptidyl transfer*. Quinupristin binds to a nearby site on the 50S ribosomal subunit and prevents elongation of the polypeptide, as well as causing incomplete chains to be released.

Ribosomal translocation

• Macrolides, clindamycin and aminoglycosides (with all these three having other potential mechanisms of action as well), have evidence of inhibition of ribosomal translocation.
• Fusidic acid prevents the turnover of elongation factor G (EF-G) from the ribosome.

Termination

• Macrolides and clindamycin (both also having other potential mechanisms) cause premature dissociation of the peptidyl-tRNA from the ribosome.

Career Avenues

• Puromycin has a structure similar to that of the tyrosinyl aminoacyl-tRNA. Thus, it binds to the ribosomal A site and participates in peptide bond formation, producing peptidyl-puromycin. However, it does not engage in translocation and quickly dissociates from the ribosome, causing a premature termination of polypeptide synthesis.
• Streptogramins also cause premature release of the peptide chain.

Protein synthesis inhibitors of unspecified mechanism

• Retapamulin

Binding site

The following antibiotics bind to the 30S subunit of the ribosome:

- Aminoglycosides
- Tetracyclines

The following antibiotics bind to the 50S ribosomal subunit:

- Chloramphenicol
- Erythromycin
- Clindamycin
- Linezolid
- Telithromycin
- Streptogramins
- Retapamulin

KEY POINTS:

■ Protein synthesis occurs on the ribosomes, which consist of protein and rRNA. Bacteria have 70S ribosomes, with a large (50S) and a small (30S) subunit. Eukaryotic ribosomes are significantly larger (80S) and contain more proteins.

■ Transfer RNAs have 73 to 93 nucleotide residues, some of which have modified bases. Each tRNA has an amino acid arm with the terminal sequence CCA(3') to which an amino acid is esterified, an anticodon arm, a TψC arm, and a D arm; some tRNAs have a fifth arm. The anticodon is responsible for the specificity of interaction between the aminoacyl-tRNA and the complementary mRNA codon.

■ The growth of polypeptides on ribosomes begins with the amino-terminal amino acid and proceeds by successive additions of new residues to the carboxyl-terminal end.

Career Avenues

GATE Questions

GATE 2008

Q.6 Match the function in Column I with organelle in Column II

Column I	Column II
(P) Protein synthesis	(1) Endoplasmic reticulum
(Q) Protein degradation	(2) Golgi body
(R) Protein glycosylation	(3) Lysosome
	(4) Peroxisome

(A)	(B)	(C)	(D)
P-3	P-1	P-1	P-4
Q-2	Q-3	Q-4	Q-1
R-1	R-2	R-3	R-2

GATE 2010

Q.7 Amino acids responsible for N-linked and O-linked glycosylation of proteins are

(A) Asparagine and Asparctic acid (B) Glutamine and Serine
(C) Glutamic acid and Serine (D) Asparagine and Threonine

Q.8 Approximate molecular weight (kDa) of the product after translation of a 390 bases mRNA will be

(A) 48 (B) 26 (C) 39 (D) 14

Answers:

Q.6 – B. Q.7 – D. Q.8 – D.

Career Avenues

REGULATORY CONTROLS IN PROKARYOTES AND EUKARYOTES

CONTROL OF GENE EXPRESSION IN PROKARYOTES

Since the structure and function of a cell are determined by the proteins it contains, the control of gene expression is a fundamental aspect of molecular cell biology. Most commonly, the "decision" to initiate transcription of the gene encoding a particular protein is the major mechanism for controlling production of the encoded protein in a cell. By controlling transcription initiation, a cell can regulate which proteins it produces and how rapidly. When transcription of a gene is repressed, the corresponding mRNA and encoded protein or proteins are synthesized at low rates. Conversely, when transcription of a gene is activated, both the mRNA and encoded protein or proteins are produced at much higher rates.

In most bacteria and other single-celled organisms, gene expression is highly regulated in order to adjust the cell's enzymatic machinery and structural components to changes in the nutritional and physical environment. Thus, at any given time, a bacterial cell normally synthesizes only those proteins of its entire proteome required for survival under the particular conditions. In multicellular organisms, control of gene expression is largely directed toward assuring that the right gene is expressed in the right cell at the right time during embryological development and tissue differentiation. Here we describe the basic features of transcription control in bacteria, using the lac operon in E. coli as our primary example. Many of the same processes, as well as others, are involved in eukaryotic transcription control also.

In *E. coli,* about half the genes are clustered into operons each of which encodes enzymes involved in a particular metabolic pathway or proteins that interact to form one multi-subunit protein. For instance, the *trp* operon mentioned earlier encodes five enzymes needed in the biosynthesis of tryptophan (see Figure 4-12). Similarly, the *lac* operon encodes three enzymes required for the metabolism of lactose, a sugar present in milk. Since a bacterial operon is transcribed from one start site into a single mRNA, all the genes within an operon are coordinately regulated; that is, they are all activated or repressed to the same extent.

Transcription of operons, as well as of isolated genes, is controlled by interplay between RNA polymerase and specific *repressor* and *activator* proteins. In order to initiate transcription, however, *E. coli* RNA polymerase must be associated with one of a small number of *(sigma) factors,* which function as initiation factors. The most common one in bacterial cells is σ^{70}.

Career Avenues

INITIATION OF *LAC* OPERON TRANSCRIPTION CAN BE REPRESSED AND ACTIVATED

When E. coli is in an environment that lacks lactose, synthesis of lac mRNA is repressed, so that cellular energy is not wasted synthesizing enzymes the cells cannot use. In an environment containing both lactose and glucose, E. coli cells preferentially metabolize glucose, the central molecule of carbohydrate metabolism. Lactose is metabolized at a high rate only when lactose is present and glucose is largely depleted from the medium. This metabolic adjustment is achieved by repressing transcription of the lac operon until lactose is present, and synthesis of only low levels of lac mRNA until the cytosolic concentration of glucose falls to low levels. Transcription of the lac operon under different conditions is controlled by lac repressor and catabolite activator protein (CAP), each of which binds to a specific DNA sequence in the lac transcription-control.

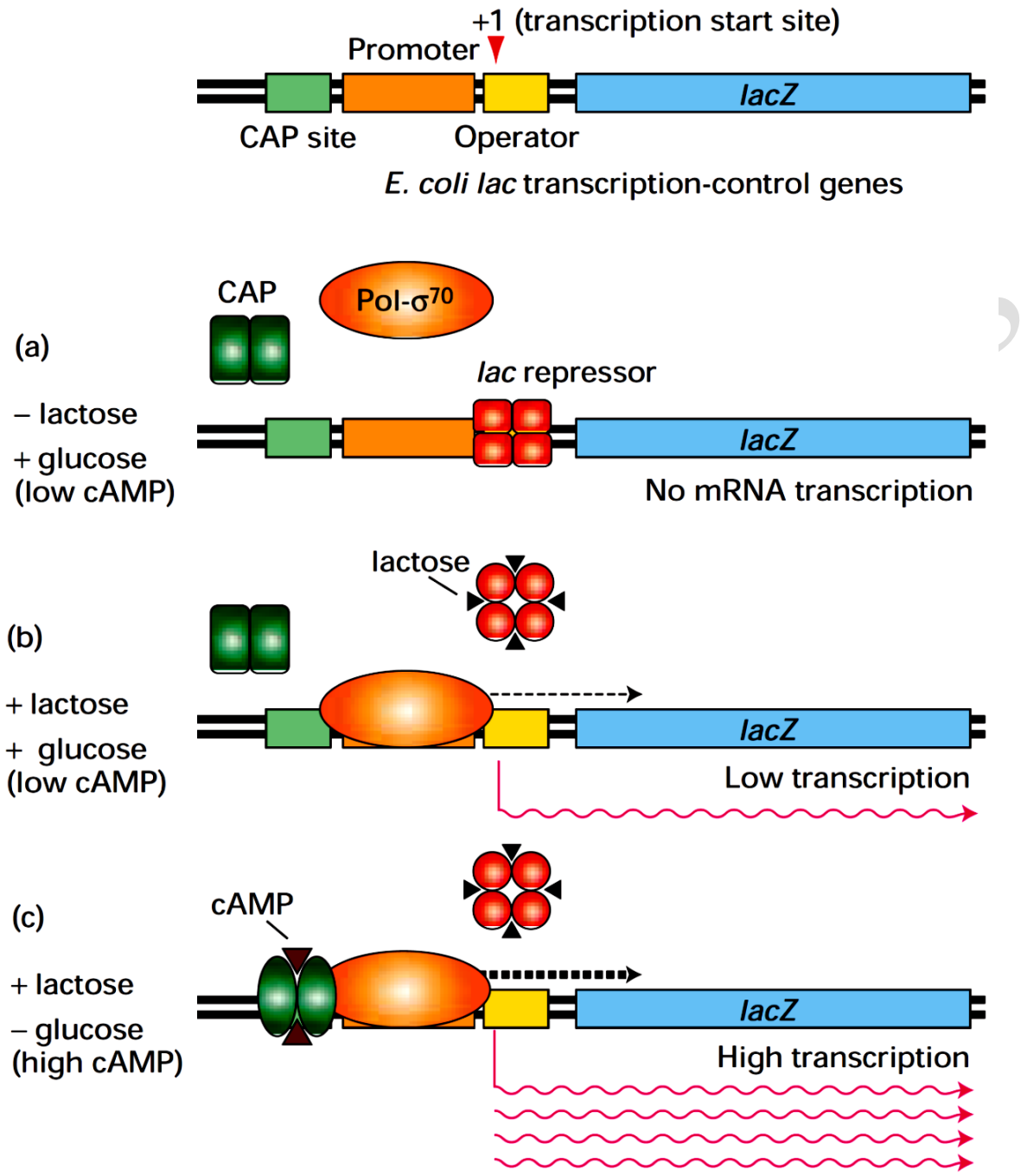
For transcription of the lac operon to begin, the 70 subunit of the RNA polymerase must bind to the lac promoter, which lies just upstream of the start site. When no lactose is present, binding of the lac repressor to a sequence called the lac operator, which overlaps the transcription start site, blocks transcription initiation by the .When lactose is present, it binds to specific binding sites in each subunit of the tetrameric lac repressor, causing a conformational change in the protein that makes it dissociate from the lac operator. As a result, the polymerase can initiate transcription of the lac operon. However, when glucose also is present, the rate of transcription initiation (i.e., the number of times per minute different polymerase molecules initiate transcription) is very low, resulting in synthesis of only low levels of lac mRNA and the proteins encoded in the lac operon.

Once glucose is depleted from the media and the intracellular glucose concentration falls, E. coli cells respond by synthesizing cyclic AMP, cAMP. As the concentration of cAMP increases, it binds to a site in each subunit of the dimeric CAP protein, causing a conformational change that allows the protein to bind to the CAP site in the lac transcription-control region. The bound CAPcAMP complex interacts with the polymerase bound to the promoter, greatly stimulating the rate of transcription initiation.

This activation leads to synthesis of high levels of lac mRNA and subsequently of the enzymes encoded by the lac operon.

Although the promoters for different E. coli genes exhibit considerable homology, their exact sequences differ. The promoter sequence determines the intrinsic rate at which an RNA polymerase – complex initiates transcription of a gene in the absence of a repressor or activator protein. Promoters that support a high rate of transcription initiation are called strong promoters. Those that support a low rate of transcription initiation are called weak promoters. The lac operon, for instance, has a weak promoter; its low intrinsic rate of initiation is further reduced by the lac repressor and substantially increased by the cAMP-CAP activator.

Career Avenues


+1 (transcription start site)
Promoter
lacZ
CAP site
Operator
E. coli lac transcription-control genes
CAP
Pol-σ70
(a)
lac repressor
– lactose
+ glucose
(low cAMP)
lacZ
No mRNA transcription
lactose
(b)
+ lactose
+ glucose
(low cAMP)
lacZ
Low transcription
cAMP
(c)
+ lactose
– glucose
(high cAMP)
lacZ
High transcription

SMALL MOLECULES REGULATE EXPRESSION OF MANY BACTERIAL GENES VIA DNA-BINDING REPRESSORS

Transcription of most *E. coli* genes is regulated by processes similar to those described for the *lac* operon. The general mechanism involves a specific repressor that binds to the operator region of a gene or operon, thereby blocking transcription initiation. A small molecule (or molecules), called an *inducer*, binds to the repressor, controlling its DNA-binding activity and consequently the rate of transcription as appropriate for the needs of the cell. For example, when the tryptophan concentration in the medium and cytosol is high, the cell does not synthesize the several enzymes encoded in the *trp* operon. Binding of tryptophan to the *trp* repressor causes a conformational change that allows the protein to bind to the *trp* operator, thereby repressing expression of the enzymes that synthesize tryptophan.

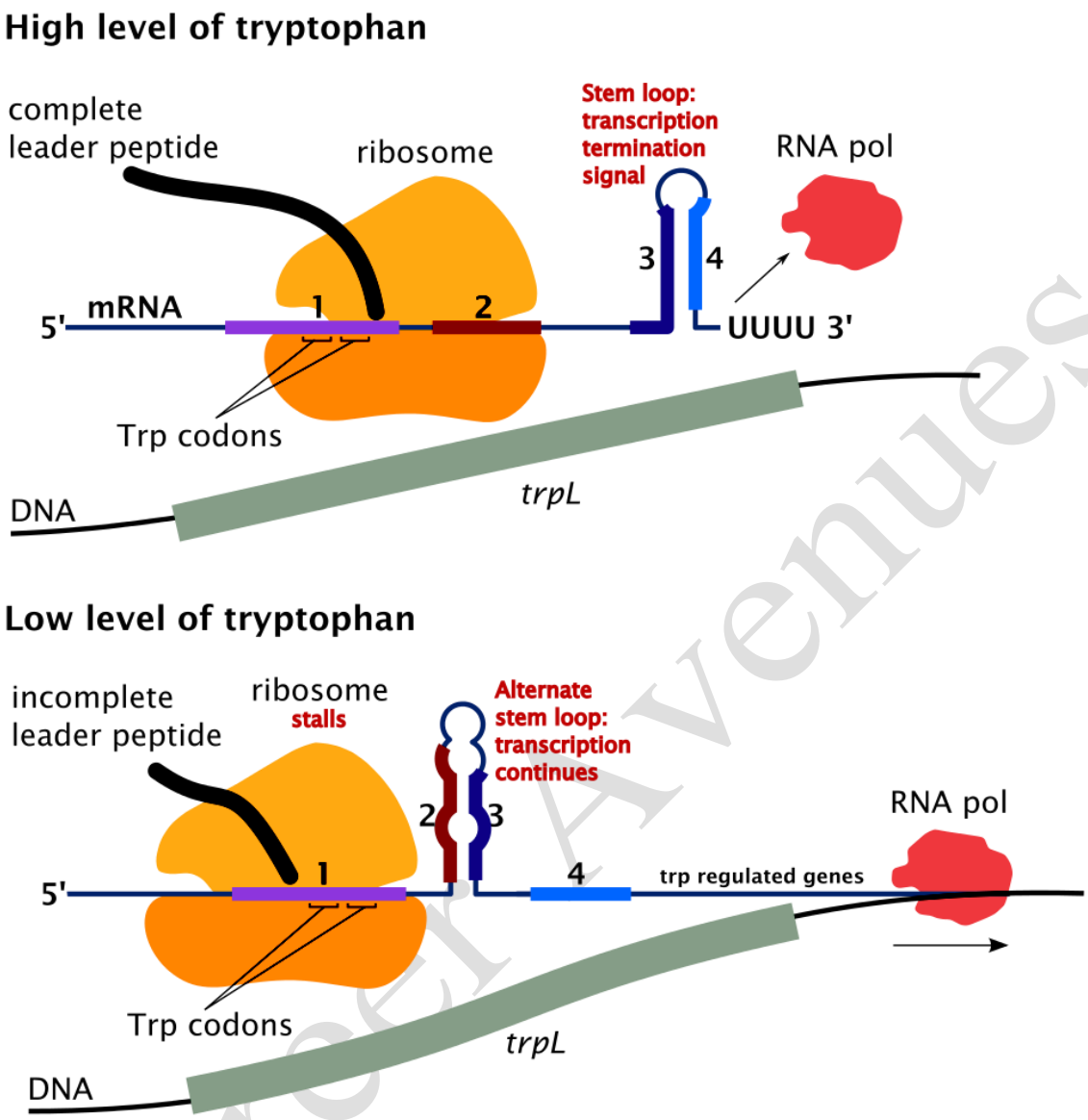
Conversely, when the tryptophan concentration in the medium and cytosol is low, tryptophan dissociates from the *trp* repressor, causing a conformational change in the protein that causes it to dissociate from the *trp* operator, allowing transcription of the *trp* operon. In the case of the *lac* operon, binding of the inducer lactose to the *lac* repressor reduces binding of the repressor to the operator, thereby promoting transcription.

Specific activator proteins, such as CAP in the lac operon, also control transcription of some but not all bacterial genes. These activators bind to DNA together with the RNA polymerase, stimulating transcription from a specific promoter. The DNA-binding activity of an activator is modulated in response to cellular needs by the binding of specific small molecules (e.g., cAMP) that alter the conformation of the activator.

TRANSCRIPTION BY 54-RNA POLYMERASE IS CONTROLLED BY ACTIVATORS THAT BIND FAR FROM THE PROMOTER

Most E. coli promoters interact with σ70-RNA polymerase, the major form of the bacterial enzyme. Transcription of certain groups of genes, however, is carried out by E. coli RNA polymerases containing one of several alternative sigma factors that recognize different consensus promoter sequences than σ70does. All but one of these is related to σ70 in sequence. Transcription initiation by RNA polymerases containing these σ70 like factors is regulated by repressors and activators that bind to DNA near the region where the polymerase binds, similar to initiation by σ70-RNA polymerase itself. The sequence of one E. coli sigma factor, σ54, is distinctly different from that of all the σ70-like factors. Transcription of genes by RNA polymerases containing σ54is regulated solely by activators whose binding sites in DNA, referred to as enhancers, generally are located 80–160 base pairs upstream from the start site. Even when enhancers are moved more than a kilobase away from a start site, σ54-activators can activate transcription.

Career Avenues



EUKARYOTIC GENE CONTROL AND RNA POLYMERASES

The primary purpose of gene control in multicellular organisms is the execution of precise developmental decisions so that the proper genes are expressed in the proper cells during development and cellular differentiation.

Transcriptional control is the primary means of regulating gene expression in eukaryotes, as it is in bacteria.

In eukaryotic genomes, DNA transcription control elements may be located many kilobases away from the promoter they regulate. Different control regions can control transcription of the same gene in different cell types.

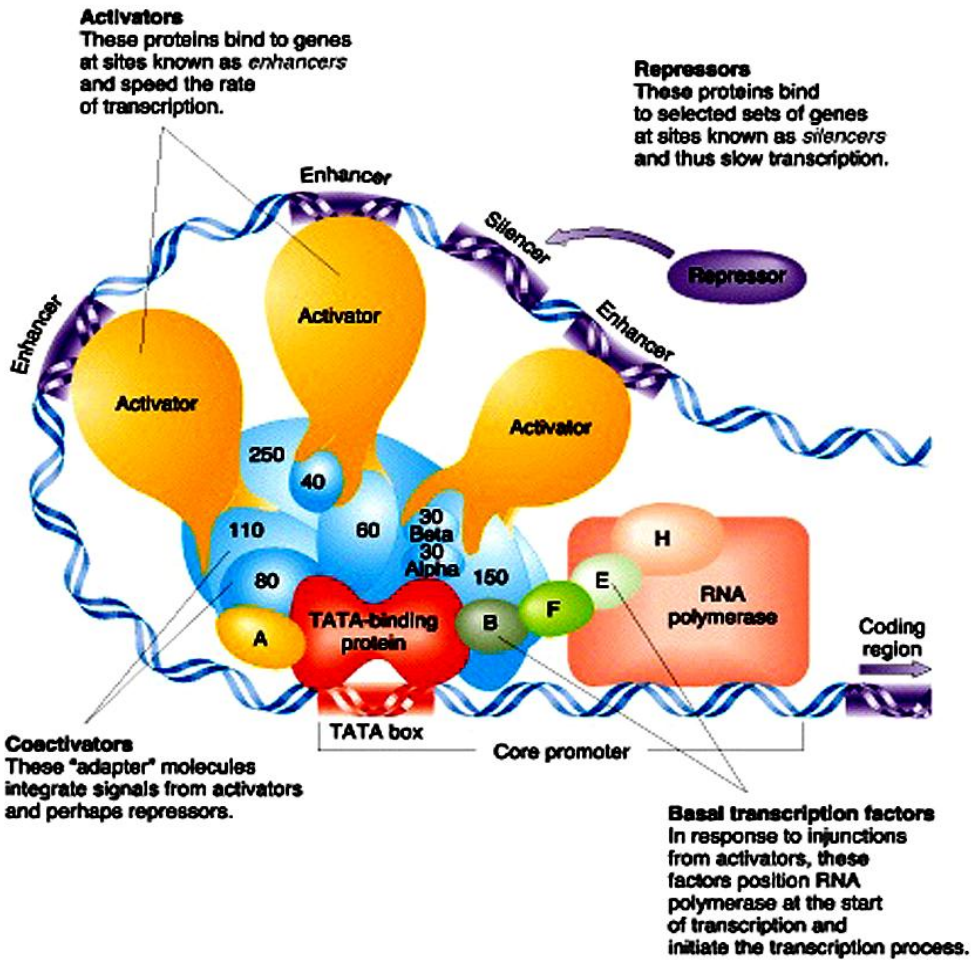
Career Avenues

Eukaryotes contain three types of nuclear RNA polymerases. All three contain two large and three smaller core subunits with homology to the β,β',α and ϖ subunits of E. coli RNA polymerase, as well several additional small subunits .

RNA polymerase I synthesizes only pre-rRNA. RNA polymerase II synthesizes mRNAs and some of the small nuclear RNAs that participate in mRNA splicing. RNA polymerase III synthesizes tRNAs, 5S rRNA, and several other relatively short, stable RNAs.

The carboxyl-terminal domain (CTD) in the largest subunit of RNA polymerase II becomes phosphorylated during transcription initiation and remains phosphorylated as the enzyme transcribes the template.

RNA polymerase II initiates transcription of genes at the nucleotide in the DNA template that corresponds to the 5' nucleotide that is capped in the encoded mRNA.



REGULATORY SEQUENCES IN PROTEIN-CODING GENES

Expression of eukaryotic protein-coding genes generally is regulated through multiple protein-binding control regions that are located close to or distant from the start site.

Promoters direct binding of RNA polymerase II to DNA, determine the site of transcription initiation, and influence transcription rate.

Three principal types of promoter sequences have been identified in eukaryotic DNA. The TATA box, the most common, is prevalent in rapidly transcribed genes. Initiator promoters are found in some genes, and CpG islands are characteristic of genes transcribed at a low rate.

Promoter-proximal elements occur within ≈200 base pairs upstream of a start site. Several such elements, containing ≈10–20 base pairs, may help regulate a particular gene.

Enhancers, which contain multiple short control elements, may be located from 200 base pairs to tens of kilobases upstream or downstream from a promoter, within an intron, or downstream from the final exon of a gene.

Promoter-proximal elements and enhancers often are cell-type-specific, functioning only in specific differentiated cell types.

ACTIVATORS AND REPRESSORS OF TRANSCRIPTION

Transcription factors, which stimulate or repress transcription, bind to promoter-proximal regulatory elements and enhancers in eukaryotic DNA.

Transcription activators and repressors are generally modular proteins containing a single DNA-binding domain and one or a few activation domains (for activators) or repression domains (for repressors). The different domains frequently are linked through flexible polypeptide regions.

Among the most common structural motifs found in the DNA-binding domains of eukaryotic transcription factors are the C2H2 zinc finger, homeodomain, basic helix-loophelix (bHLH), and basic zipper (leucine zipper). All these and many other DNA-binding motifs contain one or more α-helices that interact with major grooves in their cognate site in DNA.

The transcription-control regions of most genes contain binding sites for multiple transcription factors. Transcription of such genes varies depending on the particular repertoire of transcription factors that are expressed and activated in a particular cell at a particular time.

Combinatorial complexity in transcription control results from alternative combinations of monomers that form heterodimeric transcription factors and from cooperative binding of transcription factors to composite control sites.

Activation and repression domains in transcription factors exhibit a variety of amino acid sequences and three dimensional structures. In general, these functional domains interact with

Career Avenues

co-activators or co-repressors, which are critical to the ability of transcription factors to modulate gene expression.

Cooperative binding of multiple activators to nearby sites in an enhancer forms a multi-protein complex called an enhancesome. Assembly of enhancesomes often requires small proteins that bind to the DNA minor groove and bend the DNA sharply, allowing bound proteins on either side of the bend to interact more readily.

REGULATION OF TRANSCRIPTION-FACTOR ACTIVITY

The activities of many transcription factors are indirectly regulated by binding of extracellular proteins and peptides to cell-surface receptors. These receptors activate intracellular signal transduction pathways that regulate specific transcription factors through a variety of mechanisms.

Nuclear receptors constitute a superfamily of dimeric C4 zinc-finger transcription factors that bind lipid-soluble hormones and interact with specific response elements in DNA.

Hormone binding to nuclear receptors induces conformational changes that modify their interactions with other proteins.

Heterodimeric nuclear receptors (e.g., those for retinoids, vitamin D, and thyroid hormone) are found only in the nucleus. In the absence of hormone, they repress transcription of target genes with the corresponding response element. When bound to their ligands, they activate transcription.

Steroid hormone receptors are homodimeric nuclear receptors. In the absence of hormone, they are trapped in the cytoplasm by inhibitor proteins. When bound to their ligands, they can translocate to the nucleus and activate transcription of target genes.

REGULATED ELONGATION AND TERMINATION OF TRANSCRIPTION

Different mechanisms of transcription termination are employed by each of the eukaryotic nuclear RNA polymerases. Transcription of most protein-coding genes is not terminated until an RNA sequence is synthesized that specifies a site of cleavage and polyadenylation.

Transcription of the HIV genome by RNA polymerase II is regulated by an antitermination mechanism that requires cooperative binding by the virus-encoded Tat protein and cyclin T to the *TAR* sequence near the 5’ end of the HIV RNA.

During transcription of *Drosophila* heat-shock genes, RNA polymerase II pauses within the downstream promoterproximal region; this interruption in transcription is released when the HSTF transcription factor is activated, resulting in very rapid transcription of the heat-shock genes in response to the accumulation of denatured proteins.

Career Avenues

GATE Questions

GATE 2007

Q.9 A nonsense mutation in the gene encoding protein X leading to the synthesis of a truncated protein results in a slow growing strain. Mutagenesis of this strain towardsthe isolation of extragenic suppressors led to the isolation of a strain which grew normally and synthesized the full-length protein X. The extragenic suppressor is likely to be a gene coding for

(A) rRNA
(B) RNA polymerase
(C) tRNA
(D) Ribosomal protein

GATE 2009

Q.10 When *E.coli* is grown in glucose and lactose, the *lac* operon is not expressed. This is because glucose interferes with

(A) removal of repressor
(B) binding of activator
(C) removal of repressor and binding of activator
(D) removal of activator

GATE2010

Q.11 In tryptophan operon, tryptophan acts as

(A) Repressor (B) Activator (C) Co-repressor (D) Co-activator

Q.12 A DNA-binding motif is

(A) Helix-loop-helix
(B) Helix-turn-helix
(C) Helical wheel
(D) Loop-helix-loop

Answers			
Q.9 – C.	**Q.10 – B.**	**Q.11 – C.**	**Q.12 – B.**

Career Avenues

MENDELIAN INHERITANCE

Genetics is concerned with the transmission, expression, and evolution of genes, the molecules that control the function, development, and ultimate appearance of individuals.

CONCEPT OF HEREDITY AND VARIATION

The children or offsprings closely resemble their parents and to some extent their grandparents and great grandparents. Still the offsprings of a set of parents differ from each other and from their parents in different degrees. They have certain unique characteristics by which we can understand that they belong to the same family. The Science that deals with the mechanisms responsible for inheritance of similarities and differences in a species is called **Genetics.** It is a branch of biology that encompasses the study of the mechanism of transmission of characters from parents to offsprings. The word "genetics" is derived from the Greek word "genesis" meaning "to grow" or "to become". The Science of Genetics helps us to differentiate between heredity and variations and seeks to account for the resemblances and differences due to heredity, their source and development.

Heredity refers to the transmission of characters, resemblances as well as differences from one generation to the next. It explains how offsprings in a family resemble their parents.

Variation refers to the differences shown by individuals of the same species and also by offsprings of the same parents. It explains why offsprings even though born to the same parents differ from each other. The siblings are similar, but not identical (except in identical twins). These similarities and differences are not coincidental.

In brief, genetics is the study of heredity and variation.

Genetics: The branch of biology that deals with heredity, especially the mechanisms of hereditary transmission and the variation of inherited characteristics among similar or related organisms.

SOME COMMON TERMS

A **phenotype** is the composite of an organism's observable characteristics or traits: such as its morphology, development, biochemical or physiological properties.

- **Genotype:** it is the complete genetic constitution of an individual.
- **Alleles:** the two individual genes in a particular gene pair that occupies same locus (position) on homologous pair of chromosome.

When a gene pair (alleles) in an organism contains two identical genes controlling similar trait of a character then the organism is considered as **homozygous** for that particular character.

Career Avenues

When a gene pair (alleles) in an organism contains two different in an organism then the organism is considered as **heterozygous** for that particular character.

- **Character:** the feature or characteristic of an individual.
- **Trait:** an inherited character i.e., the feature which is normally inherited and has its variant too. Length of a plant is a character which have the variants as tall plant and dwarf plant. Here, tall and dwarf are traits.
- **Hybrid:** it is defined s an offspring which is obtained from a cross between two genetically different parents.
- **F1 generation:** first filial (= offspring) generation. i.e. the generation of hybrid which are obtained from a cross between two genetically different parents.
- **F2 generation:** second filial generation i.e. the individuals normally obtained after self crossing the individuals of F1 generation.
- **Cross:** a mating between two individual, leading to the fusion of gametes. The cross may be:
- **Monohybrid cross**: i.e. considering only one
- **Dihybrid cross:** i.e. considering two pair of contrast character.

The cross may also be:

- **Reciprocal**: i.e. the cross that could be made either way or independent of the sex of parents.
- **Back cross** i.e. the cross of hybrid of F1 generation with either of its parents.
- **Test cross** i.e. the cross of individual with its homozygous recesive parents.

Dominant and recessive traits: the trait may be dominant or recessive depending on the fact that on crossing the individuals, (which are homologous for a particular trait of same character) the trait which appears in the F1 generations called dominant and together one is called is recessive.

MENDEL'S EXPERIMENTS

Mendel conducted cross breeding experiments in the garden pea plant (*Pisum sativum*). He crossed two pea plants with contrasting character traits considering one character at a time. The resulting hybrids were crossed with each other. The data of many crosses were pooled together and the results were analysed carefully.

Reasons for Mendel's Success

A combination of luck, foresight and the aptitude of Mathematics, all contributed to the success of Mendel's experiments.

Career Avenues

Selection of Material

He chose the pea plant as it was advantageous for experimental work in many respects such as :

- It is a naturally self-fertilizing plant and so it is very easy to raise pure-breeding individuals.
- It has a short life span as it is an annual and so it is possible to follow several generations.
- It is easy to cross-pollinate the pea plant.
- It has deeply defined contrasting characters.
- The flowers are all bisexual. Therefore the pea plant proved to be an ideal experimental plant for Mendel.

Seven pairs of contrasting traits and the results of Mendel's seven monohybrid crosses of the garden pea *(Pisum sativum*).

Character	Contrasting traits		F_1 results
Seeds	round/wrinkled		all round
	yellow/green		all yellow
Pods	full/constricted		all full
	green/yellow		all green
Flower color	violet/white		all violet
Flower position	axial/terminal		all axial
Stem length	tall/dwarf		all tall

(Please note that the image is in colour but the same will not be visible as such in your book).

MONOHYBRID CROSS

(Experiments with garden pea for a single pair of contrasting characters)

P TT x tt

Tall x Dwarf

Gametes (T) (t)

F_1 Tt Monohybrid

Tall (Selfing)

Gametes (T) (t)

F_2

	T	t
T	TT	Tt
t	Tt	tt

TT - 1 Homozygous tall

Tt - 2 Heterozygous tall

tt - 1 Dwarf

Tall : Dwarf = 3:1

Mendel's Explanation of Monohybrid Cross

- **Parental Generation**: Mendel selected a pure breeding tall plant and a pure breeding dwarf plant as parents (Homozygous).
- **F1 Generation**: He crossed the parents and from the seeds obtained he raised the first filial generation. Here the plants were all tall and were called monohybrids.
- **F2 Generation**: Mendel allowed selfing of the F1 monohybrids and he obtained Tall and dwarf plants respectively in the ratio of 3:1. The actual number of tall and dwarf plants obtained by Mendel were 787 tall and 277 dwarf. The ratio of **3:1** is called **Phenotypic ratio** as it is based on external appearance of offsprings.
- **F3 Generation**: By selfing the F2 offsprings Mendel obtained the F3 generation.

He found that,

1. The F2 dwarf plants always bred true generation after generation whether self or cross pollinated.
2. Of the F2 tall plants one third bred true for tallness. The rest two thirds produced tall and dwarf in the ratio of 3:1. This meant that the F2 generation consisted of 3 types of plants.

   i. Tall homozygous (pure) - 25%
   ii. Tall heterozygous - 50%
   iii. Dwarf homozygous (pure) - 25%

Career Avenues

Thus based on the constitution of factors the ratio of a Monohybrid cross is **1:2:1** which is called the **genotypic ratio**.

MENDEL'S INTERPRETATION AND EXPLANATION

During Mendel's time structure of chromosomes or the role of meiosis was not known. So he concluded that the inheritance of characters is by particles called **hereditary units** or factors.

He explained the results of Monohybrid cross by making certain presumptions.

i. Tallness and dwarfness are determined by a pair of contrasting factors (now called as genes). A tall plant possesses a pair of determiners (represented by **T**-taking the first letter of the dominant character) and a plant is dwarf because it possesses determiners for dwarfness represented as **t**). These determiners occur in pairs and may be alike as in pure breeding tall parents (**TT**) and dwarf parents (**tt**). This is referred to as **homozygous**. They may be unlike as in the monohybrid (**Tt**) which is referred to as **heterozygous**.

ii. The two factors making up a pair of contrasting characters are called **alleles** or **allelomorphs**. One member of each pair is contributed by one parent.

iii. When two factors for alternative expression of a trait are brought together by fertilization only one expresses itself, (tallness) masking the expression of the other (dwarfness). The character which expresses itself is called **dominant** and that which is masked is called the **recessive** character.

iv. The factors are always pure and when gametes are formed, the unit factors segregate so that each gamete gets one of the two alternative factors. It means that factors for tallness (T) and dwarfness (t) are separate entities and in a gamete either T or t is present. When F1 hybrids are selfed the two entities separate and then unite independently forming tall and dwarf plants.

DIHYBRID CROSS

(Cross involving two pairs of contrasting characters)

Mendel also experimentally studied the segregation and transmission of two pairs of contrasting characters at a time. This was called the **Dihybrid cross** or **Two-factor cross**. He took up the round and wrinkled characters of seed coat along with yellow and green colour of seeds.

Mendel found that a cross between round, yellow and wrinkled green seeds (P) produced only round yellow seeds in the F1 generation, but in the F2 generation 4 types of combinations appeared of which two were different from that of the parental combinations, in the following ratio.

These are :

Round Yellow	-	9	Parental
Round Green	-	3	New combination
Wrinkled Yellow	-	3	
Wrinkled Green	-	1	Parental

Career Avenues

Therefore the offsprings of the F2 generation in a dihybrid cross were produced is a ratio of **9:3:3:1**. This ratio is called the **dihybrid ratio**.

In F2 generation, since all the four characters assorted out independent of the others, he said that a pair of contrasting characters behaves independently of the other pair; i.e. seed colour is independent of seed coat. At the time of gamete formation in the F1 dihybrid, genes for round or wrinkled character of seed coat assorted independently of the yellow or green colour of seed coat. As a result 4 types of gametes with two old and two combinations were formed namely RY, Ry, rY and ry. These 4 types of gametes on random mating produced 16 offsprings in the ratio of 9:3:3:1. The actual number of individuals got by Mendel for each of the four classes were

- 315 round yellow seeds
- 108 round green seeds
- 101 wrinkled yellow seeds
- 32 wrinkled green seeds

Mendel represented round character of seed as **R** and wrinkled as **r**, and yellow character as **Y** and green character as **y**. So the dihybrid cross was between parents having **factor constitution** as **RRYY** x **rryy**. This cross may be represented as follows:

Round Yellow Wrinkled Green

P RRYY x rryy

Gametes (RY) (ry)

F_1 RrYy (Dihybrid)

Round Yellow (selfed)

Gametes (RY) (Ry) (rY) (ry)

F_2

	RY	Ry	rY	ry
RY	RRYY Round yellow	RRYy Round Yellow	RrYY Round Yellow	RrYy Round Yellow
Ry	RRYy Round Yellow	RRyy Round Green	RrYy Round Yellow	Rryy Round Green
rY	RyYY Round Yellow	RrYy Round Yellow	rrYY Wrinkled Yellow	rrYy Wrinkled Yellow
ry	RrYy Round Yellow	Rryy Round Green	rrYy Wrinkled Yellow	rryy Wrinkled Green

Career Avenues

LAWS OF MENDEL

Based on his experiments of monohybrid and dihybrid cross, Mendel proposed three important laws which are now called Mendel's Laws of Heredity:

- Law of dominance and recessiveness
- Law of segregation or Law of purity of gametes
- Law of independent assortment

I. LAW OF DOMINANCE

The law of dominance and recessiveness states: "When two homozygous individuals with one or more sets of contrasting characters are crossed, the characters that appear in the F1 hybrid are **dominant** and those that do not appear in F1 are **recessive** characters".

II. LAW OF SEGREGATION OR LAW OF PURITY OF GAMETES

The Law of segregation states that "When a pair of contrasting factors or genes or allelomorphs are brought together in a heterozygote or hybrid, the two members of the allelic pair remain together without mixing and when gametes are formed the two separate out, so that only one enters each gamete". This law though it was a conception originally and propagated by Mendel, now it has been confirmed by cytological studies. Dominance or no dominance segregation holds good for all cases.

III. LAW OF INDEPENDENT ASSORTMENT

Law of independent assortment states: "In case of inheritance of two or more pairs of characters simultaneously, the factors or genes of one pair assort out independently of the other pairs". Mendel gave this law based on his dihybrid cross experiment. Here the total number of individuals in F2 will be sixteen which occur in a ratio of 9:3:3:1 where two parental classes and two new combinations will be produced.

BACK CROSS AND TEST CROSS

In Mendelian inheritance F2 offsprings are obtained by selfing the hybrid, but if the F1 hybrid is crossed to any of the pure breeding parents it is called a **back cross**. If the hybrid is crossed to the dominant parent, all the F2 offsprings will show dominant character. If the hybrid is crossed to recessive parent, dominant and recessive phenotypes with appear in equal proportions as shown, which is called a **test cross**.

Monohybrid Back Cross

F_1 Tt x TT

(Back cross)

Gametes (T)(t) (T)

F_2

	T	t
T	TT	Tt

All are tall

Career Avenues

Monohybrid Test Cross

F_1 Tt x tt (Test cross)

Gametes (T)(t) (t)

F_2

	T	t
t	Tt	tt

Tall : Dwarf = 1: 1

DIHYBRID TEST CROSS

In a dihybrid test cross the four types of phenotypes are obtained in equal proportions as shown. The test cross is used to determine whether segregation of alleles has taken place and to ascertain if hybrid is homozygous or heterozygous.

F_1 RrYy x rryy

dihybird recessive parent

Gametes (RY) (Ry) (rY) (ry) (ry)

F_2

	RY	Ry	rY	ry
ry	RrYy Round Yellow	Rryy Round Green	rrYy Wrinkled Yellow	rryy Wrinkled Green

RrYy Round Yellow : Rryy Round Green : rrYy Wrinkled Yellow : rryy Wrinkled Green

1 : 1 : 1 : 1

CHROMOSOMAL BASIS OF INHERITANCE

The **factors** of Mendel were called **genes** by Johansen 1909, who did not know their exact nature and structure.

GENE CONCEPT

Sutton introduced the gene concept which was elaborated by the studies of **Morgan**, **Bridges** and **Miller**.

The important features of the gene concept are:

i. Genes are transmitted from parents to offsprings and are responsible for the physical and physiological characteristics of the organism which are present inside the nucleus of the cell.

ii. The genes are present on the chromosome.

Career Avenues

iii. Since the number of genes far exceeds the number of chromosomes, several genes are located on each chromosome. In man about 40,000 genes are present in 23 pairs of chromosomes.

iv. The genes are present at a specific position on the chromosome called **locus**.

v. Genes are arranged on the chromosomes in a linear order like beads on a string.

vi. A single gene may have more than one functional state or form. These functional states are refered to as alleles.

vii. The alleles may be dominant or recessive but sometimes co-dominance or incomplete dominance may be seen.

ix. Genes may undergo sudden heritable changes called mutations, induced by chemical and physical factors.

x. Due to mutation a gene may come to possess more than two alternative states and these states of the gene are called multiple alleles.

xi. Genes undergo duplication by a phenomenon called replication.

xii. Genes are responsible for the production of proteins called enzymes by which they show their expression which brings about a change in the organism.

xiii. A gene is a particular DNA segment which contains the information to synthesize one polypeptide chain or one enzyme. The information is contained as a sequence of nucleotides which is called **genetic code**. The sequence of three nucleotides that code for an aminoacid is called **Codon**.

MOLECULAR STRUCTURE OF A GENE

A gene, is made of DNA. The gene may be subdivided into different units according to Benzer such as Recon, Muton, Cistron and Operon.

Recon

It is that smallest portion of a gene which can undergo crossing over and recombination and may be as small as a single nuclecotide pair.

Muton

It is the smallest unit of a gene that can undergo mutation and can involve a pair of nucleotides.

Cistron

It is the functional unit which can synthesize one polypeptide.

Operon

It is a group of genes having an operator a **structural gene** and other genes in sequence which all function as a unit.

Career Avenues

Exons and Introns

In Prokaryotes generally, the genes are continuous segments of DNA occurring collinearly without interruption. But in Eukaryotes, the genes on the DNA strand have coding regions called **exons** interrupted by non-coding DNA segments which do not carry genetic information called **introns**. This led to the concept of **interrupted genes** or **discontinuous genes**. Such genes while producing m-RNA will first form a primary transcript which will then cut off the introns to form the functional m-RNA and this is called **splicing**.

CHROMOSOMAL BASIS OF INHERITANCE

The chromosome basis of inheritance was put forth by **Sutton and Boveri independently** in the year **1902**. W.S. Sutton and Theodor Boveri faced and solved the problem of drawing a parallel between chromosomes and genes. Both had concluded that the genes are contained in chromosomes. Allelic genes present in a heterozygote segregate independently because the chromosomes carrying these genes segregate when the sex cells are formed. This conclusion of Sutton and Boveri was verified extensively by further studies conducted by various geneticists and cytologists.

In order to accept this conclusion we must be able to understand the behaviour of chromosomes in the light of Mendel's assumption.

- **Individuality of Chromosomes:** Every organisms has a fixed number of chromosomes. The nuclei of gametes contain haploid (n) and those of zygotes have double the number or diploid (2n) number of chromosomes.
- **Meiosis:** At the time of meiosis, for the formation of gametes, the pairs of chromosomes of the diploid sets undergo pairing.
- The chromosomes of each pair segregate independently of every other pair during their distribution into gametes. This is similar to Mendel's law of independent assortment in the segregation of factors.
- During the fusion of haploid gametes, the homologous chromosomes from two parents are brought together to form the diploid zygote. Accordingly Mendel had maintained that maternal and paternal characters mix up in the progeny.
- The chromosomes maintain the structure and uniqueness during the life time of the individual whether observable or not. Mendel had also demonstrated that the characters are never lost though they are not expressed in a particular generation.

From these points it is evident that a clear parallelism exists between Mendel's factors and chromosomes and so there is a firm basis for Mendel's Laws of heredity in the behaviour of chromosomes during meiosis and fertilisation and therefore the **Chromosomal Theory of Inheritance** has been proposed.

Career Avenues

POSTULATES OF THE CHROMOSOMAL THEORY OF INHERITANCE

- The factors described by Mendel are the genes which are the actual physical units of heredity.
- The genes are present on chromosomes in a linear order.
- Each organism has a fixed number of chromosomes which occur in two sets referred to as diploid (2n). A pair of similar chromosomes constitutes the homologous pair.
- Of this, one set is received from the male parent (paternal) and the other from the female parent (maternal).
- The maternal and paternal chromosomes are contributed by the egg and sperm respectively during zygote formation. But only sperm nucleus is involved proving that chromosomes are present within the nucleus.
- The chromosomes and therefore the genes segregate and assort independently at the time of gamete formation as explained in Mendel's law of segregation and Law of Independent Assortment.

PHYSICAL AND CHEMICAL BASIS OF HEREDITY

PHYSICAL BASIS

Gregor Johann Mendel put forward in 1866 that particles called **germinal units** or **factors** controlled heredity. These were present in both the somatic cells as well as the germinal cells. Though he was not able to actually see these particles, he did explain the pattern of inheritance of genetic characters. It was the gamete which carried these factors to the next generation and so gametes form the **physical basis of heredity.**

CHEMICAL BASIS

Now it is known that genes control heredity and these are definite segments of chromosomes and so are particulate bodies. The genes travel from one generation to the next carrying the traits and since gene is composed of DNA and protein, the DNA part functions as the **chemical basis of heredity**.

Career Avenues

GENE INTERACTION

WHAT IS GENE INTERACTION?

With the help of lot of experiments it was found that most of the characters of living organisms are controlled / influenced / governed by a collaboration of several different genes.

This condition where a single character is governed by two or more genes and every gene affect the expression of other genes involved (means these genes affect each other's expression) is known as gene interaction.

In simple way we could say that, in gene interaction, expression of one gene depends on expression (presence or absence) of another gene.

As we know, gene interactions may involve two or more pairs of genes. But all the gene interactions we have described below have the two pairs of non-allelic genes, affecting the phenotypic expression of same character. These interactions produce modified dihybrid ratios.

The genes interacting to affect a single trait, if present on different chromosomes will show independent assortment and no interference between the effects of different genes. The condition where one pair of genes reverses or inhibits the effect of another pair of genes by causing the modification of the normal phenotype is called gene interaction.

TYPES OF GENE INTERACTIONS

Gene interactions can be classified as

- Non-allelic gene interaction
- Allelic gene interaction

Non-allelic Gene Interaction

Expression of character is produced by interaction between two or more genes. These interactions occur between alleles of different genes present either on the same or different chromosome and alter the normal phenotype. Complementary gene interaction, supplementary gene interaction, duplicate factors and inhibitory factors are examples of intergenic interactions.

The interactions we have listed below, as inter and intra allelic are of this type.

- Inter-allelic
- Intra-allelic

Inter-allelic

- Without modification of normal F2 ratio
- With modification of normal F2 ratio

Career Avenues

Such kinds of interactions modify the normal F_2 ratio (9:3:3:1). Various types of such interactions are as below.

1. WITHOUT MODIFICATION OF NORMAL F_2 RATIO

Gene interaction without modification of normal F_2 ratio is explained below.

In the early part of the 20th century, Bateson and Punnett studied this type of gene interaction in case of comb type of poultry.

Comb types and breed used are as follows:

- Pea comb (brahmas breed)
- Rose comb (Wyandotte breed)
- Walnut comb
- Single comb (leghorn breed)

Cross between	F_1	F_2
Pea X Single	Pea	3 pea : 1 single
Rose X Single	Rose	3 rose : 2 single
Rose X Pea	Walnut	9 walnut : 3 rose : 3 pea : 1 single

From the results of these crosses, it can be observed that, pea type and rose type are monogenic dominant over single.

F_1 generation (F_2) of walnut type showed segregation, its behavior is based on ratio of dihybrid cross. This suggests that, walnut comb type must be governed by the pair of genes.

EXPLANATION

Suppose, gene P produces pea comb and gene R produces rose comb. Genes P and R when together give rise to walnut comb and combination of both in recessive form (p and r) produce single comb.

Intra-allelic

Lethal Gene: Genes which result in viability reduction of individual or become a cause for death of individuals carrying them are called as lethal genes. Phenomenon of action of lethal genes is called as lethality.

Some lethal genes cause death of zygote or the early embryonic stage while some express their effect in later stages of development.

Individuals carrying dominant lethal will die. Even though the dominant lethal is eliminated from the genotype of population previously carrying it, the recessive lethals are still carried in

Career Avenues

heterozygous condition. The recessive lethal in heterozygous condition reduce the viability and when they occur in homozygous condition produce lethal effect.

Usually lethal genes are dominant with respect to the phenotype they control and recessive for their lethal actions.

Example of Lethal Genes

In case of maize, albinism is example of lethal factor.

The lethal factor in heterozygous condition has no lethal effect but in homozygous condition it makes plant to die. Lethal factor in homozygous condition produces albino plant, which is unable to synthesize food due to lack of chlorophyll. Thus, the lethal factor modifies normal ratio from 3:1 to 2:1.

Allelic Gene Interaction

Expression of character is produced by interaction between alleles of a single gene. This kind of interaction occurs between alleles of the same gene pair as in the case of incomplete dominance, co dominance and multiple allelism.

- Complete dominance
- Incomplete dominance
- Co-dominance
- Over Dominance

WHAT IS DOMINANCE?

In the expression of characters genes play major role. Generally plants have two copies of each gene in their genome, one inherited from each parent.

The alternative forms of gene are known as allele. If phenotype of the organism is determined by one of the alleles by completely hiding the expression of an alternate allele, then that allele is said to be dominant.

The phenomenon is called as Dominance. The other allele is recessive; its expression is suppressed in the presence of a dominant allele of organism.

MECHANISM OF DOMINANCE

According to Mendel's beliefs, certain characters are produced in progeny. The character cannot be present as such in gametes; the only link between progeny and the parent. Therefore there

Career Avenues

must be something that represents the character and is regulates its appearance. Mendel named this something as “Factor” (today’s gene).

The gene controls expression of character by the interaction with the other genes, the cytoplasm and the environment. A diploid cell contains two sets of chromosomes which come from two opposite sexes via gametes. Homologous chromosomes pair during meiotic cell division. Each character of a pair of contrasting character is represented by allele.

Thus, homozygous tall pea plant has two identical alleles TT and homozygous dwarf plant has tt alleles on both gene loci, of the homologous chromosome. During gamete formation, the homologous chromosomes with TT or tt genes get separated and each chromosome with T or t gene is passed to gamete. The gametes from both the parents unite during fertilization and produce a new individual (heterozygote or hybrid) containing allele for both tall (T) and dwarf (t) characters.

Because the hybrids of F_1 have tall stems, the character of tallness is considered as dominant and dwarf character as recessive as it failed to express itself in F_1 generation.

TYPES OF DOMINANCE OR VARIATIONS IN DOMINANCE RELATION

Other than Simple or complete dominance other types of dominance are as follows. They are also considered as variations in dominance relation.

- Incomplete or Semidominance or Partial Dominance
- Codominance
- Over Dominance
- Dominant Negative

INCOMPLETE OR SEMIDOMINANCE OR PARTIAL DOMINANCE:

From Mendel's experiments it had been established that when two alleles are brought together from two different pure breeding parents, one of them completely dominates over the other manifesting itself in the hybrid. Researches by many investigators revealed that in a number of living organisms complete dominance was absent, the hybrid exhibited an intermediate character as both the genes of the allelomorphic pair showed partial expression.

Thus in **incomplete dominance** or **partial dominance** or **intermediate inheritance** or **blending inheritance** the F1 hybrid does not resemble either of the parents. A very good example for this is the 4 `O' clock plant *Mirabilis jalapa* studied by Correns in 1906. A similar condition is seen in *Antirrhinum majus*.

In *Mirabilis jalapa*, there are two distinctive types of flower colours namely the red and the white. Both the types are true breeding. When a pure-red flowered (r1r1) variety is crossed with a pure white flowered (r2r2) one, the F1 hybrids produce pink flower, a character which is

Career Avenues

intermediate between red and white coloured flowers of the parental generation. This is because neither red flower colour nor white is completely dominant over the other. When F1 hybrids were selfed red, pink and white flowered varieties were obtained respectively in the ratio of 1:2:1.

This is the phenotypic ratio. Here the genotypic ratio is also 1:2:1, producing one homozygous red, two heterozygous pink and one homozygous white. The red and the white varieties breed true on self-fertilisation of the F2 individuals but the pink varieties on selfing once again produce a phenotypic ratio of 1:2:1 proving the law of purity of gametes. Since neither of the parents is completely dominant over the other, the symbol for Red parent is r1r1 and the white parent r2r2, and so naturally the genotype of the hybrid is r1r2.

It has been observed in the given example that there is blending of phenotypes - not genotypes and the alleles of the genes are discrete or particulate. They appear blended in F1 but have separated out in F2 generation. Incomplete dominance is also called blending inheritance because both the characters of the parental plants are mixed to give an intermediate character which is different from that of the parents. But only the characters are mixed with each other and not the alleles. In *Mirabilis* the r1r1 always produces red coloured flowers and r2r2 produces white coloured flowers, when they are combined, the intermediate colour namely pink is produced. Because of this it is described as blending inheritance.

	Red		White
P	r_1r_1	x	r_2r_2
Gametes	(r_1)		(r_2)
$\mathbf{F_1}$		r_1r_2 (Selfed)	
		Pink	
Gametes		(r_1) (r_2)	

$\mathbf{F_2}$

	r_1	r_2
r_1	r_1r_1	r_1r_2
r_2	r_1r_2	r_2r_2

Red : Pink : White

1 : 2 : 1

r_1r_1 : r_1r_2 : r_2r_2

CODOMINANCE

Sometimes both alleles of a gene in a heterozygote lack the dominant recessive relationship, both alleles contribute to the phenotype of the heterozygote.

In codominance, heterozygote genotype gives rise to a phenotype distinctly different from either of the homozygous genotype.

Career Avenues

Example – Coat color in the short horn breed of cattle or in horses. In a cross of red coat cattle with white coat cattle, progeny of roan coat is obtained. In roan coat, the red and white hairs occur in definite patches but no hair has intermediate color of red and white.

OVER DOMINANCE

Overdominance is a condition in genetics where the phenotype of the heterozygote lies outside of the phenotypical range of both homozygote parents. Overdominance can also be described as heterozygote advantage, wherein heterozygous individuals have a higher fitness than homozygous individuals.

An example in humans is sickle cell anemia. This condition is determined by a single polymorphism. Possessors of the deleterious allele have lower life expectancy, with homozygotes rarely reaching 50 years of age. However, this allele also yields some resistance to malaria. Thus in regions where malaria exerts or has exerted a strong selective pressure, sickle cell anemia has been selected for its conferred partial resistance to the disease. While homozygotes will have either no protection from malaria or a dramatic propensity to sickle cell anemia, heterozygotes enjoy a partial resistance to both.

DOMINANT NEGATIVE

Some loss-of-function mutations are dominant. These are referred as dominant negative.

EPISTASIS

The pioneer work of Gregor Johann Mendel seemed to imply that every character was determined by a single factor or determiner or in other words a pair of genes influenced one trait. Later work by geneticists led to the idea that a character need not necessarily be due to the action of a single factor but may also be due to the action of several factors. These hereditary units or factors are now known as genes.

There are two pairs of independent non-allelic genes affecting a single trait. The suppression of the gene on one locus of a chromosome by the gene present at some other locus is called **epistasis** meaning "standing over". The gene which is suppressed is called hypostatic and the other is the epistatic or inhibiting gene which is also called the suppressing gene.

Epistasis can be of the following types

- Due to recessive gene: Recessive gene **A** masks the effect of dominant gene **B**.
- Due to dominant gene: Dominant gene A masks the effect of the dominant gene B. Apart from this, the term epistasis refers to all non-allelic interactions involving a pair of genes. Therefore epistasis may be responsible for the production of several modified dihybrid ratios as follows:
- Duplicate recessive epistasis (9:7)
- Dominant epistasis (12:3:1)
- Recessive epistasis (9:3:4)

Career Avenues

- Dominant recessive epistasis (13:3)
- Duplicate dominant epistasis (15:1)

DIFFERENCES BETWEEN EPISTASIS AND DOMINANCE

Epistasis	Dominance
i. This type of gene interaction involves two non-allelic pairs of genes.	Only one pair of genes is involved, therefore there is no interaction.
ii. One pair of genes masks the effect of another pair of genes	An allele masks the effect of another allele of the same gene pair
iii. Expression of both the dominant and recessive alleles may be suppressed by the epistatic gene	Expression of a recessive allele is masked by the dominant allele
iv. Number of phenotypes in the F_2 generation are reduced	There is no reduction in the number of phenotypes of F_2 generation

Career Avenues

LINKAGE, RECOMBINATION AND CHROMOSOME MAPPING

LINKAGE AND CROSSING-OVER

According to Mendel's principle of independent assortment, a dihybrid cross with unlinked markers ought to produce a 1:1:1:1 ratio. If a significant deviation from this ratio occurs, it may be evidence that for **linkage**, that is, that the loci are located close to each other on the same chromosome pair.

During meiosis, a pair of synapsed chromosomes is made up of four chromatids, called a **tetrad**. The phenomenon of a **cross-over** occurs when homologous chromatids in the tetrad (one from each of the two parents) exchange segments of varying length during prophase. The point of crossover is known as a **chiasma** (pl. **chiasmata**). A tetrad typically has at least one chiasma along its length. Generally, the longer the chromosome, the greater the number of chiasmata. There are two theories on the physical nature of the process. The **classical theory** proposes that cross-over and formation of the chiasma occur first, followed by breakage and reunion with the reciprocal homologues. According to this theory, chiasma formation need not be accompanied by chromosome breakage. Alternatively, according to the **chiasmatype theory**, breakage occurs first, and the broken strands then reunite. Chiasmata are thus evidence, but not the causes, of a cross-overs. Recent molecular evidence favours the latter theory, although neither is a completely satisfactory explanation of all of the evidence.

Career Avenues

Gametes produced by a dihybrid heterozygous individual with linked loci

Condition	Meiotic tetrad	Gametes	Combination
No crossover	$a^+ \ b^+$ $a^+ \ b^+$ a b a b	$a^+ \ b^+$ $a^+ \ b^+$ a b a b	**Parental** **P** **P** **P**
Single crossover: markers in *cis* **$a^+b^+//ab$**	$a^+ \ b^+$ $a^+ \ b^+$ a ↕ b a b	$a^+ \ b^+$ a^+ b a b^+ a b	**P** **Recombinant** **Recombinant** **P**
Single crossover: markers in *trans* **$a^+b//ab^+$**	a^+ b a^+ b a ↕ b^+ a b^+	a^+ b a^+ b^+ a b a b^+	**P** **Recombinant** **Recombinant** **P**

In dihybrid crosses, an arrangement in which the wild-type alleles of both loci are contributed by one parent is referred to as a ***cis* configuration**; the alternative arrangement is called a ***trans* configuration**. A gamete that shows the same configuration as the parent is referred to as a **parental** type; where the configuration is altered, the gamete is referred to as a **recombinant** type. *Cis* and *trans* configurations are altered by recombination. Linkage between loci is indicated when the recombinant phenotypes occur less frequently than the parental types. The frequency of crossing over (% recombination) between two loci is directly related to the physical distance between those two loci. Percent recombination in a test cross equals **map distance** (1 map unit = 1 % recombination).

eg. P_1 **$a^+b//a^+b$** x **$ab^+//ab^+$**

F_1 **$a^+b//ab^+$** x **ab//ab** (test cross)

$F_{2:}$ **$a^+b//ab$, $ab^+//ab$** 90% - parental combinations

$a^+b^+//ab$, ab//ab 10% - recombinant

10% recombinant indicates that loci **a** and **b** are 10 map units apart.

Gene Map **a** ——— **b**

10 m. u.

Conversely, if it is known that loci **b** and **c** are 16 map units apart, then the expected proportions of parental and recombinant phenotypes in a test cross can be predicted:

eg. P_1 **$b^+c//b^+c$** x **$bc^+//bc^+$**

F_1 **$b^+c//bc^+$** ♀ x **bc//bc** ♂ (test cross)

In a test cross

♀ / ♂	8 % **b^+c^+**	8% **bc**	42% **b^+c**	42% **bc^+**
100 % **bc**				

In a F_1 x F_1 cross

♀ / ♂	8% **b^+c^+**	8% **bc**	42% **b^+c**	42% **bc^+**
50% **b^+c**				
50% **bc^+**				

MULTIPLE CROSSOVERS

Analysis of the genetic behaviour of three or more linked loci may show evidence of multiple cross-overs. When three loci are involved, there will be two parental types, four recombinant classes with single cross-overs and two recombinant types showing cross-overs between all three loci (= double cross-over). The two parental types will be most abundant, the four single cross-over (SCO) recombinants will be next while the two double cross-over (DCO) recombinants will be least abundant.

Each cross-over situation results in two parental gametes and two recombinant gametes. The maximum recombination between any two loci is 50% (since ½ of the gametes are parental type).

Career Avenues

Gametes produced by trihybrid heterozygous individual. **a^+ b^+ c^+ / a b c**

	Meiotic tetrad	Gametes	Combination
Single Crossover **(a-b)**	a^+ b^+ c^+ a^+ b^+ c^+ a ↕ b c a b c	a^+ b^+ c^+ a^+ b c a b^+ c^+ a b c	**P** **R** **R** **P**
Single Crossover **(b-c)**	a^+ b^+ c^+ a^+ b^+ c^+ a b ↕ c a b c	a^+ b^+ c^+ a^+ b^+ c a b c^+ a b c	**P** **R** **R** **P**
Double Crossover	a^+ b^+ c^+ a^+ b^+ c^+ a ↕ b ↕ c a b c	a^+ b^+ c^+ a^+ b c^+ a b^+ c a b c	**P** **R** **R** **P**

MAKING A GENE MAP

The recombination frequency is constant for any pair of linked loci and represents the "genetic" distance between them. Each 1 m.u. is the distance that will generate 1% recombination.

It is possible to develop a gene map, showing the order of the loci and the distance between them by observing the number of offspring showing recombinant phenotypes.

Example 1: A standard problem in genetics is to determine the order of three loci known to be linked on one pair of the autosomes. Solution of the problem requires (1) a determination of the relative order of loci, and (2) the map distances between loci.

A cross is made between homozygous wild-type female *Drosophila* (**$a^+a^+b^+b^+$ c^+c^+**) and triple-mutant males (**aa bb cc**) (the order here is arbitrary). The F_1 (**a^+a b^+b c^+c**) females are test crossed back to the triple-mutant males and the F_2 phenotypic ratios are as follows:

"a^+ b c"	18
"a b^+ c"	112
"a b c"	308
"a^+ b^+ c"	66
"a b c^+"	59
"a^+ b^+ c^+"	321
"a^+ b c^+"	102
"a b^+ c^+"	15
	1000

Career Avenues

1. The gene order can be determined by examination of the relative frequencies of the F2 phenotypes.

   a. Because linked loci tend to stay together, the non-crossover (NCO) or parental phenotypes should be most frequent (and equal in number). In this case

   $$\mathbf{a^+b^+c^+}\ (321)\ \text{ and }\ \mathbf{a\ b\ c}\ (308)$$

   b. Because simultaneous crossovers between the outside and middle loci are unlikely, the double-crossover (DCO) genotypes should be the least frequent. We observe

   $$\mathbf{a^+\ b\ c}\ (18)\ \text{ and }\ \mathbf{a\ b^+c^+}\ (15)$$

   c. Then, to determined the physical order of loci, compare the parental and double crossover phenotypes. ***The marker that appears to "switch places" is in the middle*** [technically, this marker is said to be "out of phase"]. Here, the $\mathbf{a^+b^+c^+}$ NCO and $\mathbf{a\ b^+c^+}$ DCO phenotypes indicate that the **a** locus falls between the **b** and **c** loci. The correct order of the loci is **b a c**. [Note that this order is equivalent to **c a b** , and that the order of the outside markers is arbitrary].

   d. The coupling phase of the trihybrid F1 is $b^+a^+c^+$ / b a c.

2. The remaining two pairs of phenotypes correspond to single-crossovers (SCO) events in the region between either **b** and **a**, or between **a** and **c**.

   i. $\mathbf{b^+a\ c}$ (112) and $\mathbf{b\ a^+c^+}$ (102) phenotypes indicate crossovers between **b** & **a.**
   ii. $\mathbf{b^+a^+c}$ (66) and $\mathbf{b\ a\ c^+}$ (59) phenotypes indicate crossovers between **c** & **a**.

3. The percent recombination between two markers indicates the map distance between them: 1% recombination = 1 map unit (m.u.). To determine the map distance between a pair of loci, ***count the number of SCO and DCO events***, and use the following formula [the most common error is to neglect the DCO classes].

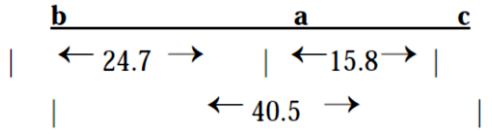
$$\text{Map distance} = \%\ \text{recombination} = \frac{(\#\ \text{in SCO phenotypes} + \#\ \text{in DCO phenotypes} \times 100)}{(\text{total \# progeny})}$$

($\mathbf{b \leftrightarrow a}$) Map distance $= \frac{112 + 102 + 18 + 15}{1000} \times 100 = 24.7\% = 24.7$ m.u.

($\mathbf{a \leftrightarrow c}$) Map distance $= \frac{66 + 59 + 18 + 15}{1000} \times 100 = 15.8\% = 15.8$ m.u.

($\mathbf{b \leftrightarrow c}$) Map distance = 24.7 m.u. + 15.8 m.u. = 40.5 m.u.

4. We can now draw a map segment showing order and distances among loci. Again, note that the orders **b-a-c** and **c-a-b** are equivalent and that the left/right the orientation of this map is arbitrary]



USING A GENE MAP

The gene map can be used as a table of probabilities to predict the expected amount of recombination between certain loci. In a test cross the male contributes only recessive alleles. Recombination occurs in the formation of the female gametes. Therefore whatever alleles present in the female gamete will be expressed in the phenotype of the offspring.

There is a certain probability that a cross-over will form between **a** and **b** loci (= map distance between **a** and **b**) and another independent probability that a cross-over will occur between **b** and **c** loci (= map distance between **b** and **c**). The probability of a double cross-over is the product of these two independent probabilities.

Example 2: Given the map segment **cn** **vg** **sm**

|← 9.5 →| ← 24.5 → |

In a test cross of **cn$^+$vg$^+$sm$^+$ // cn vg sm**
Expected DCO = (% recomb. **cn-vg**) (% recomb. **vg-sm**)
= 0.095 x 0.245 = 2.3%

Therefore we expect to find 2.3% of the female gametes to be the results of double crossovers

1.15% **cn$^+$vg sm$^+$**
1.15% **cn vg$^+$ sm**

Expected SCO (cn-vg)

From the gene map 9.5% of the gametes would be expected to have crossovers between **cn** and **vg**, however this includes the 2.3% of double crossovers. Therefore 9.5 -2.3 = 7.2% of the female gametes should have single crossovers:

3.6% **cn vg$^+$ sm$^+$** &. 3.6% **cn$^+$ vg sm**

Expected SCO (vg-sm)

From the gene map 24.5% of the gametes would be expected to have crossovers between vg and sm, this includes the 2.3% of double crossovers. Therefore 22.2% of the female gametes should have single crossovers

11.1% **cn$^+$vg$^+$sm** & 11.1% **cn vg sm$^+$**

Total crossovers = 2.3% + 7.2% + 22.2% = 31.7%
Expect 68.3% parental gametes (34.15% of each).

Career Avenues

♀ gametes		♂ gametes = 100 % **cn vg sm**
34.15 %	**cn$^+$vg$^+$sm$^+$**	
34.15 %	**cn vg sm**	
11.1 %	**cn vg sm$^+$**	
11.1 %	**cn$^+$vg$^+$sm**	
3.6 %	**cn vg$^+$sm$^+$**	
3.6 %	**cn$^+$ vg sm**	
1.15%	**cn vg$^+$ sm**	
1.15 %	**cn$^+$ vg sm$^+$**	

These percentages can then be used to determine an expected ratio.

wild-type	:	**cn vg sm**	:	**cn vg**	:	**sm**	:	**cn**	:	**vg sm**	:	**vg**	:	**cn sm**
34.15	:	34.15	:	11.1	:	11.1	:	3.6	:	3.6	:	1.15	:	1.15
29.7	:	29.70	:	9.7	:	9.7	:	3.1	:	3.1	:	1	:	1

Career Avenues

EXTRACHROMOSOMAL INHERITANCE

PRINCIPLES

Some inherited traits are determined by DNA molecules present in organelles, principally mitochondria and chloroplasts. A hallmark of such traits is uniparental inheritance, either maternal or paternal.

Progeny resembling the mother is characteristic of both maternal inheritance and maternal effects. Although maternal inheritance results from the transmission of non-nuclear genetic factors, maternal effects result from nuclear genes expressed in the mother that determine the phenotype of the progeny.

Mitochondria, photo synthetic plastids, and certain other cellular organelles are widely believed to have originated through independent episodes of symbiosis in which a prokaryotic cell, the progenitor of the present organelle, became an intracellular symbiont of the proto-eukaryotic cell.

Evidence from DNA sequence variation in human mitochondrial DNA suggests that the mitochondrial DNA present in all modern human beings originated from a single female who lived approximately 150,000 to 300,000 years ago, probably in Africa.

Most traits in eukaryotes show Mendelian inheritance because they are determined by genes in the nucleus that segregate in meiosis. Less common are traits determined by factors located outside the nucleus. These include traits determined by the DNA in mitochondria and in chloroplasts — the self-replicating cytoplasmic organelles specialized for respiration and for photosynthesis, respectively. Each of these organelles contains its own DNA that codes for numerous proteins and RNA molecules. The DNA is replicated within the organelles and transmitted to daughter organelles as they form. The organelle genetic system is separate from that of the nucleus, and traits determined by organelle genes show patterns of inheritance quite different from the familiar Mendelian ratios. Many organisms also contain intracellular parasites or symbionts, including cytoplasmic bacteria, viruses, and other elements. In some cases, inherited traits of the infected organism are determined by such cytoplasmic entities.

RECOGNITION OF EXTRANUCLEAR INHERITANCE

Other than a non-Mendelian pattern of inheritance, no criterion is universally applicable to distinguish extranuclear from nuclear inheritance. In higher organisms, the transmission of a trait through only one parent, uniparental inheritance, usually indicates extranuclear inheritance. Genetic transmission of cytoplasmic factors, such as mitochondria or chloroplasts, is determined by maternal and paternal contributions at the time of fertilization, by mechanisms of elimination from the zygote, and by the irregular sorting of cytoplasmic elements during cell division.

Career Avenues

Uniparental transmission through the mother constitutes maternal inheritance; uniparental transmission through the father is paternal inheritance.

MITOCHONDRIAL GENETIC DISEASES

In higher animals, mitochondria are typically inherited through the mother because the egg is the major contributor of cytoplasm to the zygote. Therefore, mitochondria usually show maternal inheritance in which the progeny of a mutant mother and a normal father are mutant, whereas the progeny of a normal mother and a mutant father are normal.

HETEROPLASMY

The condition in which two or more genetically different types of mitochondria (or other organelle) are present in the same cell is known as heteroplasmy. Heteroplasmy of mitochondria is unusual among animals. For example, a typical human cell contains from 1000 to 10,000 mitochondria, and all of them are normally genetically identical.

MATERNAL INHERITANCE AND MATERNAL EFFECTS

A maternal pattern of inheritance usually indicates extra-nuclear inheritance, but not always. The difficulty is in distinguishing between maternal inheritance and maternal effects. The influence of the mother's nuclear genotype on the phenotype of the progeny is referred to as a maternal effect. Such effects may be mediated either through substances present in the egg that affect early development or through the effects of nurture. Examples of developmental and nurturing effects are, respectively, the intrauterine environment of female mammals and their ability to produce milk. The distinction between maternal inheritance and maternal effects is as follows:

- In maternal inheritance, the hereditary determinants of a trait are extra-nuclear, and genetic transmission is only through the maternal cytoplasm; Mendelian segregation is not observed.
- In a maternal effect, the nuclear genotype of the mother determines the phenotype of the progeny. The hereditary determinants are nuclear genes transmitted by both sexes, and in suitable crosses the trait undergoes Mendelian segregation.

In higher plants, organelle inheritance is often uniparental, but the particular type of uniparental inheritance depends on the organelle and the organism. For example, among angiosperms, mitochondria typically show maternal inheritance, but chloroplasts may be inherited maternally (the predominant mode), paternally, or from both parents, depending on the species. In conifers, chloroplast DNA is usually inherited paternally, but a few offspring result from maternal transmission. The red-wood Sequoia sempervirens shows paternal transmission of both mitochondrial and chloroplast DNA. Whatever the pattern of cytoplasmic transmission, most species have a few exceptional progeny, which indicates that the predominant mode of transmission is usually not absolute.

Career Avenues

ORGANELLE HEREDITY

As noted earlier, mitochondria are respiratory organelles, and chloroplasts are photo synthetic organelles. The DNA of these organelles is usually in the form of super-coiled circles of double-stranded molecules (Figure).

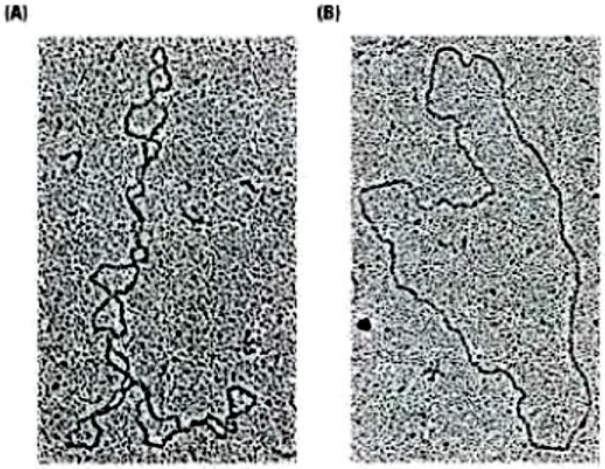


Figure: (A) DNA from rat liver mitochondria. (B) Native supercoiled configuration.

Relaxed configuration produced by a nick in the DNA strand in the process of isolation. The size of each molecule is 16.298 nucleotide pairs.

The chloroplast DNA in most plants ranges in size from 120 to 160 kb. Mitochondrial genomes are usually smaller and have a greater size range. For example, the mitochondrial genome in mammals is about 16.5 kb, that in D. melanogaster is about 18.5 kb, and that in some higher plants is more than 100 kb. The mitochondrial genomes of higher plants are exceptional in consisting of two or more circular DNA molecules of different sizes. Several copies of the genome are present in each organelle. There are typically from 2 to 10 copies in mitochondria and from 20 to 100 copies in chloroplasts. In addition, most cells contain multiple copies of each organelle.

Organelle genes code for DNA polymerases that replicate the organelle DNA. They also code for other components essential to their function or replication, but many organelle components are determined by nuclear genes. These components are synthesized in the cytoplasm and transported into the organelle. Mitochondrial DNA contains a relatively small number of genes. For example, both the human mitochondrial genome and that of yeast contain approximately 40 genes. Chloroplast genomes are generally larger than those of mitochondria and contain more genes. Many genes in mitochondria and chloroplasts code for the ribosomal RNA and transfer RNA components used in protein synthesis.

Career Avenues

RESPIRATION-DEFECTIVE MITOCHONDRIAL MUTANTS

Very small colonies of the yeast Saccharomyces cerevisiae are occasionally observed when the cells are grown on solid medium that contains glucose. These are called petite mutants. Microscopic examination indicates that the cells are of normal size. Physiological studies show that the cells grow normally during early stages of colony formation when they are fermenting the glucose to ethanol but cease to grow thereafter because of a defect in oxygen-requiring respiration that is needed for further metabolism of the ethanol. The colonies are quite small, because the yield of ATP is much higher during respiration than during fermentation. There are several types of petite mutants. We will discuss two of them. Genetic analysis of crosses between petites and wildtype distinguishes the two major types of petites illustrated in Figure.

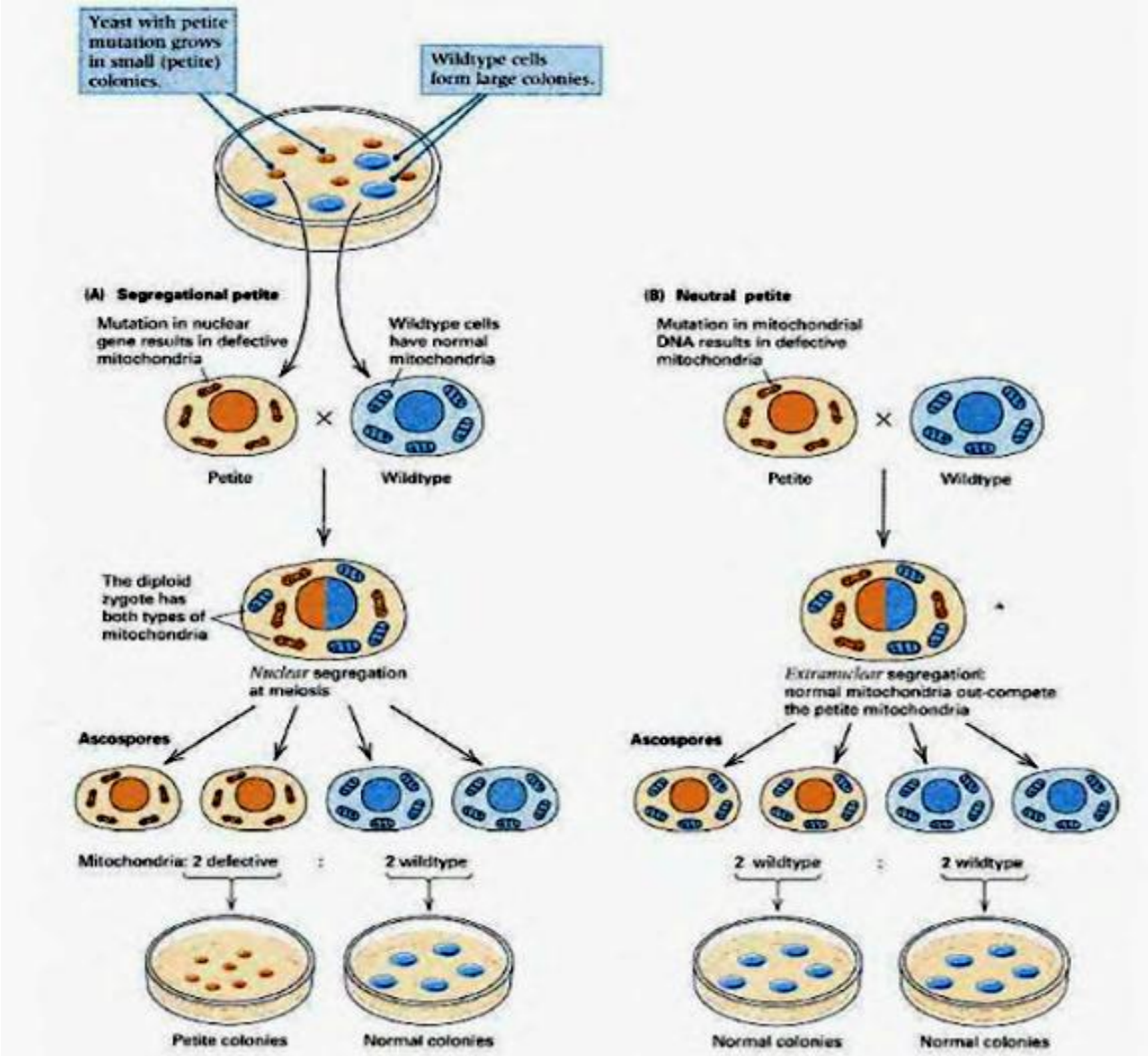


Figure in previous page: Behavior of petite mutations in genetic crosses. The brown circles represent petite cells. The brown nucleus in part A contains an allele resulting in petite colonies. The blue cells have normal mitochondria. (Colours may appear as different shades of grey in your book.)

One type, called segregational petites, exhibits the typical segregation of a simple Mendelian recessive. In a cross between a segregational petite and wildtype, the diploid zygotes are normal, and when the diploids undergo meiosis, half of the ascospores in an ascus produce petite colonies and half form wildtype colonies. The 1 : 1 ratio (actually 2 : 2) indicates that these petites are the result of a nuclear mutation and that the allele for petite is recessive.

The second type, neutral petites, is quite different: In a cross with wildtype, all diploid zygotes produce wildtype ascospores (4: 0 segregation). The same pattern is found when the progeny from such a cross are backcrossed with neutral petites. In both cases, the phenotype that is inherited is that of the normal parent, so the inheritance is uniparental. The explanation for these results is that the majority of neutral petites lack most or all mitochondrial

DNA, in which many of the genes determining oxidative respiration are encoded. When a neutral-petite cell mates with a wildtype cell, the cytoplasm of the latter is the source of the normal mitochondrial DNA in the resulting progeny spores. Petites arise at a frequency of roughly 10^{-5} per generation. The neutral petite phenotype is the result of large deletions in mitochondrial DNA. For unknown reasons, yeast mitochondria often fuse and fragment, which may cause occasional deletions of DNA. The production of neutral petites is apparently a result of the segregation of aberrant mitochondrial DNA from normal DNA and the further sorting out of mutant mitochondria in the course of cell division.

CYTOPLASMIC MALE STERILITY IN PLANTS

An example of extra-nuclear inheritance that is important in agriculture is cytoplasmic male sterility in plants, a condition in which the plant does not produce functional pollen, but the female reproductive organs and fertility are normal. This type of sterility is used extensively in the production of hybrid corn seed, because it circumvents the need for manual "detasseling" of the plants to be used as females to produce the hybrid. Because the tassel is the pollen-bearing organ, the detasseled female plants cannot produce pollen and so cannot fertilize themselves. The detasseled female plants must therefore use pollen from the still-tasseled male plants used to produce the hybrid.

The genetically transmitted male sterility is not controlled by nuclear genes but is transmitted through the egg cytoplasm from generation to generation. The pattern of inheritance of cytoplasmic male sterility in corn is summarized in Figure.

Career Avenues

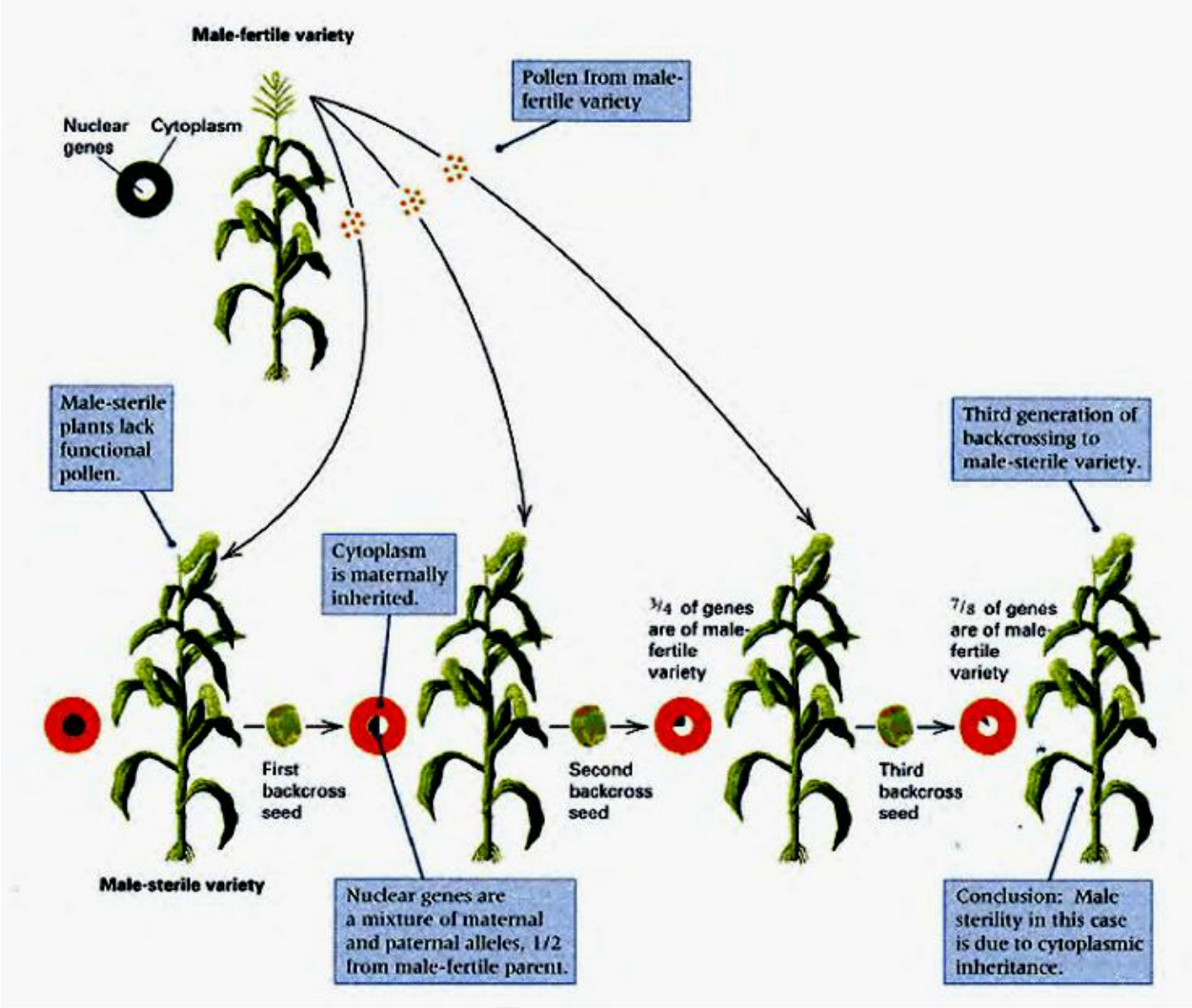


Figure: Diagram of an experiment demonstrating maternal inheritance of male sterility (orange cytoplasm) in maize.

The key observation is that repeated backcrossing of a male-sterile variety, using pollen from a male-fertile variety, does not restore male fertility. The resulting plants, in which nearly all nuclear genes from the male-sterile variety have been replaced with those from the male-fertile variety, remain male sterile. Because a small amount of functional pollen is produced by the male-sterile variety, it is also possible to carry out the reciprocal cross:

male-fertile ♀ × male-sterile ♂

In this case, the progeny are fully male-fertile. The difference between the reciprocal crosses means that male sterility or fertility in these varieties is maternally inherited. Cytoplasmic male sterility in maize results from rearrangements in mitochondrial DNA. For example, one DNA rearrangement fuses two mitochondrial genes and creates a novel protein that causes male sterility. Although cytoplasmic male sterility is maternally inherited, certain nuclear genes called

fertility restorers can suppress the male-sterilizing effect of the cytoplasm. Different types of cytoplasmic male sterility can be classified on the basis of their response to restorer genes. For example, in one case, restoration of fertility requires the presence of dominant alleles of two different nuclear genes, and plants with male-sterile cytoplasm and both restorer alleles produce normal pollen. In another case, suppression requires the dominant allele of a different restorer gene, but this gene acts only in the gametophyte; hence plants with male-sterile cytoplasm that are heterozygous for the restorer allele produce a 1 : 1 ratio of normal to aborted pollen grains.

THE CYTOPLASMIC TRANSMISSION OF SYMBIONTS

In eukaryotes, a variety of cytoplasmically transmitted traits result from the presence of bacteria and viruses living in the cytoplasm of certain cells. One of the best known is the killer phenomenon found in certain strains of the protozoan Paramecium aurelia. Killer strains of Paramecium release to the surrounding medium a substance that is lethal to many other strains of the protozoan. The killer phenotype requires the presence of cytoplasmic bacteria referred to as kappa particles, whose maintenance is dependent on a dominant nuclear gene, K. The kappa particles are cells of Caedobacter taeniospiralis, which produce the killer substance. Why killer strains are immune to the substance has not yet been determined.

Paramecium is a diploid protozoan that undergoes sexual exchange through a mating process called conjugation.

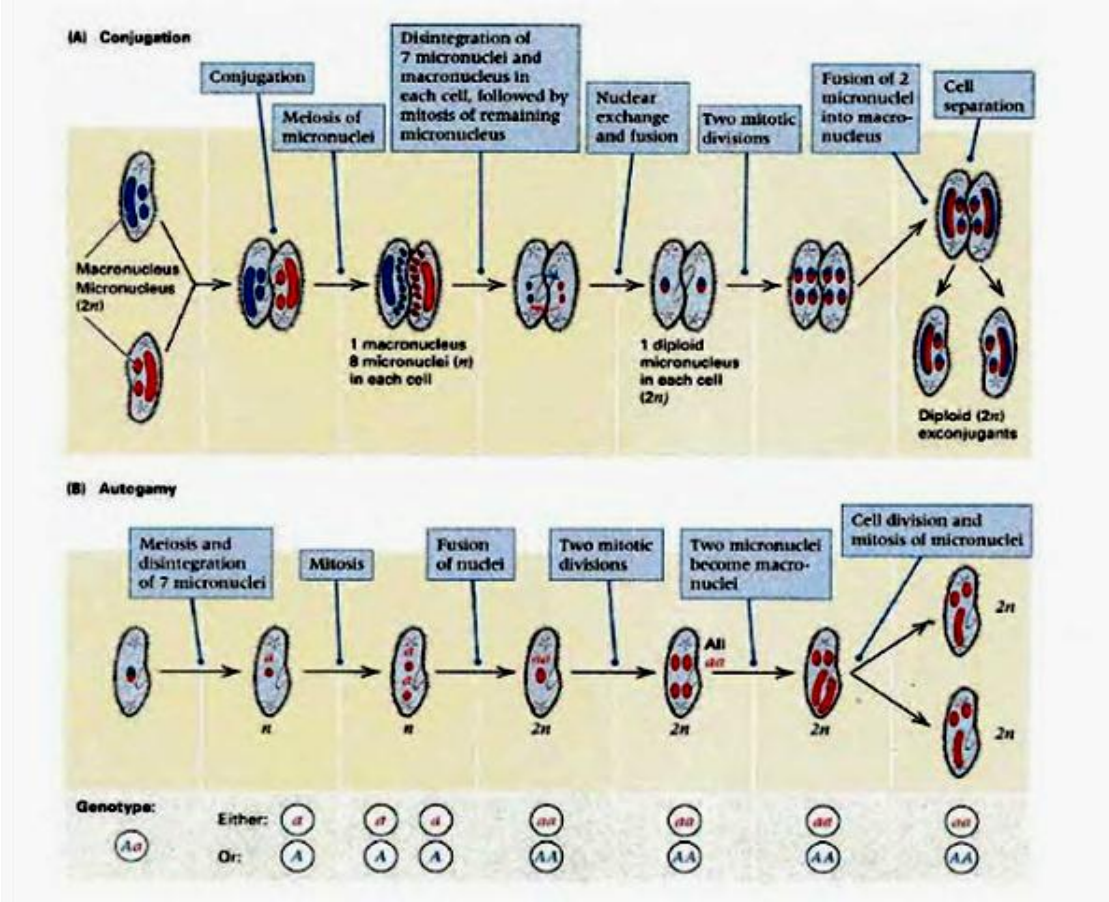


Figure in previous page: Conjugation in Paramecium results in reciprocal fertilization and the formation of two genetically identical exconjugant cells. (B) Autogamy in Paramecium. The alleles in the heterozygous Aa cell segregate in meiosis, with the result that the progeny cells become homozygous for either A or a. The production of aa progeny cells is shown in the diagram.

The initial steps resemble those in conjugation in part A: The micronuclei undergo meiosis, after which seven of the products and the macronucleus degenerate. Initially, each cell has two diploid micronuclei. Many protozoans contain micronuclei (small nuclei) and a macronucleus (a large nucleus with specialized function), but only the micronuclei are relevant to the genetic processes described here. When two cells come into contact for conjugation, the two micronuclei in each cell undergo meiosis, forming eight micronuclei in each cell. Seven of the micronuclei and the macronucleus disintegrate, leaving each cell with one micronucleus, which then undergoes a mitotic division. The cell membrane between the two cells breaks down slightly, and the cells exchange a single micronucleus; then the nuclei fuse to form a diploid nucleus. After nuclear exchange, the two new cells (the exconjugants) are genotypically identical. In this sequence of events, only the micronuclei are exchanged, without any mixing of cytoplasm. Single cells of Paramecium also occasionally undergo an unusual nuclear phenomenon called autogamy. Meiosis takes place and again seven of the micronuclei disintegrate. The surviving nucleus then undergoes mitosis and nuclear fusion to recreate the diploid state.

The genetic importance of autogamy is that even if the initial cell is heterozygous, the newly formed diploid nucleus becomes homozygous because it is derived from a single haploid meiotic product. In a population of cells undergoing autogamy, the surviving micronucleus is selected randomly, so for any pair of alleles, half of the new cells contain one allele and half contain the other.

Conjugation usually allows an exchange only of micronuclei at the surface of contact between the two cells. Occasionally, small amounts of cytoplasm also are exchanged. A comparison of the phenotypes following conjugation of killer (KK) cells and sensitive (kk) cells, with and without cytoplasmic exchange, yields evidence for cytoplasmic inheritance of the killer phenotype. Without cytoplasmic mixing (see figure), the expected 1 : 1 ratio of killer to sensitive exconjugant cells is observed; with cytoplasmic mixing (Figure B), both exconjugants are killer cells because each cell has kappa particles derived from the cytoplasm of the killer parent. Conjugation is eventually followed by autogamy of each of the exconjugants. Autogamy of Kk killer cells results in an equal number of KK and kk cells, but the kk cells cannot maintain the kappa particle and so become sensitive.

Career Avenues

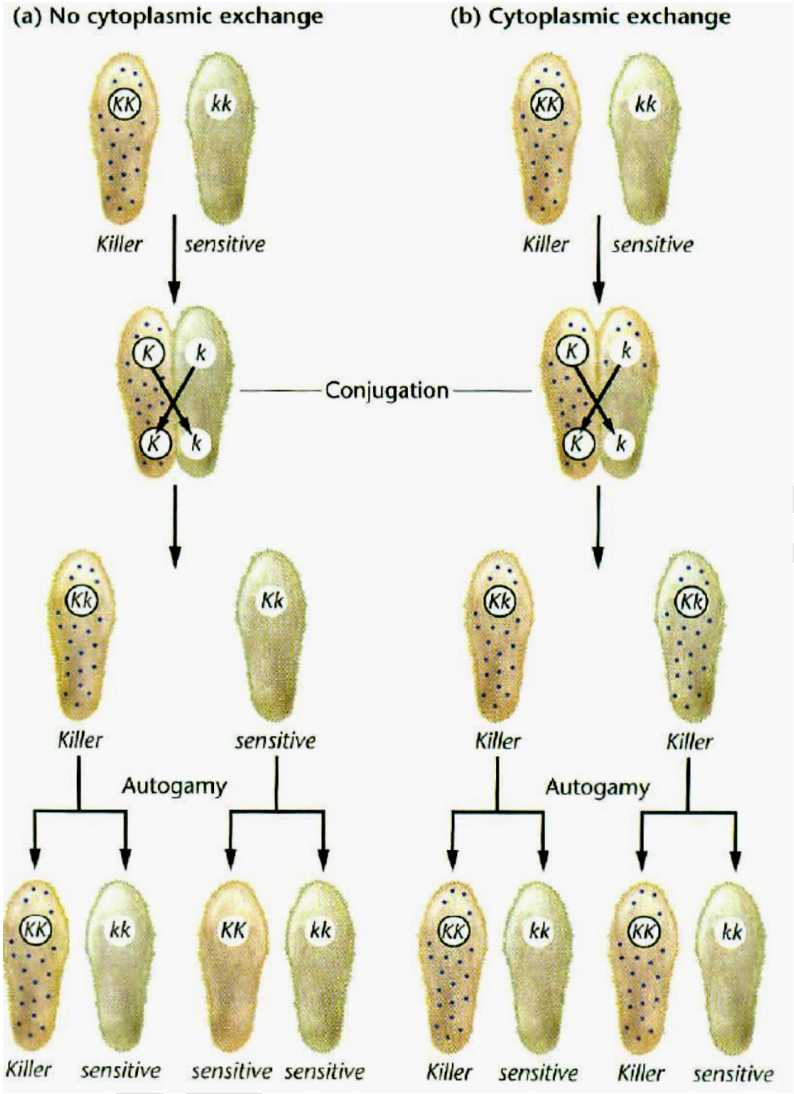


Fig. Results of crosses between Killer (KK) and sensitive (kk) strains of Paramecium, with and without cytoplasmic exchange during conjugation. The kappa particles(dots) are maintained only when a K allele is present. The genotype and phenotype of each cell are indicated.

Another example of a symbiont-related cytoplasmic effect is a condition in Drosophila called maternal sex ratio, which is characterized by the production of almost no male progeny. The daughters of sex-ratio females pass on the trait, whereas the occasional sons that are produced do not. Cytoplasm taken from the eggs of sex-ratio females transmits the condition when injected into the female embryos from unaffected cultures of the same or other Drosophila species. A species of bacteria has been isolated from the cytoplasm of sex-ratio females; when these bacteria are allowed to infect other Drosophila females, these acquire the sex-ratio trait. The causative agent of the sex-ratio condition is not the bacteria themselves but rather a virus that multiplies in them. When released by the bacterial cells, this virus kills most male Drosophila embryos. Why female embryos are not killed by the virus is still a mystery.

MATERNAL EFFECT IN SNAIL-SHELL COILING

There are also some examples in which maternal transmission takes place, but the maternal transmission is not mediated by cytoplasmic organelles or symbionts but rather through the cytoplasmic transmission of the products of nuclear genes. An example is the determination of the direction of coiling of the shell of the snail Limnaea peregra.

The direction of coiling, as viewed by looking into the opening of the shell, may be either to the right (dextral coiling) or to the left (sinistral coiling). Reciprocal crosses between homozygous strains give the following results:

dextral ♀ × sinistral ♂ → all F_1 dextral
sinistral ♀ × dextral ♂ → all F_1 sinistral

In these crosses, the F1 snails have the same genotype, but the reciprocal crosses give different results in that the direction of coiling is the same as that of the mother. This result is typical of traits with maternal inheritance. However, in this case, all F2 progeny from both crosses exhibit dextral coiling, a result that is inconsistent with cytoplasmic inheritance. The F3 generation provides the explanation.

The F3 generation, obtained by self-fertilization of the F2 snails (the snail is hermaphroditic), indicates that the inheritance of coiling direction depends on nuclear genes rather than extranuclear factors. Three-fourths of the F3 progeny have dextrally coiled shells and one-fourth have sinistrally coiled shells. (Figure).

Career Avenues

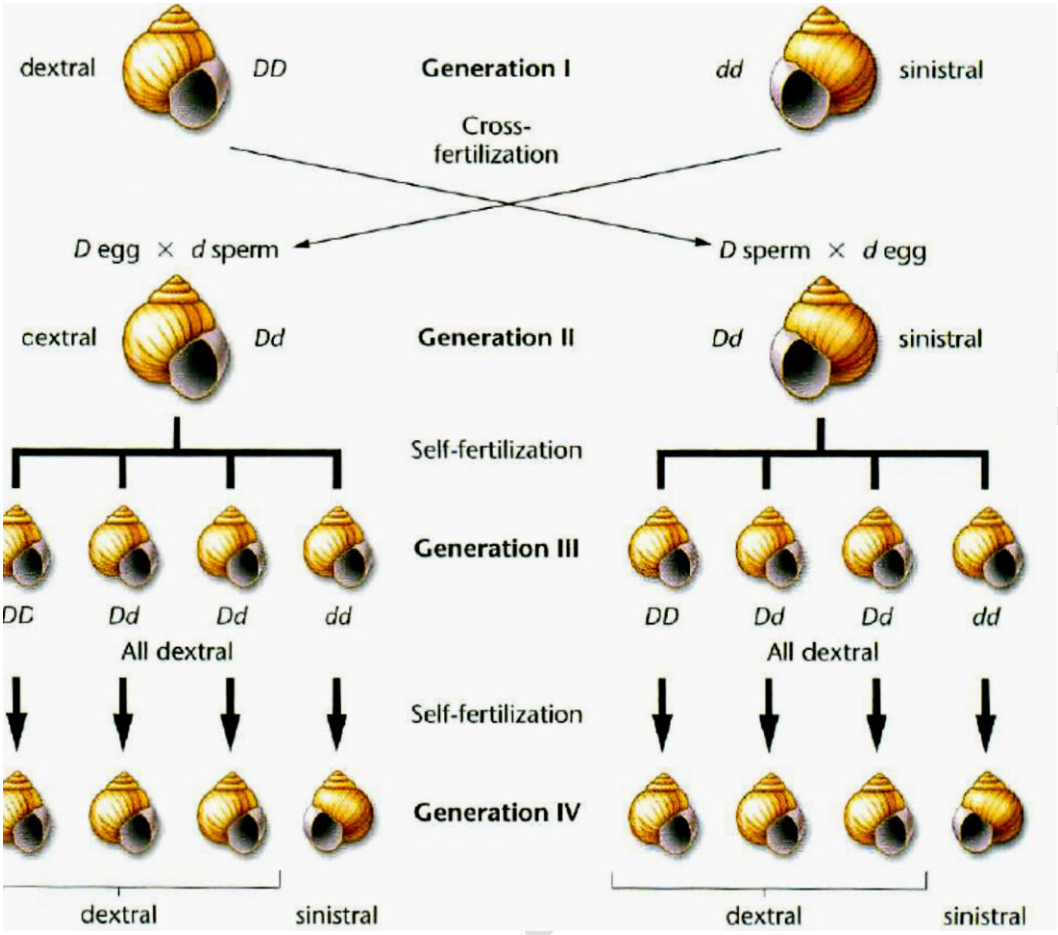


Figure: *Inheritance of the direction of shell coiling in the snail Limnaea. Sinistral coiling is determined by the recessive allele s, dextral coiling by the wild-type + allele. However, the direction of coiling is determined by the nuclear genotype of the mother, not by the genotype of the snail itself or by extra-nuclear inheritance.*

The F2 and F3 generations can be obtained by self-fertilization because the snail is hermaphroditic and can undergo either self-fertilization or cross-fertilization.

This is a typical 3 : 1 ratio, which indicates Mendelian segregation in the F2. Dextral coiling (+/+ and s/+) is dominant to sinistral coiling (s/s). The reason why reciprocal crosses give different results, and why the production of the 3 : 1 Mendelian ratio is delayed for a generation, is as follows:

- The coiling phenotype of an individual snail is determined by the genotype of its mother. This means that the direction of shell coiling is not a case of maternal inheritance but rather a case of maternal effect. Cytological analysis of developing eggs has provided the explanation:
- The genotype of the mother determines the orientation of the spindle in the initial mitotic division after fertilization, and this in turn controls the direction of shell coiling of the offspring. This classic example of maternal effect indicates that more than a single generation of crosses is needed to provide conclusive evidence of extra-nuclear inheritance of a trait.

CHROMOSOMAL VARIATION

- Duplication of the entire chromosome complement present in a species—or in a hybrid between species — is a major process in the evolution of higher plants.
- The genetic unbalance caused by a single chromosome that is extra or missing may have a more serious phenotypic effect than an entire extra set of chromosomes.
- Chromosome abnormalities are an important cause of human genetic disease and are a major factor in spontaneous abortions.
- Aneuploid (unbalanced) chromosome rearrangements usually have greater phenotypic effects than euploid (balanced) chromosome rearrangements.
- By a process of mispairing and unequal crossing-over, genes that are duplicated in tandem along the chromosome can give rise to chromosomes with even more copies.
- Gene-for-gene pairing between a wildtype chromosome and one that contains an inversion of a segment of genes results in the formation of a loop in one of the chromosomes; crossing-over within the "inversion loop" leads to chromosomal abnormalities.
- Reciprocal translocations result in abnormal gametes because they upset segregation.
- Some types of cancer are associated with particular chromosome rearrangements.

ANEUPLOIDY

Aneuploidy is an abnormal number of chromosomes, and is a type of chromosome abnormality. An extra or missing chromosome is a common cause of genetic disorders (birth defects). Some cancer cells also have abnormal numbers of chromosomes. Aneuploidy occurs during cell division when the chromosomes do not separate properly between the two cells.

MONOSOMY

Monosomy refers to lack of one chromosome of the normal complement. Partial monosomy can occur in unbalanced translocations or deletions, in which only a portion of the chromosome is present in a single copy. Monosomy of the sex chromosomes (45,X) causes Turner syndrome.

DISOMY

Disomy is the presence of two copies of a chromosome. For organisms such as humans that have two copies of each chromosome (those that are diploid), it is the normal condition. For organisms that normally have three or more copies of each chromosome (those that are triploid or above), disomy is an aneuploid chromosome complement. In uniparental disomy, both copies of a chromosome come from the same parent (with no contribution from the other parent).

Career Avenues

TRISOMY

Trisomy refers to the presence of three copies, instead of the normal two, of a particular chromosome. The presence of an extra chromosome 21, which is found in Down syndrome, is called trisomy 21. Trisomy 18 and Trisomy 13, known as Edwards and Patau Syndrome, respectively, are the two other autosomal trisomies recognized in live-born humans. Trisomy of the sex chromosomes is possible, such as in (47, XXX), (47, XXY), and (47, XYY).

CHROMOSOMAL ABERRATIONS

Chromosomal aberrations are disruptions in the normal chromosomal content of a cell and are a major cause of genetic conditions in humans, such as Down syndrome, although most aberrations have little to no effect. Some chromosome abnormalities do not cause disease in carriers, such as translocations, or chromosomal inversions, although they may lead to a higher chance of birthing a child with a chromosome disorder. Abnormal numbers of chromosomes or chromosome sets, called aneuploidy, may be lethal or may give rise to genetic disorders. Genetic counseling is offered for families that may carry a chromosome rearrangement.

The gain or loss of DNA from chromosomes can lead to a variety of genetic disorders. Human examples include:

- **Cri du chat,** which is caused by the deletion of part of the short arm of chromosome 5. "Cri du chat" means "cry of the cat" in French; the condition was so-named because affected babies make high-pitched cries that sound like those of a cat. Affected individuals have wide-set eyes, a small head and jaw, moderate to severe mental health issues, and are very short.
- **Down syndrome**, the most common trisomy, usually caused by an extra copy of chromosome 21 (trisomy 21). Characteristics include decreased muscle tone, stockier build, asymmetrical skull, slanting eyes and mild to moderate developmental disability.
- **Edwards syndrome**, or trisomy-18, the second-most-common trisomy. Symptoms include motor retardation, developmental disability and numerous congenital anomalies causing serious health problems. Ninety percent of those affected die in infancy. They have characteristic clenched hands and overlapping fingers.
- **Isodicentric 15**, also called idic(15), partial tetrasomy 15q, or inverted duplication 15 (inv dup 15).
- **Jacobsen syndrome,** which is very rare. It is also called the terminal 11q deletion disorder. Those affected have normal intelligence or mild developmental disability, with poor expressive language skills. Most have a bleeding disorder called Paris-Trousseau syndrome.
- **Klinefelter's syndrome (XXY).** Men with Klinefelter syndrome are usually sterile, and tend to be taller and have longer arms and legs than their peers. Boys with the syndrome are often shy and quiet, and have a higher incidence of speech delay and dyslexia. Without testosterone treatment, some may develop gynecomastia during puberty.

Career Avenues

- **Patau Syndrome**, also called D-Syndrome or trisomy-13. Symptoms are somewhat similar to those of trisomy-18, without the characteristic folded hand.
- Small supernumerary marker chromosome. This means there is an extra, abnormal chromosome. Features depend on the origin of the extra genetic material. Cat-eye syndrome and isodicentric chromosome 15 syndrome (or Idic15) are both caused by a supernumerary marker chromosome, as is Pallister-Killian syndrome.
- **Triple-X syndrome** (XXX). XXX girls tend to be tall and thin and have a higher incidence of dyslexia.
- **Turner syndrome** (X instead of XX or XY). In Turner syndrome, female sexual characteristics are present but underdeveloped. Females with Turner syndrome often have a short stature, low hairline, abnormal eye features and bone development and a "caved-in" appearance to the chest.
- **XYY syndrome.** XYY boys are usually taller than their siblings. Like XXY boys and XXX girls, they are more likely to have learning difficulties.
- **Wolf-Hirschhorn syndrome,** which is caused by partial deletion of the short arm of chromosome 4. It is characterized by severe growth retardation and severe to profound mental health issues.

NOTE

DOSAGE COMPENSATION

A mechanism by which species with sex chromosomes ensure that the homogametic sex does not have too much or the heterogametic sex have too little activity of loci on the homogametic sex chromosome. The process, in organisms using a chromosomal sex determination mechanism (such as XX versus XY), that allows standard structural genes on the homogametic sex chromosome to be expressed at the same levels in females and males, regardless of the number of homogametic sex chromosomes. In mammals, dosage compensation operates by maintaining only a single active X chromosome in each cell (see Lyon hypothesis), while in Drosophila it operates by hyperactivating the single male X chromosome.

WHAT IS BARR BODY?

It is a condensed, inactivated X chromosome present in most female mammals in the nuclei of all cells except the germ cells: used, as in sports competitions, to verify that an individual is genetically female.

Career Avenues

POPULATION GENETICS

Population genetics explores the genetic composition of groups of individual members of the same species (populations) and how that composition changes over time and space. Because evolution is genetic change, population genetics is fundamentally the study of evolution. The focus of population genetics is the group of genes found in a population. It is convenient and traditional to divide the study of genetics into these three groups, but we should recognize that the fields overlap and that each major subdivision can be further divided into a number of more specialized fields, such as chromosomal genetics, biochemical genetics, quantitative genetics, and so forth. Genetics can alternatively be subdivided by organism (fruit fly, corn, or bacterial genetics), and each of these organisms can be studied at the level of transmission, molecular, and population genetics. Modern genetics is an extremely broad field, encompassing many interrelated subdisciplines and specializations.

GENETIC VARIATION

An obvious and pervasive feature of life is variability. Consider a group of students in a typical college class, the members of which vary in eye color, hair color, skin pigmentation, height, weight, facial features, blood type, and susceptibility to numerous diseases and disorders. No two students in the class are likely to be even remotely similar in appearance.

Humans are not unique in their extensive variability; almost all organisms exhibit variation in phenotype. For instance, lady beetles are highly variable in their patterns of spots , mice vary in body size, snails have different numbers of stripes on their shells, and plants vary in their susceptibility to pests. Much of this phenotypic variation is hereditary. Recognition of the extent of phenotypic variation and its genetic basis led Charles Darwin to the idea of evolution through natural selection. In fact, even more genetic variation exists in populations than is visible in the phenotype. Much variation exists at the molecular level owing to the redundancy of the genetic code, which allows different codons to specify the same amino acids. Thus two individuals can produce the same protein even if their DNA sequences are different.

DNA sequences between the genes and introns within genes do not encode proteins; so much of the variation in these sequences also has little effect on the phenotype. The amount of genetic variation within natural populations and the forces that limit and shape it are of primary interest to population geneticists. Genetic variation is the basis of all evolution, and the extent of genetic variation within a population affects its potential to adapt to environmental change.

An important, but frequently misunderstood, tool used in population genetics is the mathematical model. Let's take a moment to consider what a model is and how it can be used. A mathematical model usually describes a process in terms of an equation. Factors that may influence the process are represented by variables in the equation; the equation defines the way in which the variables influence the process. Most models are simplified representations of a process, because it is

Career Avenues

impossible to simultaneously consider all of the influencing factors; some must be ignored in order to examine the effects of others.

CALCULATION OF GENOTYPIC FREQUENCIES

A frequency is simply a proportion or a percentage, usually expressed as a decimal fraction. For example, if 20% of the alleles at a particular locus in a population are *A*, we would say that the frequency of the *A* allele in the population is .20. For large populations, where it is not practical to determine the genes of all individuals, a sample of individuals from the population is usually taken and the genotypic and allelic frequencies are calculated for this sample .The genotypic and allelic frequencies of the sample are then used to represent the gene pool of the population.

To calculate a **genotypic frequency,** we simply add up the number of individuals possessing the genotype and divide by the total number of individuals in the sample (*N*). For a locus with three genotypes *AA, Aa,* and *aa,* the frequency (*f*) of each genotype is:

$$f(AA) = \frac{\text{number of } AA \text{ individuals}}{N}$$

$$f(Aa) = \frac{\text{number of } Aa \text{ individuals}}{N}$$

$$f(aa) = \frac{\text{number of } aa \text{ individuals}}{N}$$

The sum of all the genotypic frequencies always equals 1.

CALCULATION OF ALLELIC FREQUENCIES

The gene pool of a population can also be described in terms of the allelic frequencies. There are always fewer alleles than genotypes; so the gene pool of a population can be described in fewer terms when the allelic frequencies are used. In a sexually reproducing population, the genotypes are only temporary assemblages of the alleles: the genotypes break down each generation when individual alleles are passed to the next generation through the gametes, and so it is the types and numbers of alleles, not genotypes, that have real continuity from one generation to the next and that make up the gene pool of a population. Allelic frequencies can be calculated from (1) the numbers or (2) the frequencies of the genotypes. To calculate the **allelic frequency** from the numbers of genotypes, we count the number of copies of a particular allele present in a sample and divide by the total number of all alleles in the sample:

$$\text{frequency of an allele} = \frac{\text{number of copies of the allele}}{\text{number of copies of all alleles at the locus}}$$

For a locus with only two alleles (*A* and *a*), the frequencies of the alleles are usually represented by the symbols *p* and *q*, and can be calculated as follows:

Career Avenues

$$p = f(A) = \frac{2n_{AA} + n_{Aa}}{2N}$$

$$q = f(a) = \frac{2n_{aa} + n_{Aa}}{2N}$$

where *nAA*, *nAa*, and *naa* represent the numbers of *AA*, *Aa*, and *aa* individuals, and *N* represents the total number of individuals in the sample. We divide by 2*N* because each diploid individual has two alleles at a locus. The sum of the allelic frequencies always equals 1 ($p + q = 1$); so after *p* has been obtained, *q* can be determined by subtraction: $q = 1 - p$. Alternately, allelic frequencies can also be calculated from the genotypic frequencies. To do so, we add the frequency of the homozygote for each allele to half the frequency of the heterozygote (because half of the heterozygote's alleles are of each type):

$$p = f(A) = f(AA) + \tfrac{1}{2} f(Aa)$$

$$q = f(a) = f(aa) + \tfrac{1}{2} f(Aa)$$

THE HARDY-WEINBERG LAW

The primary goal of population genetics is to understand the processes that shape a population's gene pool. First, we must ask what effects reproduction and Mendelian principles have on the genotypic and allelic frequencies: How do the segregation of alleles in gamete formation and the combining of alleles in fertilization influence the gene pool? The answer to this question lies in the **Hardy-Weinberg law,** one of the most important principles of population genetics. The Hardy-Weinberg law was formulated independently by both Godfrey H. Hardy and Wilhelm Weinberg in 1908. (Similar conclusions were reached by several other geneticists about the same time.) The law is actually a mathematical model that evaluates the effect of reproduction on the genotypic and allelic frequencies of a population. It makes several simplifying assumptions about the population and provides two key predictions if these assumptions are met. For an autosomal locus with two alleles, the Hardy- Weinberg law can be stated as follows: **Assumptions** — If a population is large, randomly mating, and not affected by mutation, migration, or natural selection, then:

- **Prediction 1** — the allelic frequencies of a population do not change; and
- **Prediction 2** — the genotypic frequencies stabilize (will not change) after one generation in the proportions *p*2 (the frequency of *AA*), 2*pq* (the frequency of *Aa*), and *q*2 (the frequency of *aa*), where *p* equals the frequency of allele *A* and *q* equals the frequency of allele *a*.

The Hardy-Weinberg law indicates that, when the assumptions are met, reproduction alone does not alter allelic or genotypic frequencies and the allelic frequencies determine the frequencies of genotypes. The statement that genotypic frequencies stabilize after one generation means that they may change in the first generation after random mating, because one generation of random mating is required to produce Hardy-Weinberg proportions of the genotypes. Afterward, the genotypic frequencies, like allelic frequencies, do not change as long as the population continues

Career Avenues

to meet the assumptions of the Hardy-Weinberg law. When genotypes are in the expected proportions of p^2, $2pq$, and q^2, the population is said to be in Hardy-Weinberg equilibrium. The Hardy-Weinberg law describes how reproduction and Mendelian principles affect the allelic and genotypic frequencies of a population.

Estimating Allelic Frequencies with the Hardy-Weinberg Law

A practical use of the Hardy-Weinberg law is that it allows us to calculate allelic frequencies when dominance is present. For example, cystic fibrosis is an autosomal recessive disorder characterized by respiratory infections, incomplete digestion, and abnormal sweating (see p. 000 in Chapter 6). Among North American Caucasians, the incidence of the disease is approximately 1 person in 2000. The formula for calculating allelic frequency (Equation 23.3) requires that we know the numbers of homozygotes and heterozygotes, but cystic fibrosis is a recessive disease, and so we cannot easily distinguish between homozygous normal persons and heterozygous carriers. Although molecular tests are available for identifying heterozygous carriers of the cystic fibrosis gene, the low frequency of the disease makes widespread screening impractical. In such situations, the Hardy-Weinberg law can be used to estimate the allelic frequencies. If we assume that a population is in Hardy-Weinberg equilibrium with regard to this locus, then the frequency of the recessive genotype (*aa*) will be q^2, and the allelic frequency is the square root of the genotypic frequency:

$$q = \sqrt{f(aa)}$$

The frequency of cystic fibrosis in North American Caucasians is approximately 1 in 2000, or .0005; so $q = \sqrt{0.0005} = .02$. Thus, about 2% of the alleles in the Caucasian population encode cystic fibrosis. We can calculate the frequency of the normal allele by subtracting: $p = 1 - q = 1 - .02 = .98$. After we have calculated p and q, we can use the Hardy-Weinberg law to determine the frequencies of homozygous normal people and heterozygous carriers of the gene:

$$f(AA) = p^2 = (.98)^2 = .960$$

$$f(Aa) = 2pq = 2(.02)(.98) = .0392$$

Thus about 4% (1 of 25) of Caucasians are heterozygous carriers of the allele that causes cystic fibrosis.

CHANGES IN ALLELIC FREQUENCIES

The Hardy-Weinberg law indicates that allelic frequencies do not change as a result of reproduction; thus, other processes must cause alleles to increase or decrease in frequency.

Processes that bring about change in allelic frequency include mutation, migration, genetic drift (random effects due to small population size), and natural selection.

MUTATION

Before evolution can occur, genetic variation must exist within a population; consequently, all evolution depends on processes that generate genetic variation. Although new *combinations* of existing genes may arise through recombination in meiosis, all genetic variants ultimately arise through mutation.

MIGRATION

Another process that may bring about change in the allelic frequencies is the influx of genes from other populations, commonly called **migration** or **gene flow.** One of the assumptions of the Hardy-Weinberg law is that migration does not take place, but many natural populations do experience migration from other populations. The overall effect of migration is twofold: (1) it prevents genetic divergence *between* populations and (2) it increases genetic variation *within* populations.

GENETIC DRIFT

The Hardy-Weinberg law assumes random mating in an infinitely large population; only when population size is infinite will the gametes carry genes that perfectly represent the parental gene pool. But no real population is infinitely large, and when population size is limited, the gametes that unite to form individuals of the next generation carry a sample of alleles present in the parental gene pool. Just by chance, the composition of this sample may deviate from that of the parental gene pool, and this deviation may cause allelic frequencies to change. The smaller the gametic sample, the greater the chance that its composition will deviate from that of the entire gene pool.

The role of chance in altering allelic frequencies is analogous to flipping a coin. Each time we flip a coin, we have a 50% chance of getting a head and a 50% chance of getting a tail. If we flip a coin 1000 times, the observed ratio of heads to tails will be very close to the expected 50:50 ratio. If, however, we flip a coin only 10 times, there is a good chance that we will obtain not exactly 5 heads and 5 tails, but rather maybe 7 heads and 3 tails or 8 tails and 2 heads. This kind of deviation from an expected ratio due to limited sample size is referred to as **sampling error.**

Sampling error occurs when gametes unite to produce progeny. Many organisms produce a large number of gametes but, when population size is small, a limited number of gametes unite to produce the individuals of the next generation. Chance influences which alleles are present in this limited sample and, in this way, sampling error may lead to changes in allelic frequency, which is called **genetic drift.** Because the deviations from the expected ratios are random, the direction of change is unpredictable. We can nevertheless predict the magnitude of the changes.

Career Avenues

NATURAL SELECTION

A final process that brings about changes in allelic frequencies is natural selection, the differential reproduction of genotypes. Natural selection takes place when individuals with adaptive traits produce more offspring. If the adaptive traits have a genetic basis, they are inherited by the offspring and appear with greater frequency in the next generation. A trait that provides a reproductive advantage thereby increases over time, enabling populations to become better suited to their environments — to become better adapted. Natural selection is unique among evolutionary forces in that it promotes adaptation.

FITNESS AND SELECTION COEFFICIENT

The effect of natural selection on the gene pool of a population depends on the fitness values of the genotypes in the population. **Fitness** is defined as the relative reproductive success of a genotype. Here the term *relative* is critically important: fitness is the reproductive success of one genotype compared with the reproductive successes of other genotypes in the population. Fitness (*W*) ranges from 0 to 1.

A related variable is the **selection coefficient** (*s*), which is the relative intensity of selection against a genotype. The selection coefficient is equal to 1 -*W*.

Career Avenues

TRANSPOSABLE ELEMENTS

THE NATURE OF TRANSPOSABLE ELEMENTS

Transposable elements are mobile DNA sequences found in the genomes of all organisms. In many genomes, they are quite abundant: for example, they make up at least 50% of human DNA. Most transposable elements are able to insert at many different locations, relying on mechanisms that are distinct from homologous recombination. They often cause mutations, either by inserting into another gene and disrupting it or by promoting DNA rearrangements such as deletions, duplications, and inversions.

GENERAL CHARACTERISTICS OF TRANSPOSABLE ELEMENTS

There are many different types of transposable elements: some have simple structures, encompassing only those sequences necessary for their own transposition (movement), whereas others have complex structures and encode a number of functions not directly related to transposition. Despite this variation, many transposable elements have certain features in common.

Short, flanking direct repeats of 3 to 12 base pairs are present on both sides of most transposable elements. They are not a part of a transposable element and do not travel with it. Rather, they are generated in the process of transposition, at the point of insertion. The sequences of these repeats vary, but the length is constant for each type of transposable element.

The presence of flanking direct repeats indicates that staggered cuts are made in the target DNA when a transposable element inserts itself, as shown in FIGURE.

Career Avenues

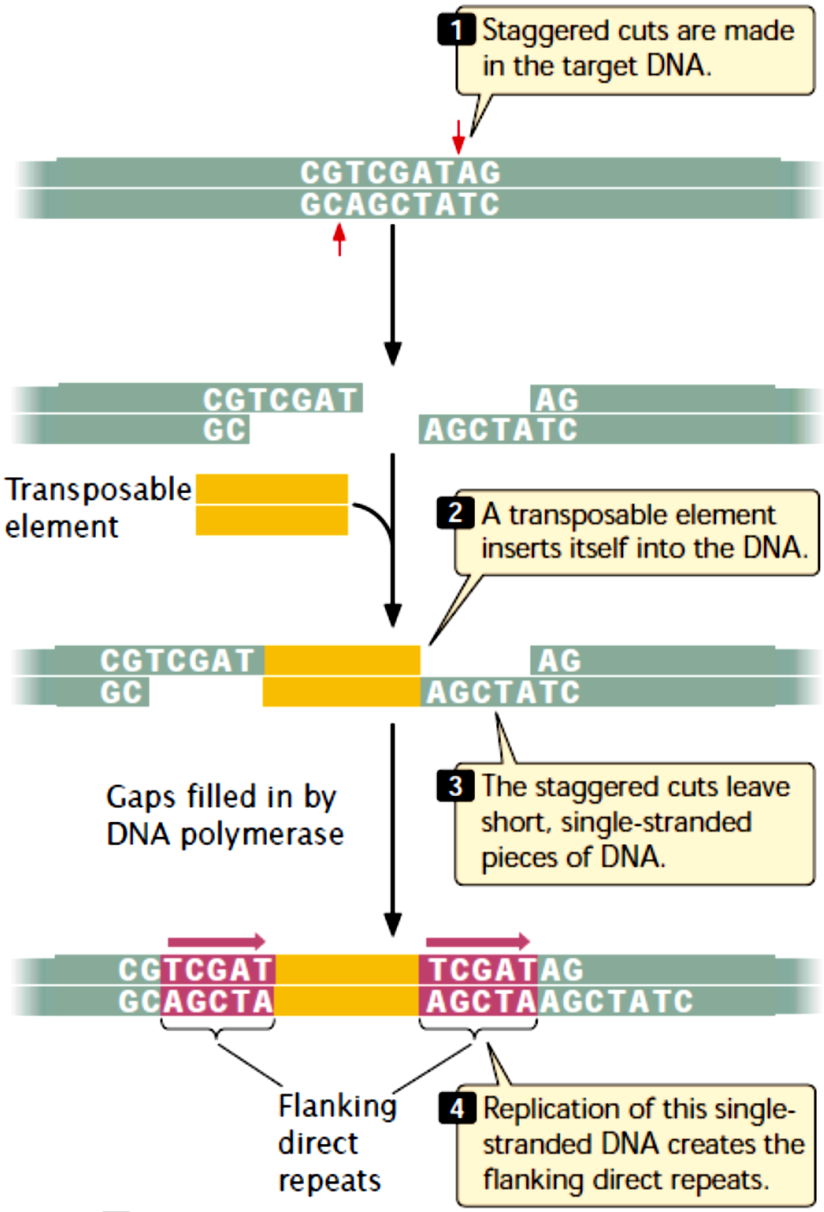


Fig: *Flanking direct repeats are generated when a transposable element inserts into DNA.*

The staggered cuts leave short, single-stranded pieces of DNA on either side of the transposable element. Replication of the single-stranded DNA then creates the flanking direct repeats. At the ends of many, but not all, transposable elements are **terminal inverted repeats,** which are sequences from 9 to 40 bp in length that are inverted complements of one another.

TRANSPOSITION

Transposition is the movement of a transposable element from one location to another. Although our understanding of transposition is still incomplete, it's clear that, rather than a single mechanism, several different mechanisms are required for transposition in both prokaryotic and eukaryotic cells. Nevertheless, all types of transposition have several features in common: (1) staggered breaks are made in the target DNA ; (2) the transposable element is joined to single-stranded ends of the target DNA; and (3) DNA is replicated at the single strand gaps.

MECHANISMS OF TRANSPOSITION

Some transposable elements transpose through DNA intermediates, whereas others use RNA intermediates. Among those that transpose through DNA, transposition may be replicative or non-replicative. In **replicative transposition,** a new copy of the transposable element is introduced at a new site while the old copy remains behind at the original site; the number of copies of the transposable element increases. In **non-replicative transposition,** the transposable element excises from the old site and inserts at a new site without any increase in the number of its copies. Non-replicative transposition requires replication of only the few nucleotides that constitute the direct repeats.

Replicative transposition Replicative transposition, sometimes called copy-and-paste transposition, can be either between two different DNA molecules or between two parts of the same DNA molecule. FIGURE summarizes the steps of transposition between two circular DNA molecules.

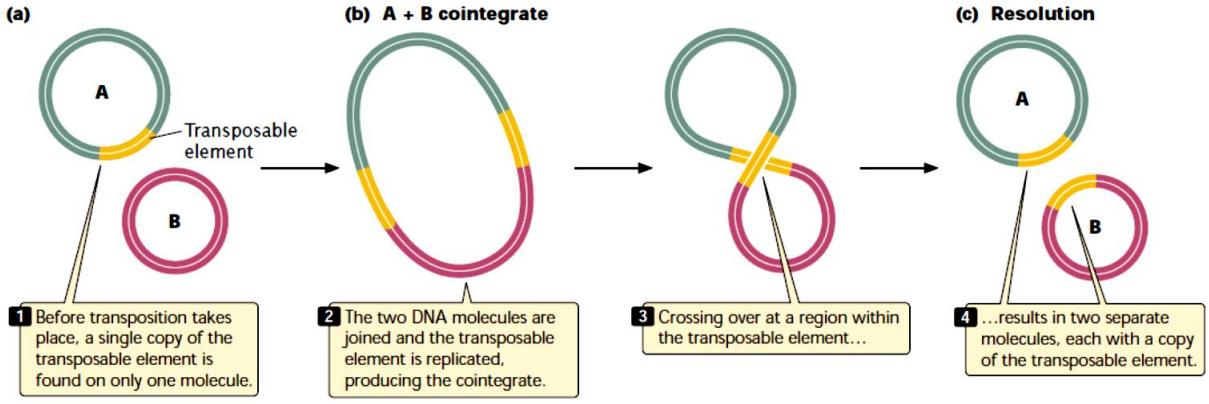


Fig: Replicative transposition increases the number of copies of the transposable element.

Before transposition (see Figure), the transposable element is on one molecule. In the first step, the two DNA molecules are joined, and the transposable element is replicated, producing the **cointegrate structure** that consists of molecules A +B fused together with two copies of the transposable element (see Figure). In a moment, we'll see how the copy is produced, but let's first look at the second step of the replicative transposition process. After cointegrate has formed, crossing over at regions within the transposable elements produces two molecules, each with a copy of the transposable element. This second step is known as resolution of cointegrate.

How are the steps of replicative transposition (cointegrate formation and resolution) brought about? Cointegrate formation requires four events. First, a **transposase** enzyme (often encoded by the transposable element) makes single strand breaks at each end of the transposable element and on either side of the target sequence where the element inserts**.** Second, the free ends of the transposable element attach to the free ends of the target sequence**.** Third, replication takes place on the single-stranded templates, beginning at the 3' ends of the single strands and proceeding through the transposable element**.** This replication creates the cointegrate, with its two copies of both the transposable element and the sequence at the target site, which is now on one side of each copy**.**

The enzymes that perform the replication and ligation functions are cellular enzymes that function in replication and DNA repair. Fourth, after cointegrate has formed, it undergoes resolution, which requires crossing over between sites located within the transposon. Resolution gives rise to two copies of the transposable element**.** The resolution step is brought about by **resolvase** enzymes (encoded in some cases by the transposable element and in other cases by a cellular gene) that function in homologous recombination.

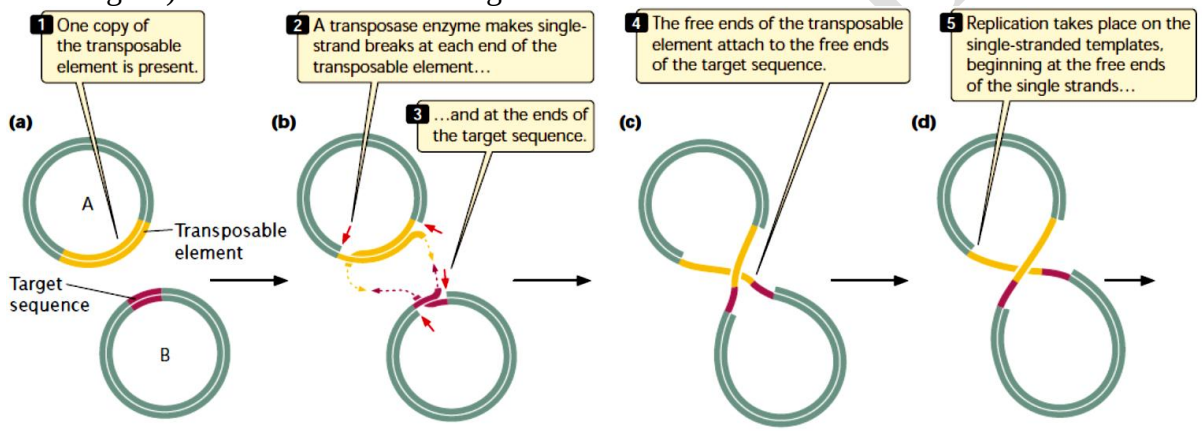


Fig: Replicative transposition requires single-strand breaks, replication, and resolution.

NON-REPLICATIVE TRANSPOSITION

In non-replicative transposition, the transposable element moves from one site to another without replication of the entire transposable element, although short sequences in the target DNA are replicated, generating flanking direct repeats. Sometimes referred to as cut-and-paste transposition, non-replicative transposition requires only that the transposable element and the target DNA be cleaved and joined together.

Cleavage requires a transposase enzyme produced by the transposable element. The joining of the transposable element and target DNA is probably carried out by normal replication and repair enzymes. If a transposable element moves by non-replicative transposition, how does it increase in copy number in the genome? The answer comes from examining the fate of the original site of the element. After excision, a break will be left at the original insertion site. Such breaks are harmful to the cell, and so they are repaired efficiently. One common method of repair is to copy sequence information from a homologous template; the sister chromatid is the preferred template

for this type of repair. Before transposition, both sisters will have a copy of the transposable element. After excision from one chromatid, repair of the break can result in copying the transposable element sequence off the sister chromatid. Thus, the transposable element is moved from the original site to a new site, but a copy is restored to the original site by DNA repair mechanisms.

TRANSPOSITION THROUGH A RNA

Intermediate Eukaryotic transposable elements that transpose through RNA intermediates are called **retrotransposons.** A retrotransposon in DNA is first transcribed into an RNA sequence**,** which may be processed. The processed RNA undergoes reverse transcription by a reverse transcriptase enzyme to produce a double-stranded DNA copy of the RNA**.** Staggered cuts are made in the target DNA**,** and the DNA copy of the retrotransposon inserts into the genome**.** Replication fills in the short gaps produced by the staggered cuts, generating flanking direct repeats on both sides of the retrotransposon.

TRANSPOSABLE ELEMENTS IN BACTERIA

The two major groups of bacterial transposable elements are:

- Simple transposable elements that carry only the information required for movement; and
- More-complex transposable elements that contain DNA sequences not directly related to transposition.

INSERTION SEQUENCES

The simplest type of transposable element in bacterial chromosomes and plasmids is an insertion sequence (IS). This type of element carries only the genetic information necessary for its movement. Insertion sequences are common constituents of bacteria and plasmids. They are designated by IS, followed by an identifying number. For example, IS1 is a common insertion sequence found in E. coli. Insertion sequences are typically from 800 to 2000 bp in length and possess the two hallmarks of transposable elements: terminal inverted repeats and the generation of flanking direct repeats at the site of insertion. Most insertion sequences contain one or two genes that code for transposase. IS1, a typical insertion sequence, is 768 nucleotide pairs long and has terminal inverted repeats of 23 bp at each end (FIGURE).

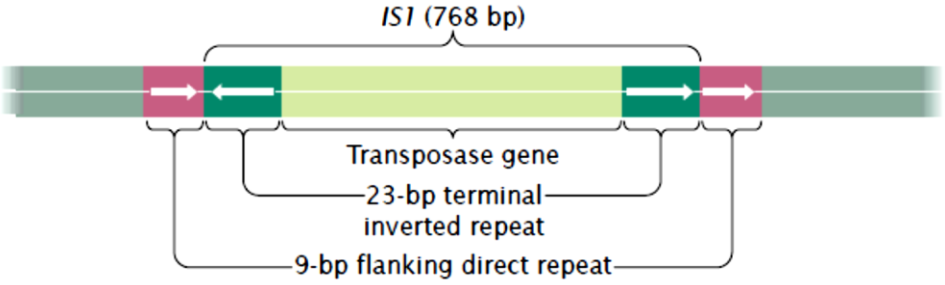


Fig: Insertion sequences are simple transposable elements found in bacteria.

Career Avenues

The flanking direct repeats created by *IS1* are each 9 bp long — the most common length for flanking direct repeats. Table 11.4 summarizes these features for several bacterial insertion sequences.

COMPOSITE TRANSPOSONS

Any segment of DNA that becomes flanked by two copies of an insertion sequence may itself transpose and is called a **composite transposon.** Each type of composite transposon is designated by the abbreviation *Tn*, followed by a number. *Tn10* is a composite transposon of about 9300 bp that carries a gene (about 6500 bp) for tetracycline resistance between two *IS10* insertion sequences. The insertion sequences have terminal inverted repeats; so the composite transposon also ends in inverted repeats. Composite transposons also generate flanking direct repeats at their sites of insertion. The insertion sequences at the ends of a composite transposon may be in the same orientation or they may be inverted relative to one other (as in *Tn10*).

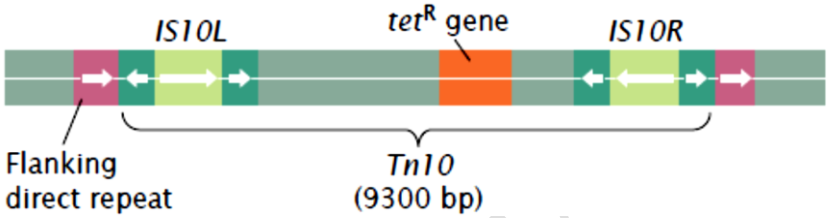


Fig: Tn10 is a composite transposon in bacteria.

NON-COMPOSITE TRANSPOSONS

As already stated, insertion sequences carry only information for their own movement, whereas bacterial transposons are more complex. Some transposable elements in bacteria lack insertion sequences and are referred to as non-composite transposons. For instance, *Tn3* is a non-composite transposon that is about 5000 bp long, possesses terminal inverted repeats of 38 bp, and generates flanking direct repeats that are 5 bp in length. *Tn3* carries genes for transposase and resolvase (mentioned earlier in this chapter), plus a gene that codes for the enzyme -lactamase, which provides resistance to ampicillin.

TRANSPOSABLE ELEMENTS IN EUKARYOTES

Eukaryotic transposable elements can be divided into two groups. One group is structurally similar to transposable elements found in bacteria, typically ending in short inverted repeats and transposing through DNA intermediates. The other group comprises retrotransposons; they use RNA intermediates and are similar in structure and movement to retroviruses. On the basis of their structure, function, and genomic sequences, it is clear that some retrotransposons are evolutionarily related to retroviruses. Although their mechanism of movement is fundamentally different from that of other transposable elements, retrotransposons also generate direct repeats at the point of insertion. Retrotransposons include the *Ty* elements in yeast, the *copia* elements and *P* elements in *Drosophila*, and the *Alu* sequences in humans., Ac and Ds elements in maize Transposable elements were first identified in maize (corn), more than 50 years ago by Barbara McClintock.

Career Avenues

TRANSPOSABLE ELEMENTS IN HUMANS

Almost 50% of the human genome consists of sequences derived from transposable elements, although most of these elements are now inactive and no longer capable of transposing. One of the most common transposable elements in the human genome is *Alu*, named after a restriction enzyme (AluI), which cleaves the element into two parts. Every human cell contains more than 1 million related, but not identical, copies of *Alu* in its chromosomes. Unlike the retrotransposons we have described earlier (*Ty* elements from yeast and *copia* elements from *Drosophila*), *Alu* sequences are not similar to retroviruses. They do not have genes resembling *gag* and *pol*, and are therefore non-autonomous. Rather, *Alu* sequences are similar to the gene that encodes the 7S RNA molecule, which transports newly synthesized proteins across the endoplasmic reticulum. *Alu* sequences create short flanking direct repeats when they insert into DNA and have characteristics that suggest that they have transposed through an RNA intermediate.

Alu belongs to a class of repetitive sequences found frequently in mammalian and some other genomes. These sequences are collectively referred to as SINEs, (short interspersed sequences). The human genome also has many LINEs (long interspersed sequences), which are somewhat more similar in structure to retroviruses, but not as similar as *Ty* or *copia*. The human genome contains evidence for several classes of transposable elements that transpose through a DNA intermediate, by the cut-and-paste mechanism. However, these all appear to have been inactive for about 50 million years; the nonfunctional sequences that remain have been referred to as DNA fossils.

Career Avenues

MOLECULAR BASIS OF GENETIC DISEASES AND APPLICATIONS

GENETIC DISEASES AND FAULTY DNA REPAIR

Several human diseases are connected to defects in DNA repair. These diseases are often associated with high incidences of specific cancers, because defects in DNA repair lead to increased rates of mutation. Among the best studied of the human DNA repair diseases is xeroderma pigmentosum, a rare autosomal recessive condition that includes abnormal skin pigmentation and acute sensitivity to sunlight. Persons who have this disease also have a strong predisposition to skin cancer, with an incidence from 1000 to 2000 times that found in unaffected people. Sunlight includes a strong UV component; so exposure to sunlight produces pyrimidine dimers in the DNA of skin cells. Although human cells lack photolyase (the enzyme that repairs pyrimidine dimers in bacteria), most pyrimidine dimers in humans can be corrected by nucleotideexcision repair. However, the cells of most people with xeroderma pigmentosum are defective in nucleotideexcision repair, and many of their pyrimidine dimers go uncorrected and may lead to cancer.

Xeroderma pigmentosum can result from defects in several different genes; studies have identified at least seven different xeroderma pigmentosum complementation groups, meaning that at least seven genes are required for nucleotide-excision repair in humans. Recent molecular research has led to the identification of genetic defects of nucleotide-excision repair associated with these complementation groups. Some persons with xeroderma pigmentosum have mutations in a gene encoding the protein that recognizes and binds to damaged DNA; others have mutations in a gene encoding helicase. Still others have defects in the genes that play a role in cutting the damaged strand on the 5’or 3’ sides of the pyrimidine dimer. Some persons have a slightly different form of the disease (xeroderma pigmentosum variant) owing to mutations in the gene encoding polymerase ή, the DNA polymerase that bypasses pyrimidine dimers by inserting AA.

Two other genetic diseases due to defects in nucleotideexcision repair are Cockayne syndrome and trichothiodystrophy (also known as brittle-hair syndrome). Persons who have either of these diseases do not have an increased risk of cancer but do exhibit multiple developmental and neurological problems. Both diseases result from mutations in some of the same genes that cause xeroderma pigmentosum. Several of the genes taking part in nucleotide-excision repair produce proteins that also play a role in recombination and the initiation of transcription. These other functions may account for the developmental symptoms seen in Cockayne syndrome and trichothiodystrophy. Another genetic disease caused by faulty DNA repair is an inherited form of colon cancer called hereditary non-polyposis colon cancer (HNPCC). This cancer is one of the most common hereditary cancers, accounting for about 15% of colon cancers. Research indicates that HNPCC arises from mutations in the proteins that carry out mismatch repair.

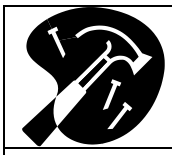
Career Avenues

Li-Fraumeni syndrome is caused by mutations in a gene called *p53*, which plays an important role in regulating the cell cycle. The product encoded by the *p53* gene can halt cell division until damage to DNA has been repaired; it can also directly stimulate DNA repair. The *p53* gene product may actually cause cells with damaged DNA to self-destruct, preventing their mutated genetic instructions from being passed on. Patients who have Li-Fraumeni syndrome exhibit multiple independent cancers in different tissues.

Table: Genetic diseases associated with defects in DNA repair systems

Disease	Symptoms	Genetic Defect
Xeroderma pigmentosum	Frecklelike spots on skin, sensitivity to sunlight, predisposition to skin cancer	Defects in nucleotide-excision repair
Cockayne syndrome	Dwarfism, sensitivity to sunlight, premature aging, deafness, mental retardation	Defects in nucleotide-excision repair
Trichothiodystrophy	Brittle hair, skin abnormalities, short stature, immature sexual development, characteristic facial features	Defects in nucleotide-excision repair
Hereditary nonpolyposis colon cancer	Predisposition to colon cancer	Defects in mismatch repair
Fanconi anemia	Increased skin pigmentation, abnormalities of skeleton, heart, and kidneys, predisposition to leukemia	Possibly defects in the repair of interstrand cross-links
Ataxia telangiectasia	Defective muscle coordination, dilation of blood vessels in skin and eyes, immune deficiencies, sensitivity to ionizing radiation, predisposition to cancer	Defects in DNA damage detection and response
Li-Fraumeni syndrome	Predisposition to cancer in many different tissues	Defects in DNA damage response

Career



GATE Questions

GATE BT 2010

Q.13 Determine the correctness or otherwise of the following Assertion (a) and the Reason (r)
Assertion: Cytoplasmic male sterility (cms) is invariably due to defect(s) in mitochondrial function.
Reason: cms can be overcome by pollinating a fertility restoring (Rf) plant with pollen from a non cms plant.
(A) both (a) and (r) are true but (r) is the correct reason for (a).
(B) both (a) and (r) are true but (r) is not the correct reason for (a).
(C) (a) is false but (r) is true.
(D) (a) is true but (r) is false.

Q.14. True breeding *Drosophila* flies with curved wings and dark bodies were mated with true breeding short wings and tan body *Drosophila*. The F_1 progeny was observed to be with curved wings and tan body. The F_1 progeny was again allowed to breed and produced flies of the following phenotype, 45 curved wings tan body, 15 short wings tan body, 16 curved wings dark body and, 6 short wings dark body.
The mode of inheritance is

(A) Typical Mendelian with curved wings and tan body being dominant.
(B) Typical non-Mendelian with curved wings and tan body not following any pattern.
(C) Mendelian with suppression of phenotypes.
(D) Mendelian with single crossover.

GATE BT 2011

Q.15 Diploid *Drosophila* has eight chromosomes. Which one of the following terms should NOT be used to describe *Drosophila* with sixteen number of chromosomes?
(A) Polyploid (B) Aneuploid. (C) Euploid (D) Tetraploid

Q.16 The study of evolutionary relationships is known as
(A) Genomics (B) Proteomics (C) Phylogenetics (D) Genetics

Common Data Questions 17 and 18: Red-green colour blindness is inherited as a recessive X-linked trait.

Q.17 what will be the probability of having the colour-blind son to a woman with phenotypically normal parents and a colour-blind brother, and married to a normal man? (Assume that she has no previous children)
(A) 100% (B) 50% (C) 25% (D) 12.5%

Career Avenues

Q.18 What will be the probability of having the colour-blind daughter to a phenotypically normal woman, who already had one colour-blind son, and is married to a colour-blind man?

(A) 75% (B) 50% (C) 25% (D) 15%

GATE BT 2012

Q.19 The direction of shell coiling in the snail *Linnaea peregra* is a classic example of
(A) chromosomal inheritance. (B) extra- chromosomal inheritance.
(C) chromosomal translocation. (D) homologous recombination.

Q.20 Match the terms in Group I with the ploidy in Group II.

Group I	Group II
P. Disome	1. 2n + 1
Q. Monosome	2. 2n - 1
R. Nullisome	3. n - 1
S. Trisome	4. n + 1

(A) P-4,Q-2,R-3,S-1 (B) P-4,Q-3,R-1,S-2
(C) P-2,Q-3,R-4,S-1 (D) P-1,Q-4,R-3,S-2

Answers

13 – D. 14 – B. 15 – B. 16 – C. 17 – B. 18 – B. 19 – B. 20 – C.

Career Avenues